IDEA for teachers

Tutorial for Justice: Kolibri, Nextcloud, Wikipedia offline, Files Disk, Backup, Console

For Justice and the teachers he trains. Draft, October 2026

IDEA - offline school computing. Prepared for Koen Swings's team.
Screenshots from the IDEA demo box with demo accounts. No passwords are shown.

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1. What IDEA is

IDEA (Initiative for Digital Education in Africa) is a charity based in Belgium. It puts offline computer systems in rural schools. Everything works from a small server in the school, without internet.

What is in the school

- **The IDEA box (Engine):** a small Raspberry Pi computer that runs the school's apps. On the demo box it is called `wizardly-hugle`.
- **Disks:** the box has its own **System Disk**. Extra USB disks can be plugged in: a **Backup Disk** that saves the apps' data (chapter 9) and a **Files Disk** for your own files (chapter 8).
- **The IDEA Wi-Fi:** teachers and learners connect their phones, tablets or laptops to it. You only need a web browser.
- **The Console:** a web page that lists the apps and opens them (chapter 2). Operators also use it to look after disks and backups (chapters 10 and 11).

The apps

App	What you use it for	Chapters
Kolibri	Ready-made lessons, videos and exercises (for example Khan Academy). You make classes, lessons and quizzes and see each learner's progress.	3, 4
Nextcloud	Your own files: share worksheets with a class, write documents together, collect homework.	5, 6, 8
Kiwix	Wikipedia and other reference books, offline.	7

Who does what

- **Teachers** open apps from the Console and use Kolibri, Nextcloud and Kiwix (chapters 2 to 7).
- **Learners** sign in with the accounts the teacher gives them (chapters 4 and 6).
- **The school operator** (with the IDEA contact on the phone) looks after disks, backups and restores (chapters 8 to 11).

How to use this guide

Each chapter has numbered steps with a screenshot. The screenshots come from the IDEA demo box, using demo accounts whose names start with `demo` (see the appendix). Your school's names and addresses will be different. Text marked TO CONFIRM still has to be checked by the IDEA team.

> **No internet needed.** Everything in this guide works inside the school. Nothing you do is sent to the internet.

2. Getting connected and opening an app

You need: a phone, tablet or laptop with a web browser, and the school's IDEA Wi-Fi.

1. Connect your device to the IDEA Wi-Fi (name and password: **TO CONFIRM** per school; ask your IDEA contact).
2. Open the browser and go to the Console address your IDEA contact gives you. The Console opens on the **Apps** page: one card per app (Kiwix, Kolibri, Nextcloud, …). The green dot at the top shows the Console is connected to the IDEA box.

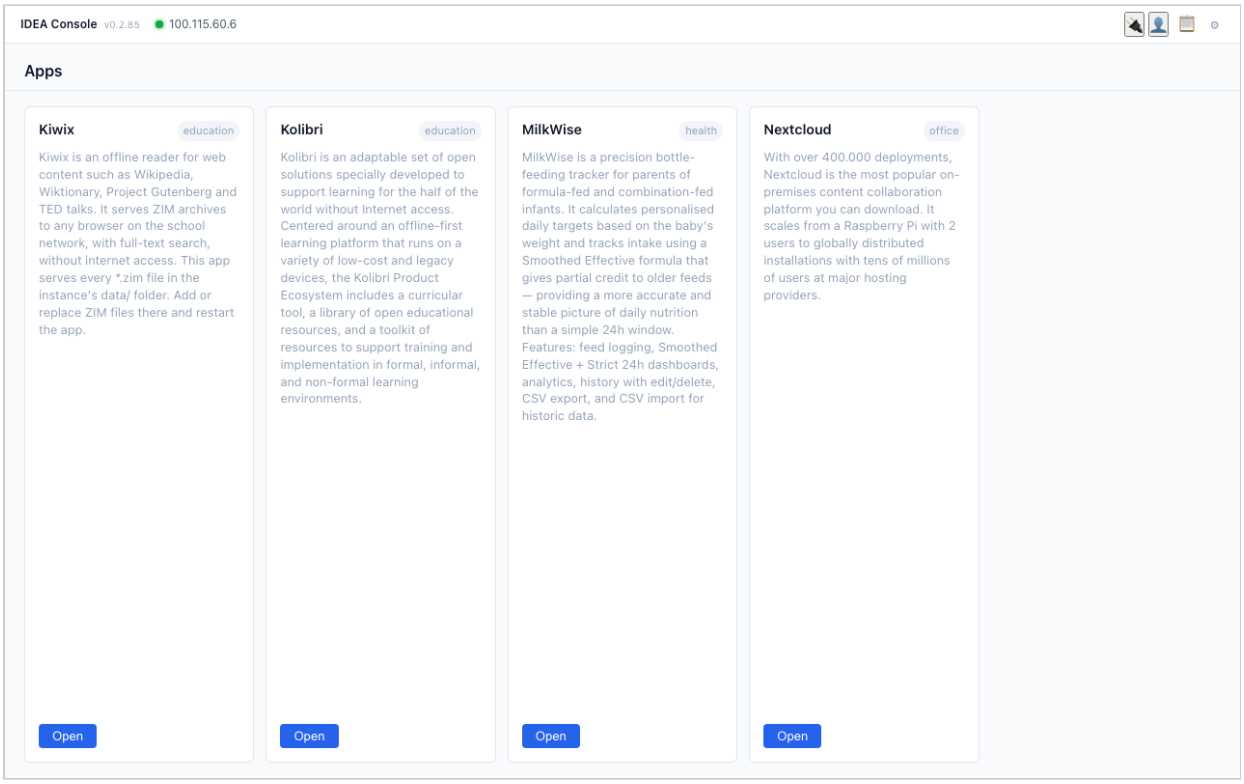


Figure 1. Console Apps page

3. Find the card of the app you need and click its blue **Open** button. The app opens in a new tab. Here: Kolibri's sign-in page.

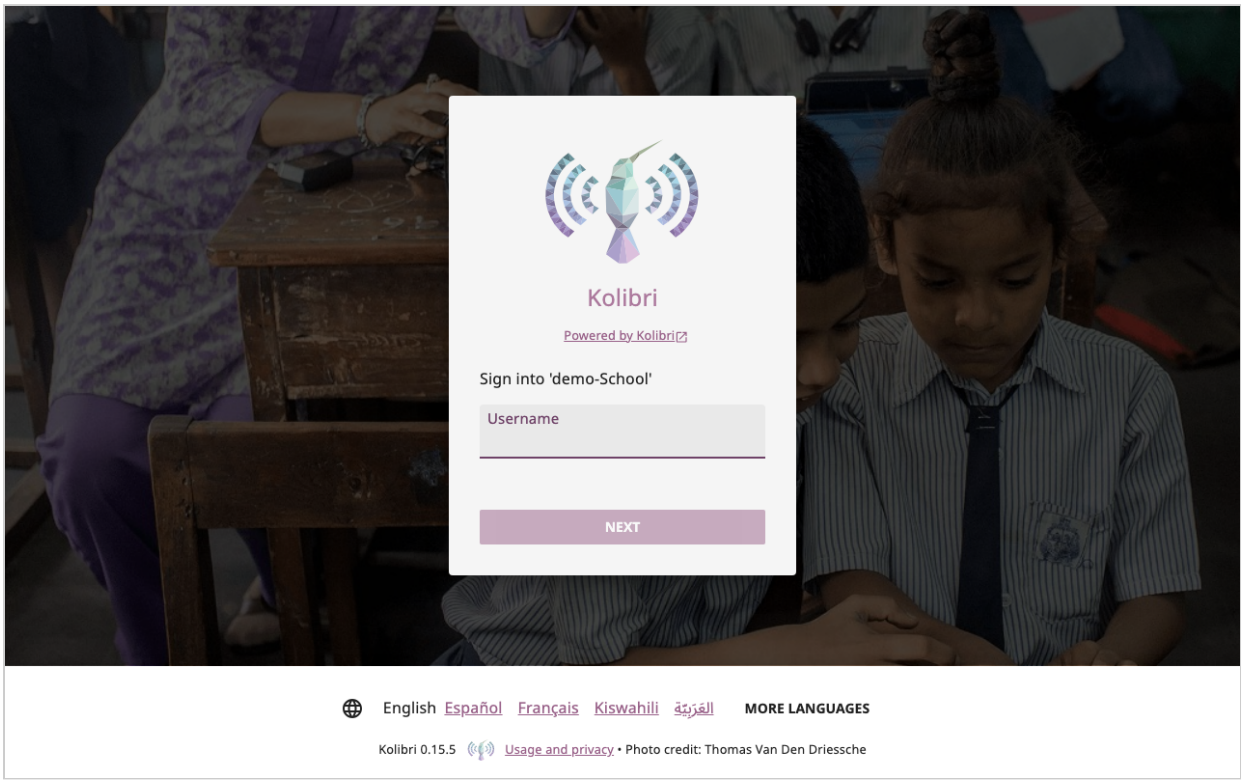


Figure 2. Kolibri opened from the Console

Important:

- A card that is greyed out and says **Not available** is not running. Do not try to fix it: tell your IDEA contact or the school operator.
- Do not type numbers (ports) into the address bar. Always use **Open** in the Console.
- The Console has an operator area (person icon, **Operator Login**). Teachers do not need it.

Tip: you can bookmark the page that opens. If the bookmark stops working, go back to the Console and use **Open** again.

3. Kolibri for teachers

Demo: school (facility) `demo-School`, teacher `demo_teacher1`, class `demo-Grade5A`, channel **Khan Academy (English - US curriculum)** > Math by grade > 5th grade > Add and subtract fractions.

Signing in

1. Open Kolibri from the Console (chapter 2). Type your username and click **NEXT**, then type your password.

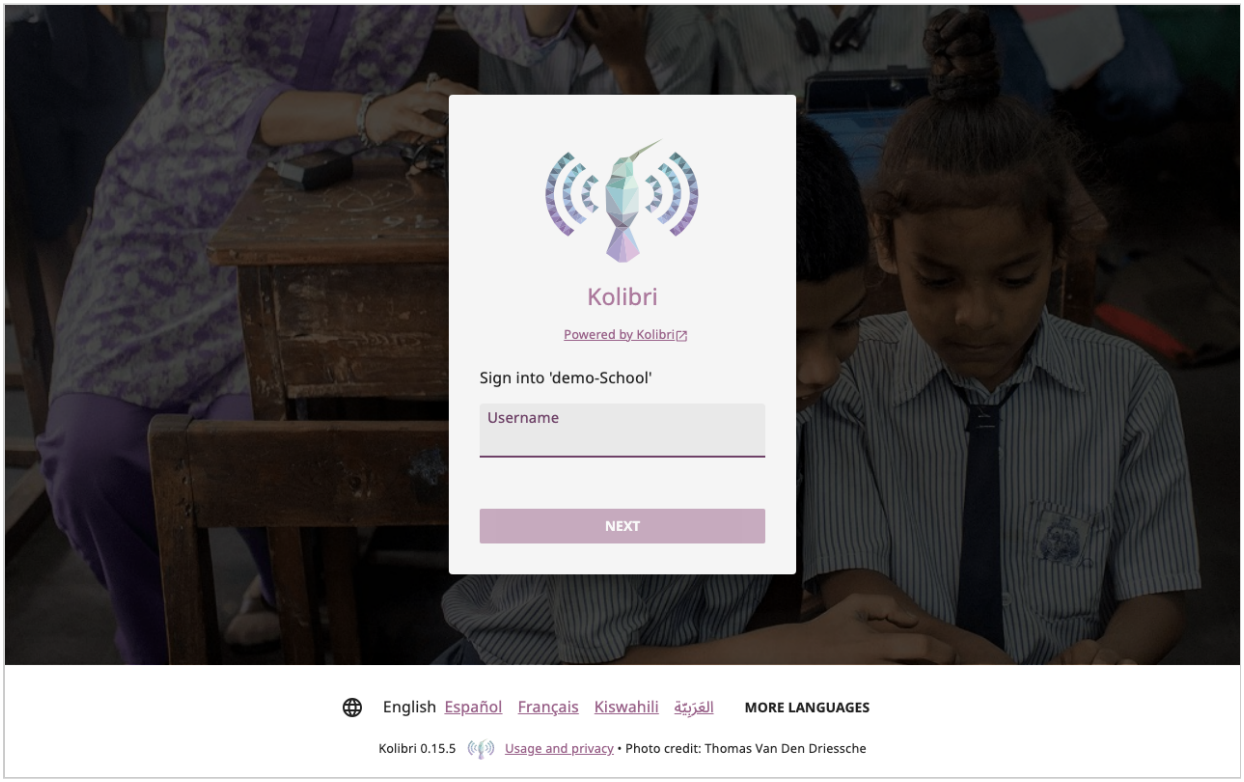


Figure 3. Sign in

K1. Create a class

Creating classes needs a Kolibri admin account (in the demo, `demo_teacher1` is admin and coach).

1. Open **Facility** > **CLASSES**. You see the list of classes.

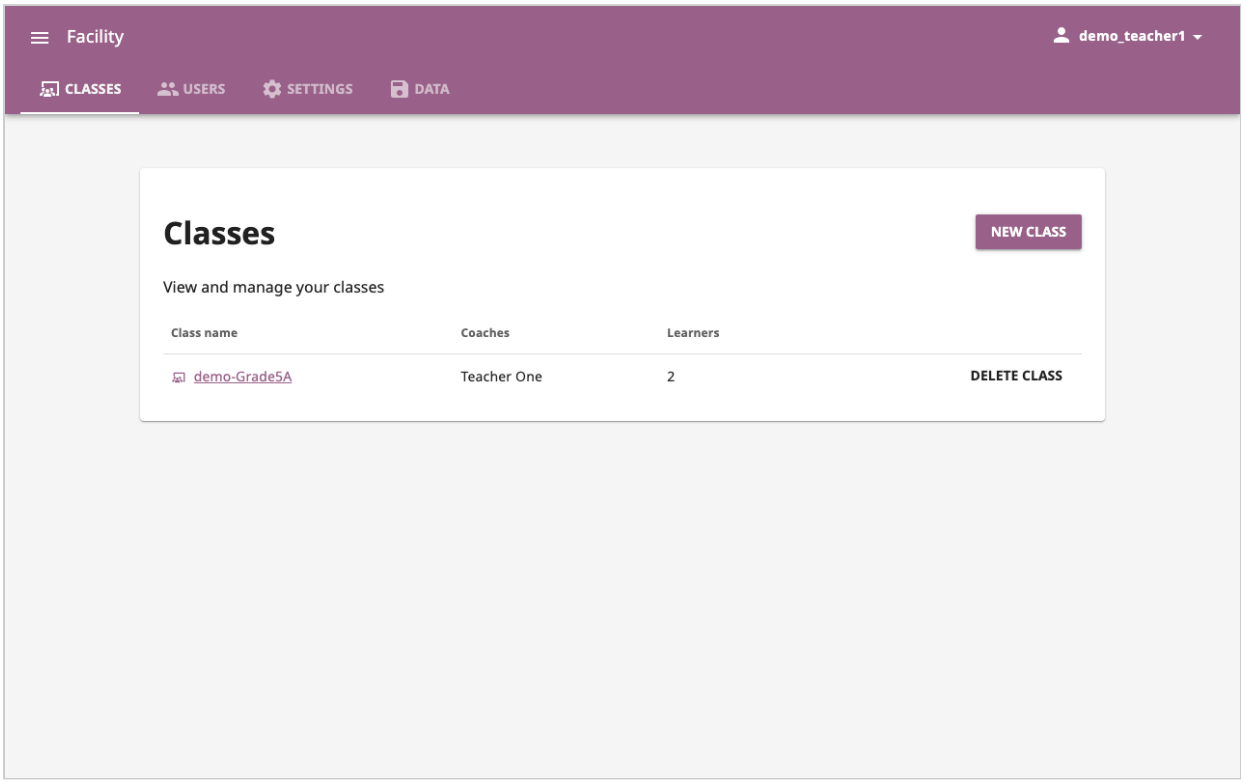


Figure 4. Classes list

2. Click **NEW CLASS**, type the class name, click **SAVE**. (In the demo we pressed CANCEL because `demo-Grade5A` already exists.)

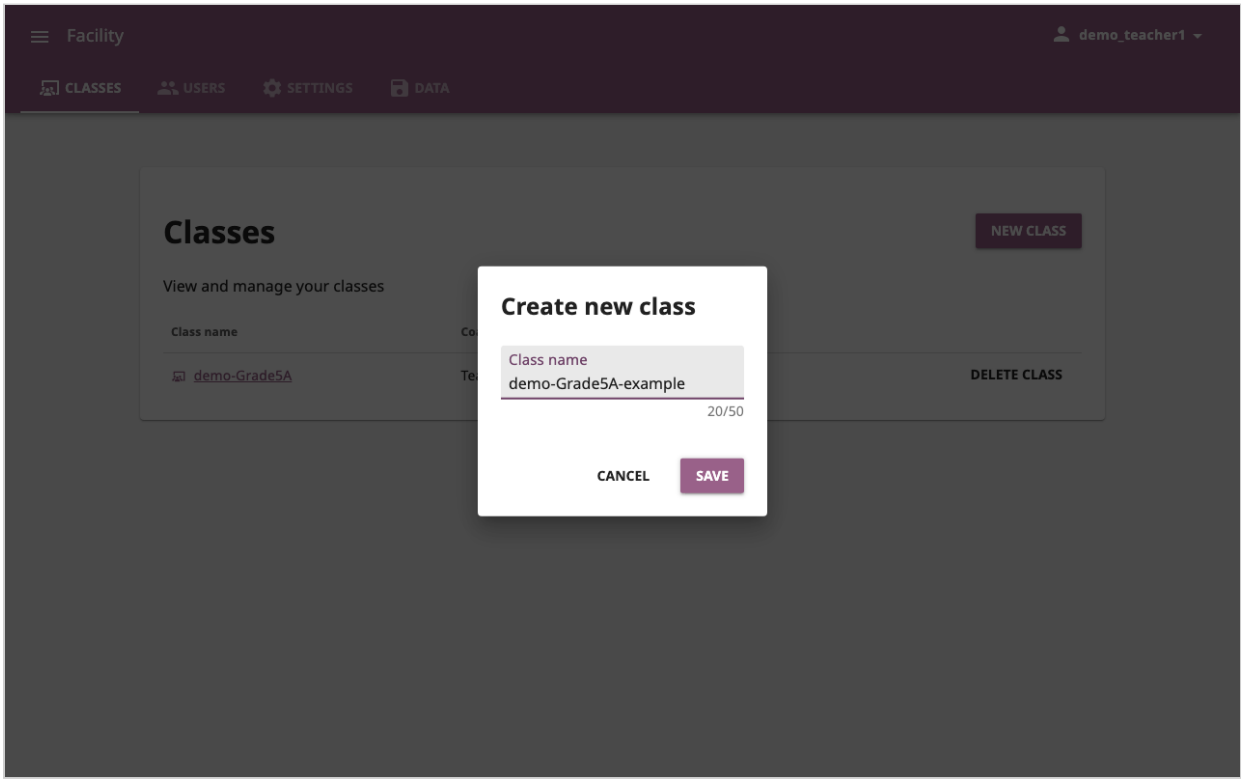


Figure 5. Create new class

K2. Enroll learners

1. Click the class name. The class page lists **Coaches** and **Learners**. Click **ENROLL LEARNERS**.

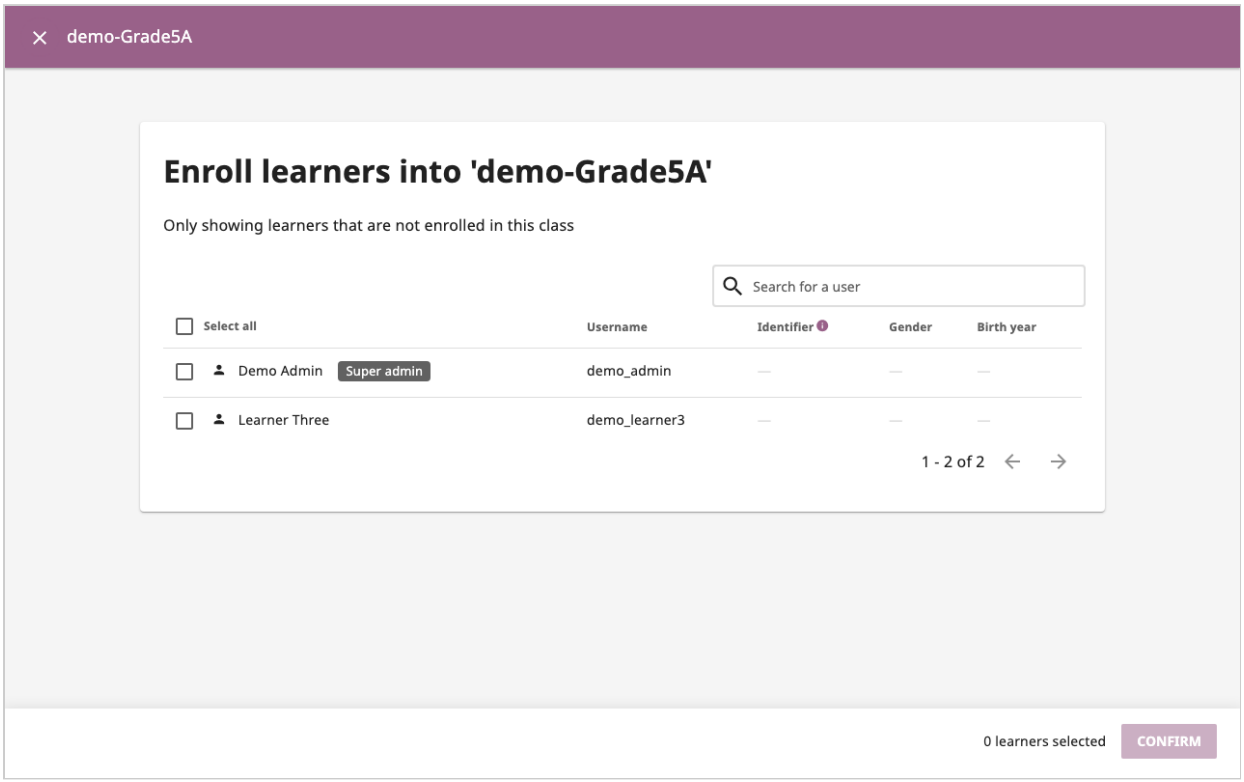


Figure 6. Enroll learners page

2. The list only shows learners who are not yet in the class. Tick each learner (here: Learner Three), then click **CONFIRM**.

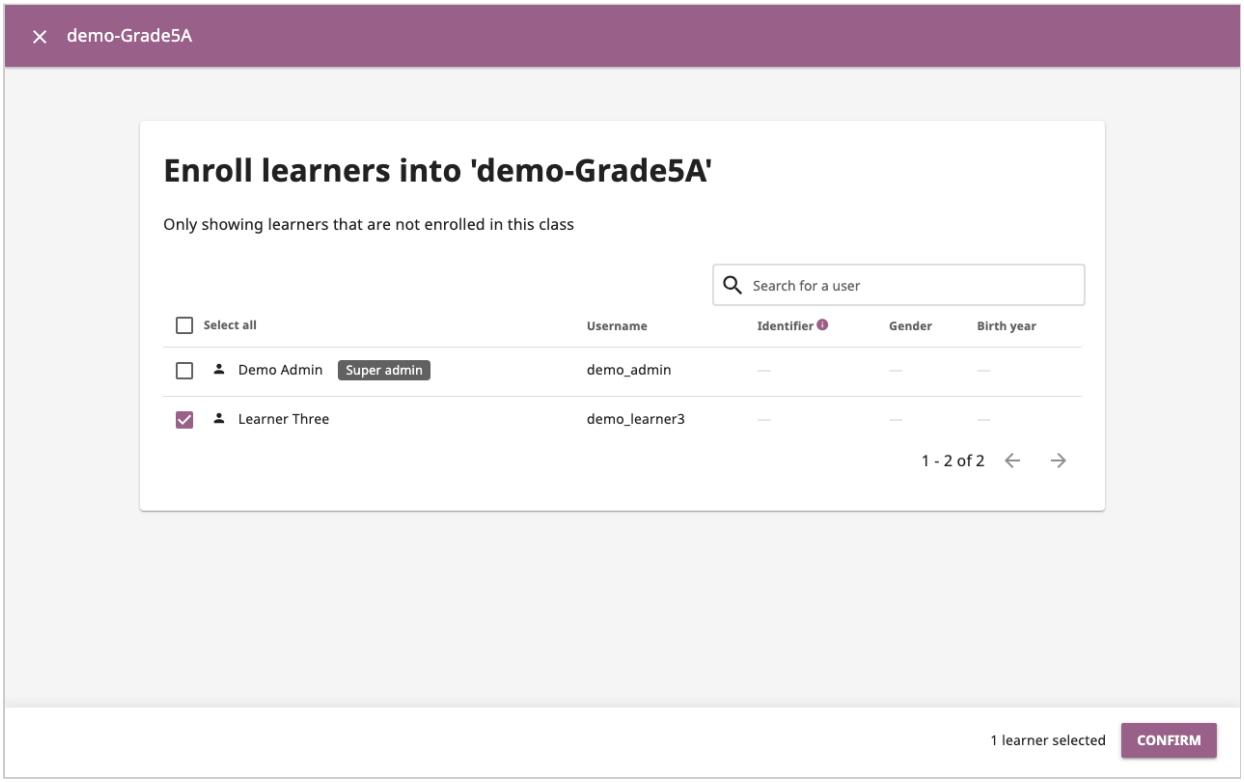


Figure 7. Learner ticked

3. The learner now appears under Learners.

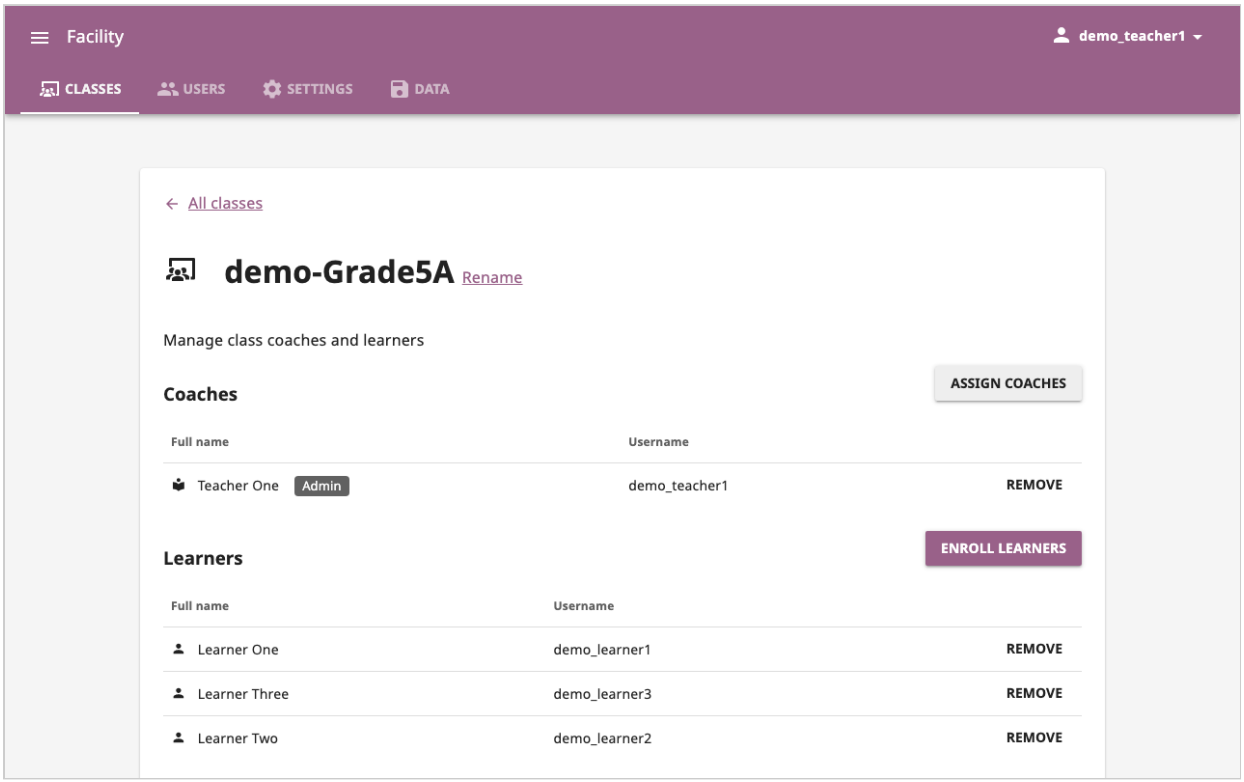


Figure 8. Learner enrolled

K3. Make a lesson and show it to the class

A good lesson has 3 videos (explain) and 3 exercises (practise).

1. Open **Coach**, choose the class, click **PLAN** > **LESSONS** > **NEW LESSON**.

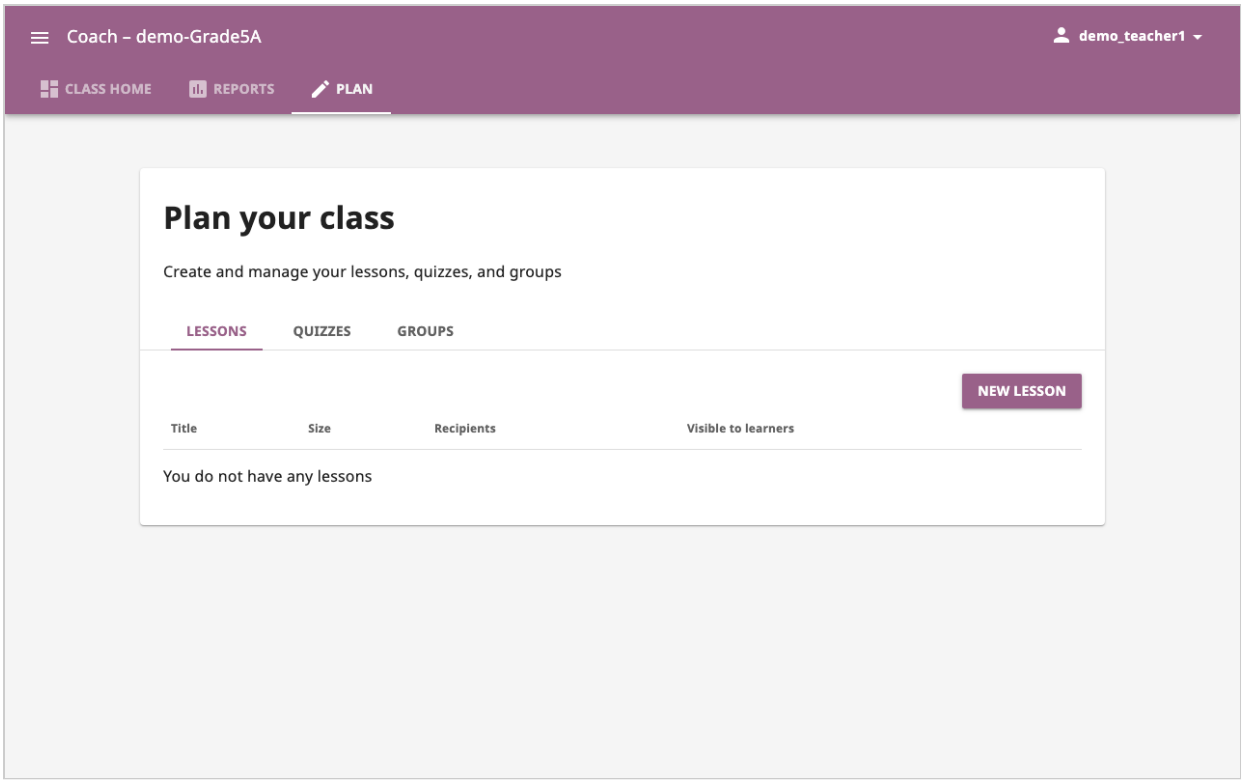


Figure 9. Plan your class

2. Type a title (`demo-Fractions lesson`). Under **Recipients** keep **Entire class**. Click **SAVE CHANGES**.

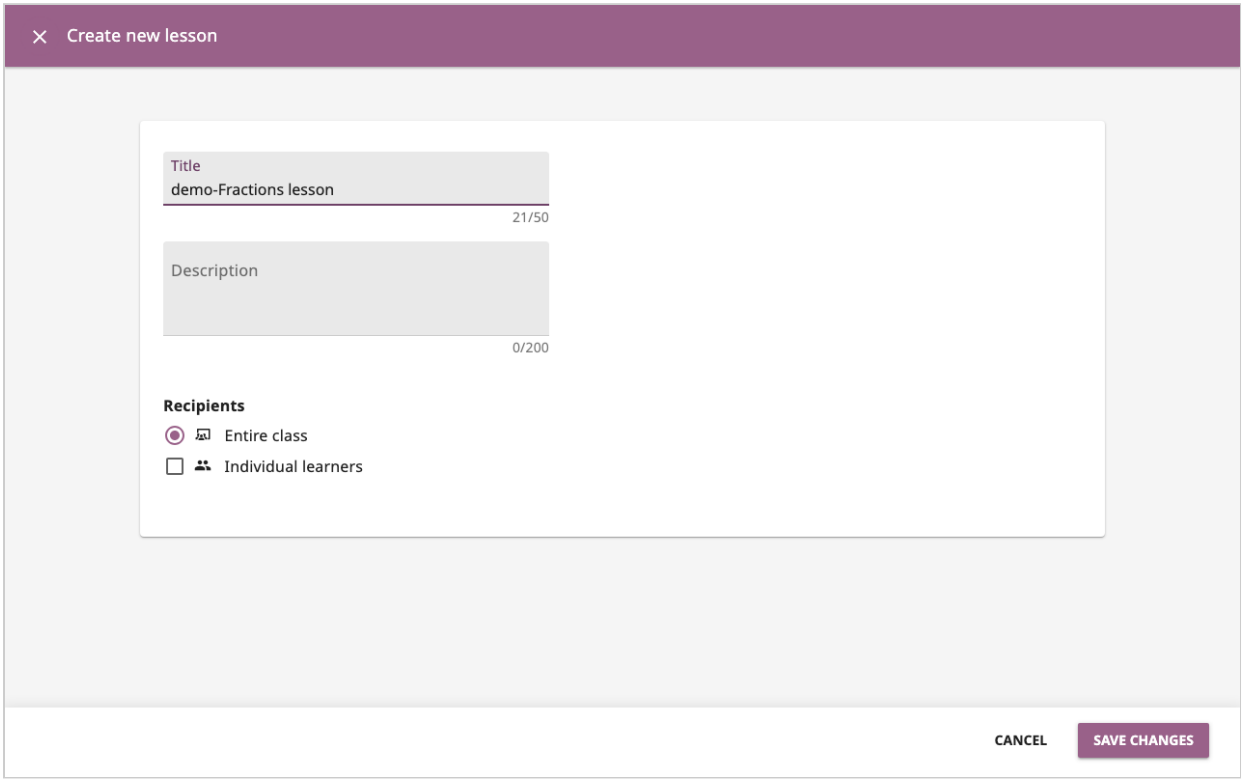


Figure 10. New lesson

3. On the lesson page, click **MANAGE RESOURCES**.

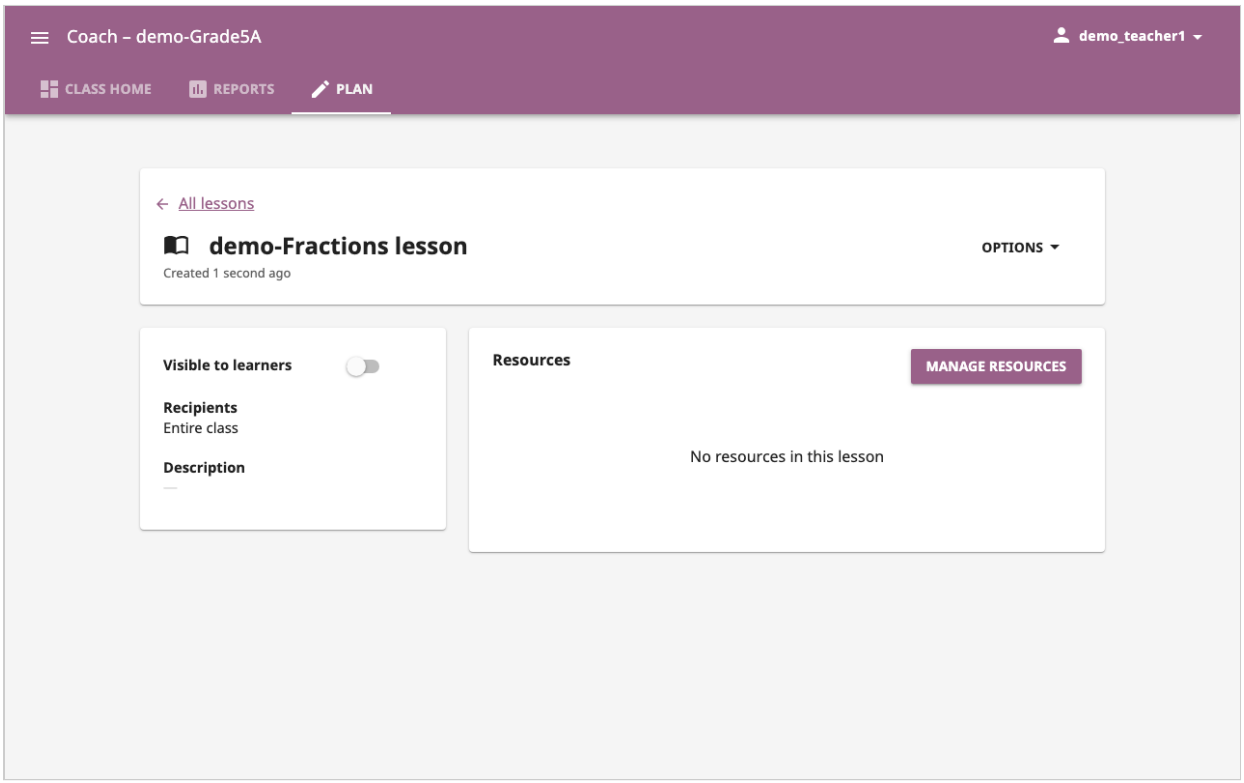


Figure 11. Empty lesson

4. Choose the channel, then click through Math by grade > 5th grade > Add and subtract fractions.

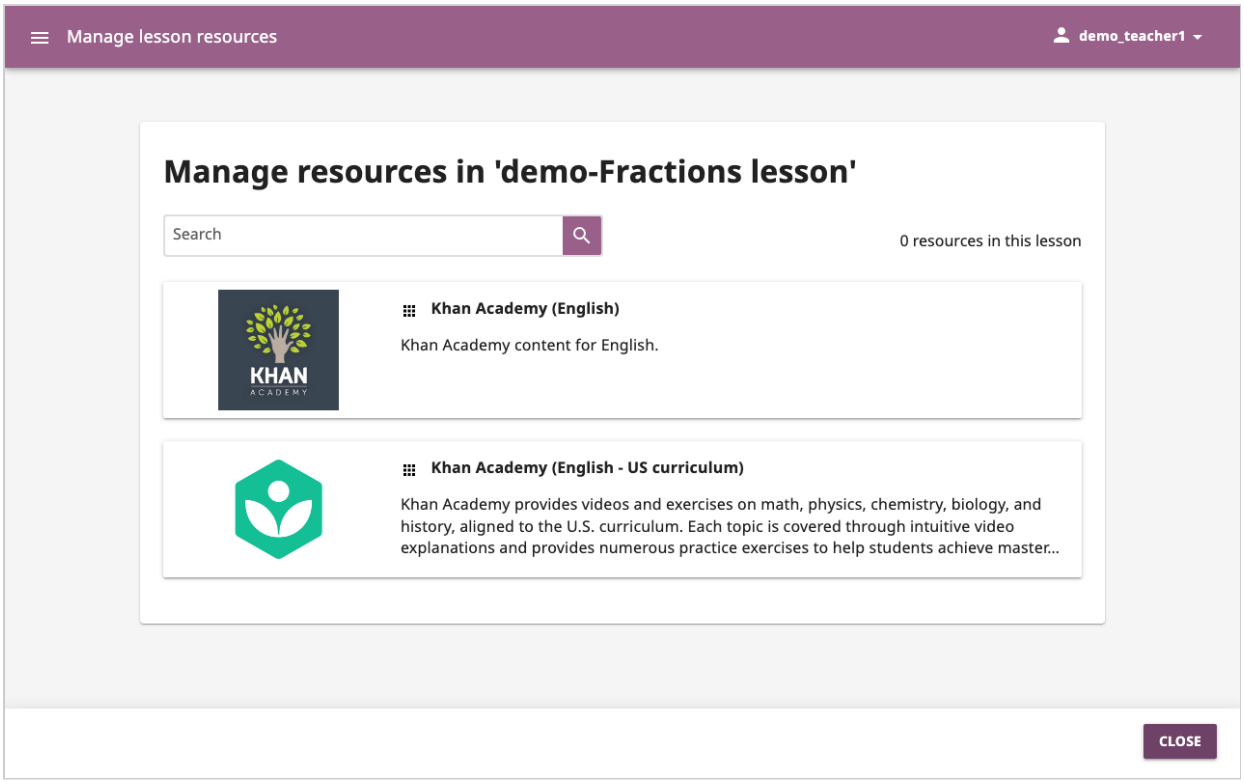


Figure 12. Channels

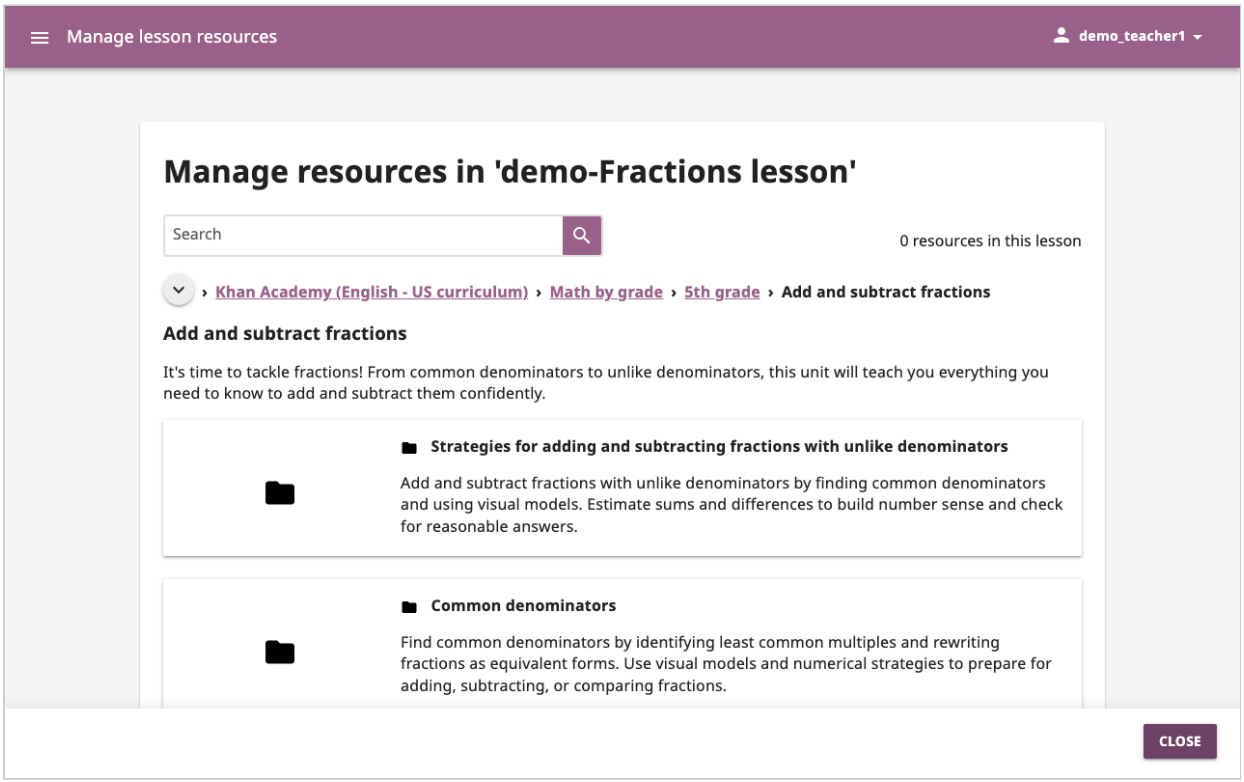


Figure 13. Topic

5. Open "Adding and subtracting fractions with unlike denominators". Tick 3 videos and 3 exercises (exercises say "Practice ..."). Each tick is saved straight away; a message "resources added" appears. Click **CLOSE** when done.

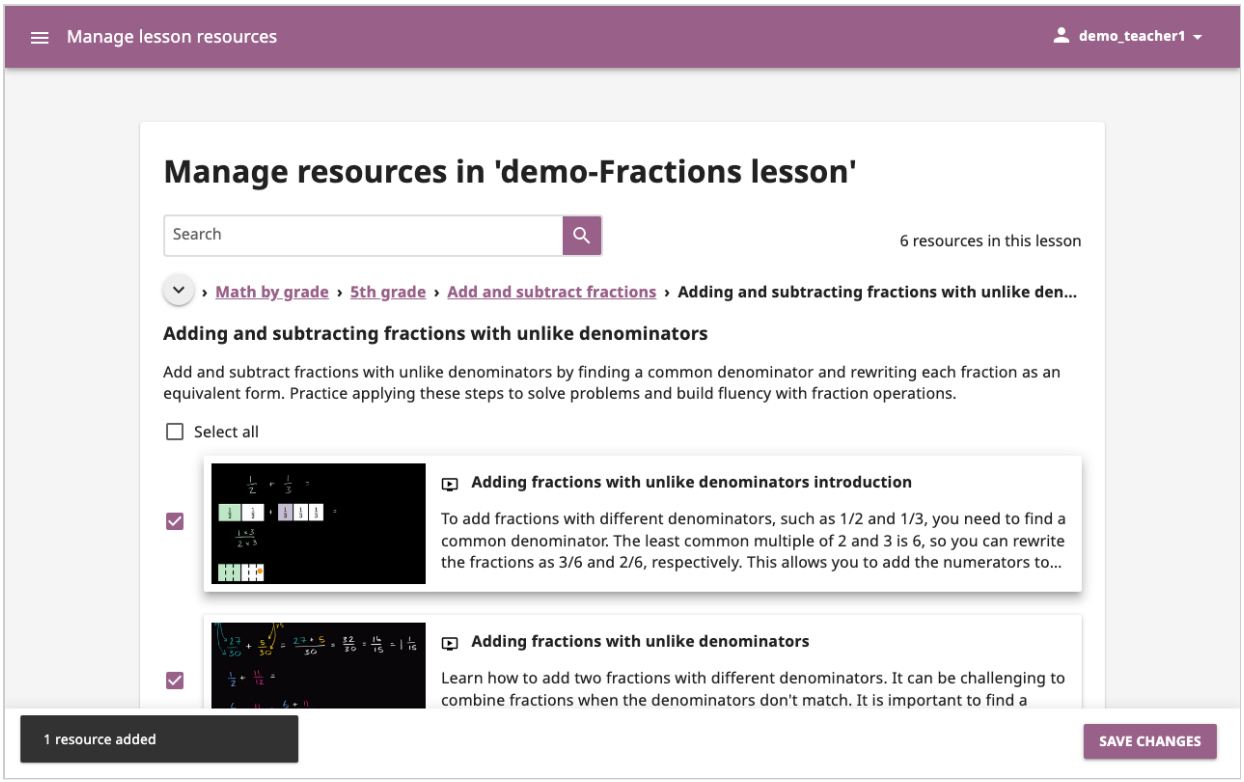


Figure 14. Resources ticked

6. Back on the lesson page, check the 6 resources and switch on **Visible to learners**.

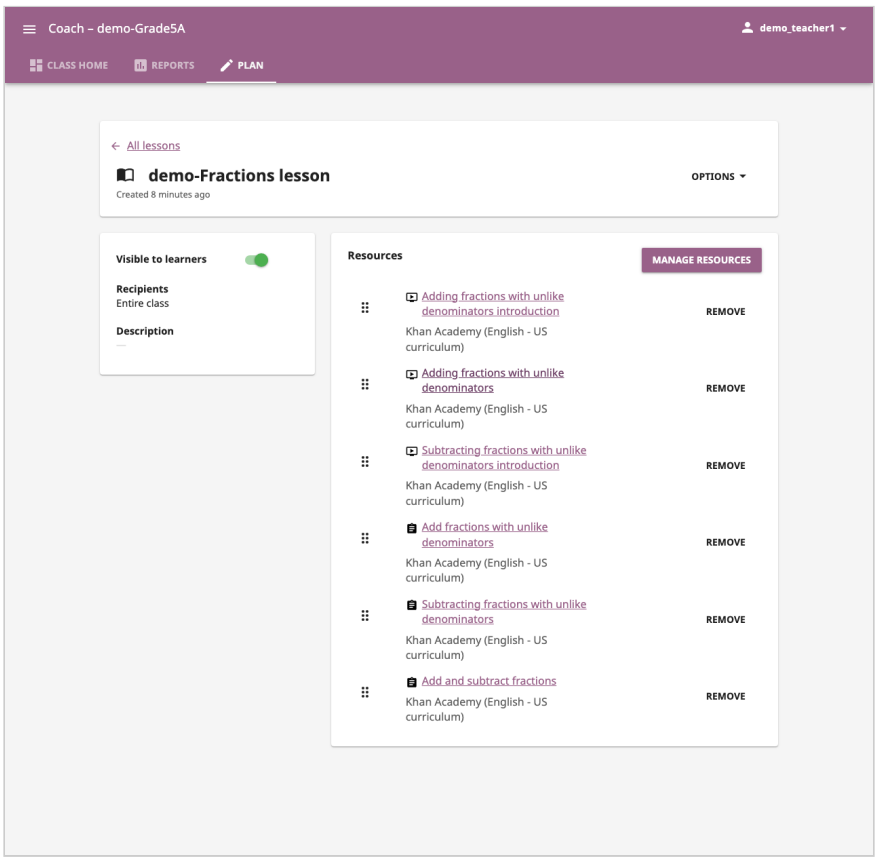


Figure 15. Lesson with 6 resources, visible

K4. Make a quiz

Quizzes are built from exercises only.

1. **PLAN** > **QUIZZES** > **NEW QUIZ**.

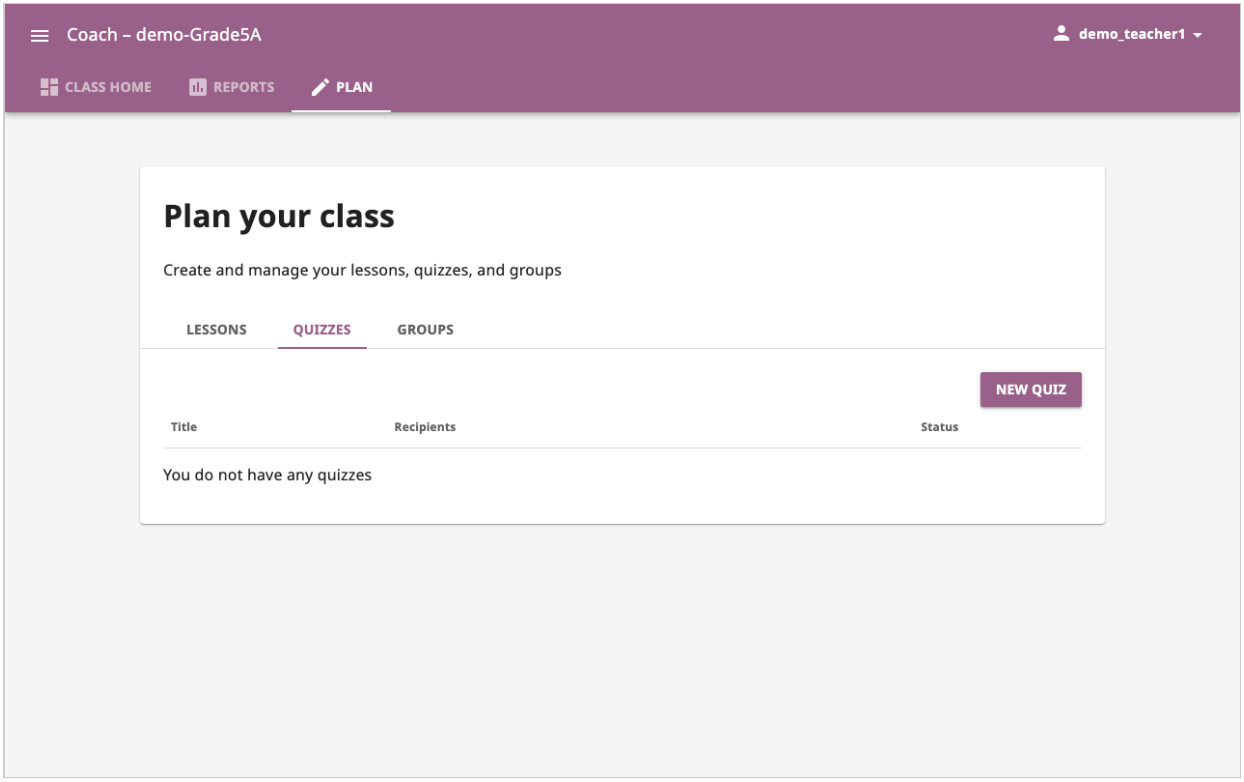


Figure 16. Quizzes tab

2. Type a title (`demo-Fractions quiz`) and the **Number of questions** (6). Browse to Add and subtract fractions and tick a folder (or single exercises). Click **CONTINUE**.

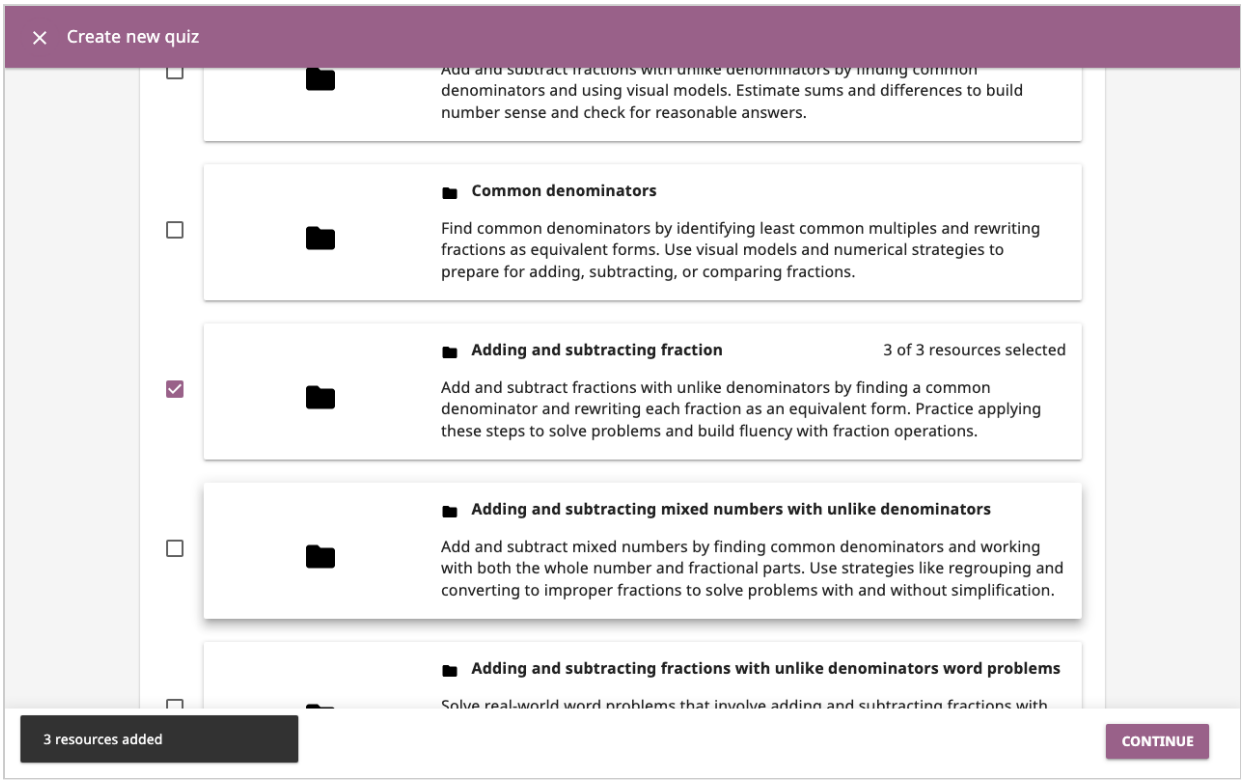


Figure 17. Create new quiz

3. **Preview quiz**: check the questions, choose **Randomized** or **Fixed** order, click **FINISH**.

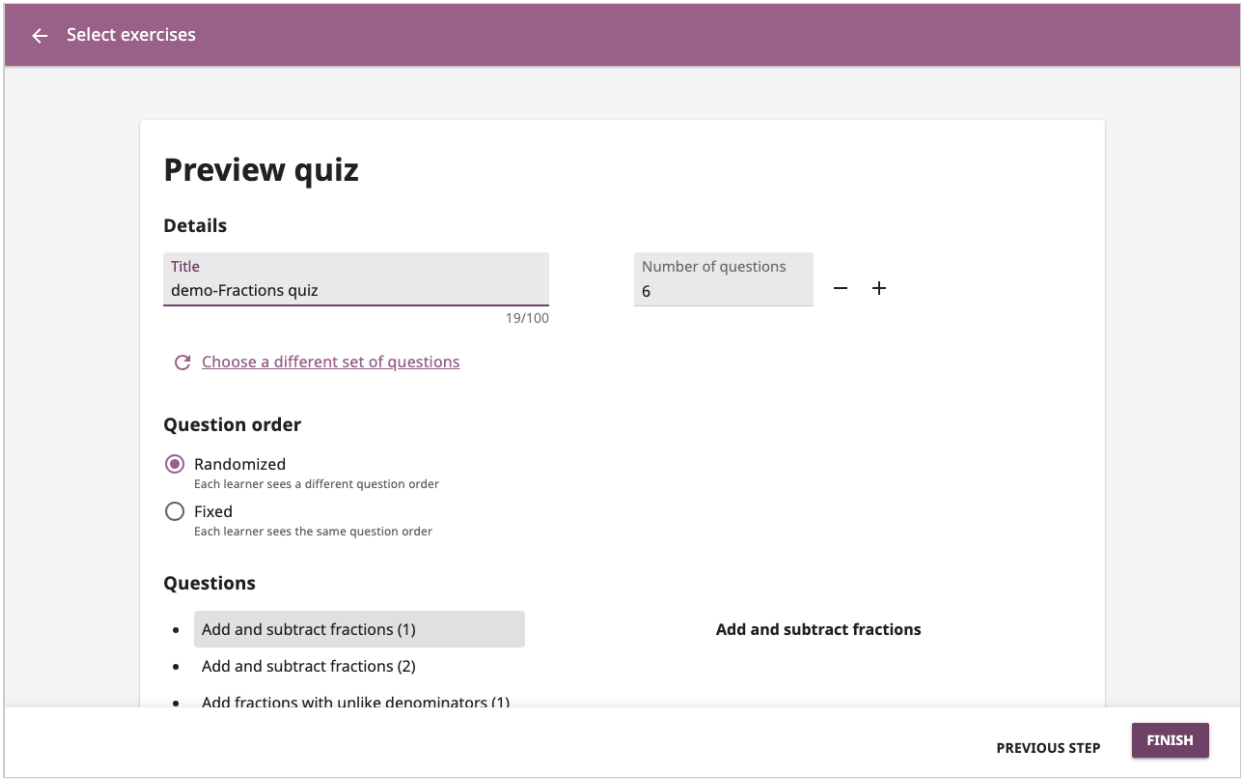


Figure 18. Preview quiz

4. The quiz is not open yet. When the class is ready, click **START QUIZ** and then **CONTINUE**. Later use **End quiz** (Options; **TO CONFIRM** wording) to close it.

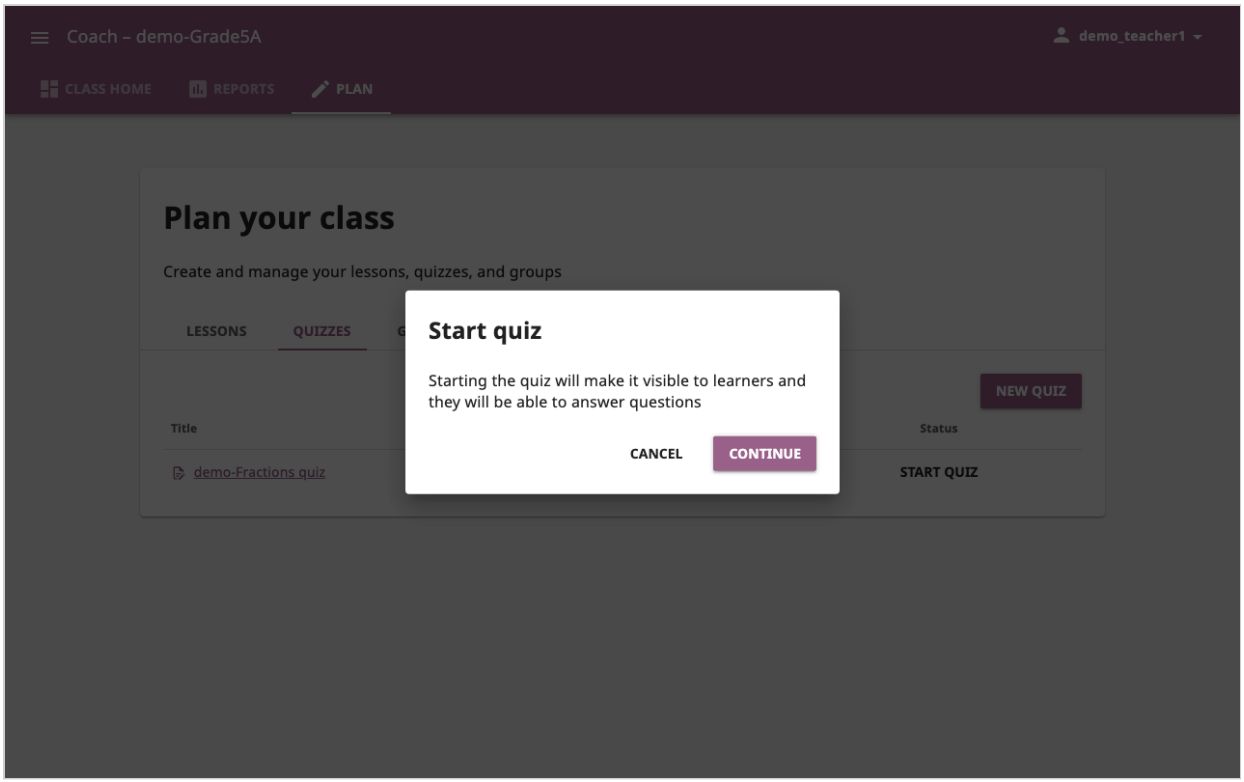


Figure 19. Start quiz

K5. See progress (Reports)

Do this after learners have worked (chapter 4).

1. Click **REPORTS**. Tabs: LESSONS, QUIZZES, GROUPS, LEARNERS.

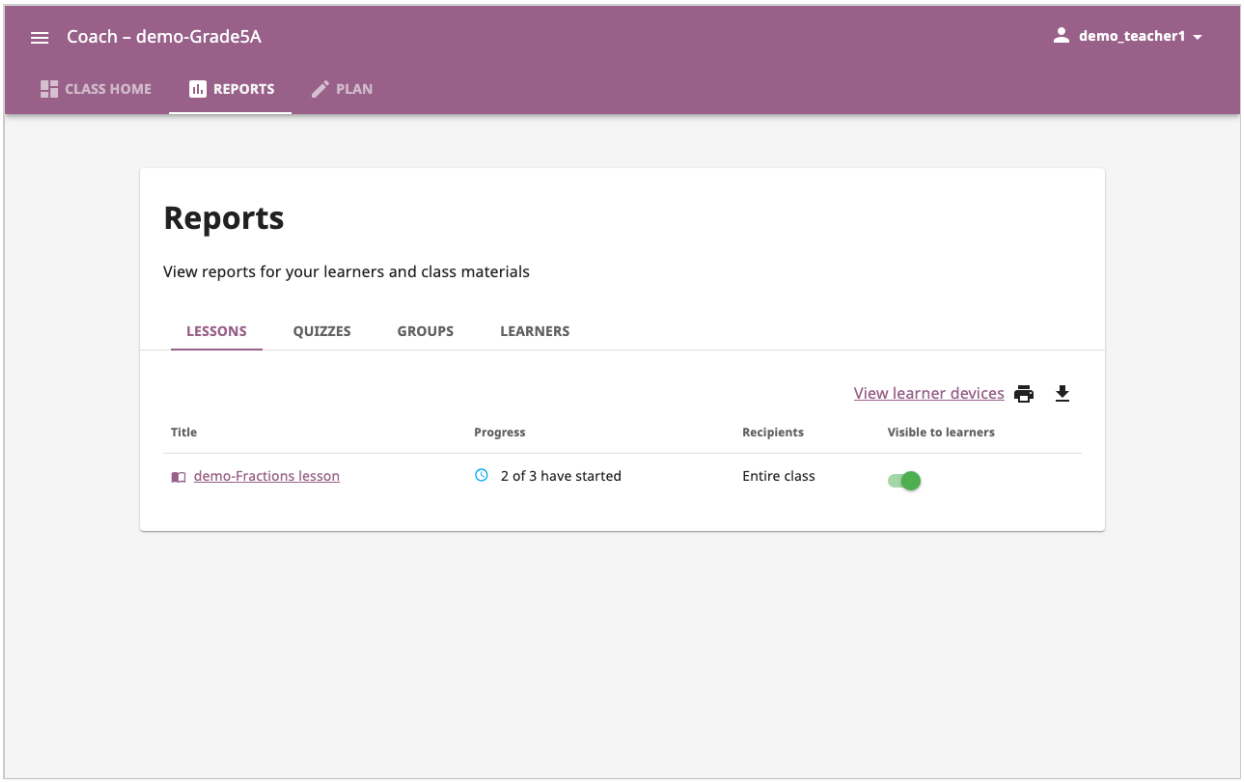


Figure 20. Reports

2. **LESSONS** > the lesson > **LEARNERS**: who completed, started or did not start.

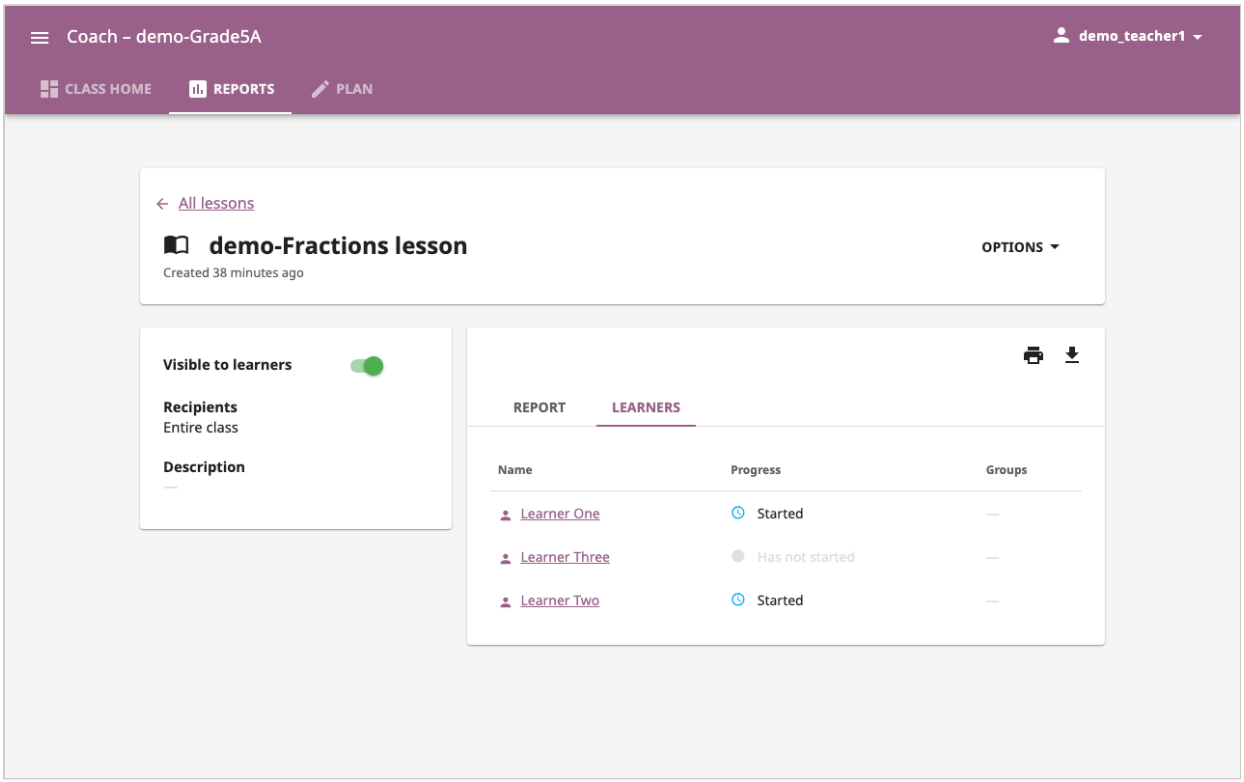


Figure 21. Lesson report

3. **QUIZZES** > the quiz: status and score per learner; **DIFFICULT QUESTIONS** shows what to re-teach.

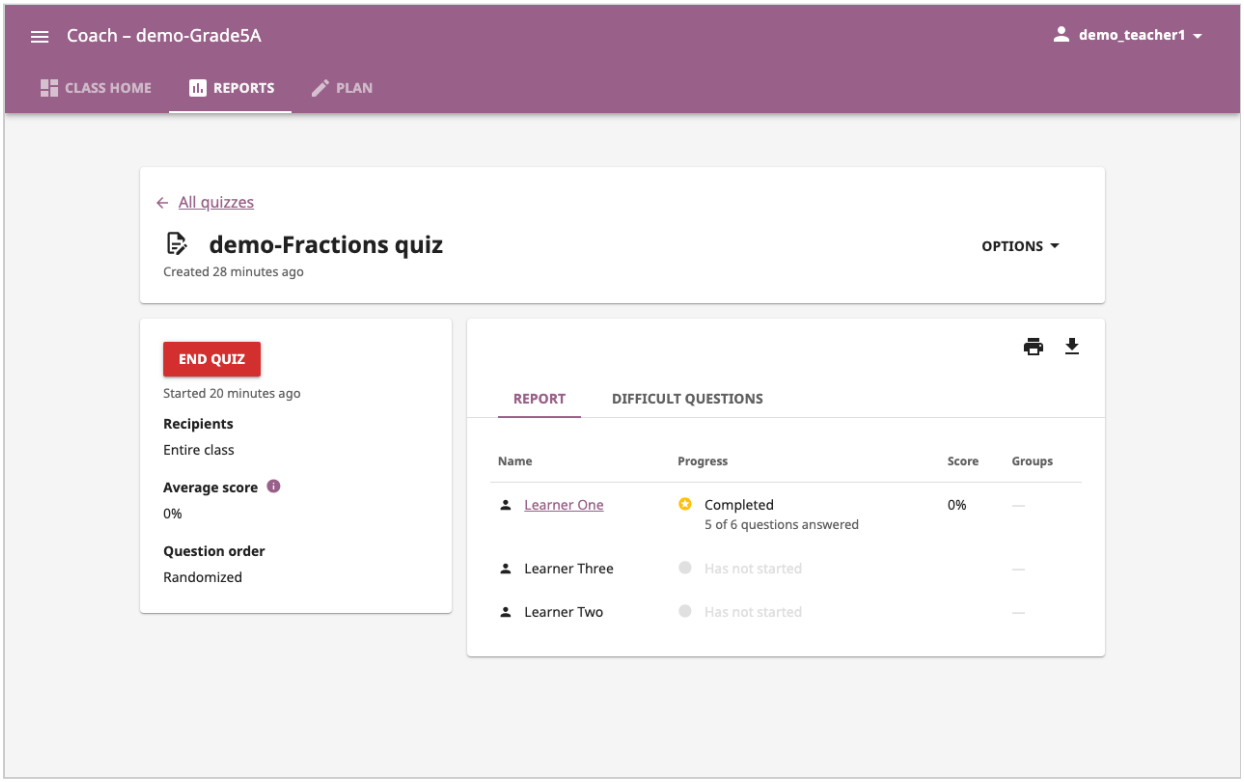


Figure 22. Quiz report

4. **LEARNERS** > a name: that learner's lessons and quizzes.

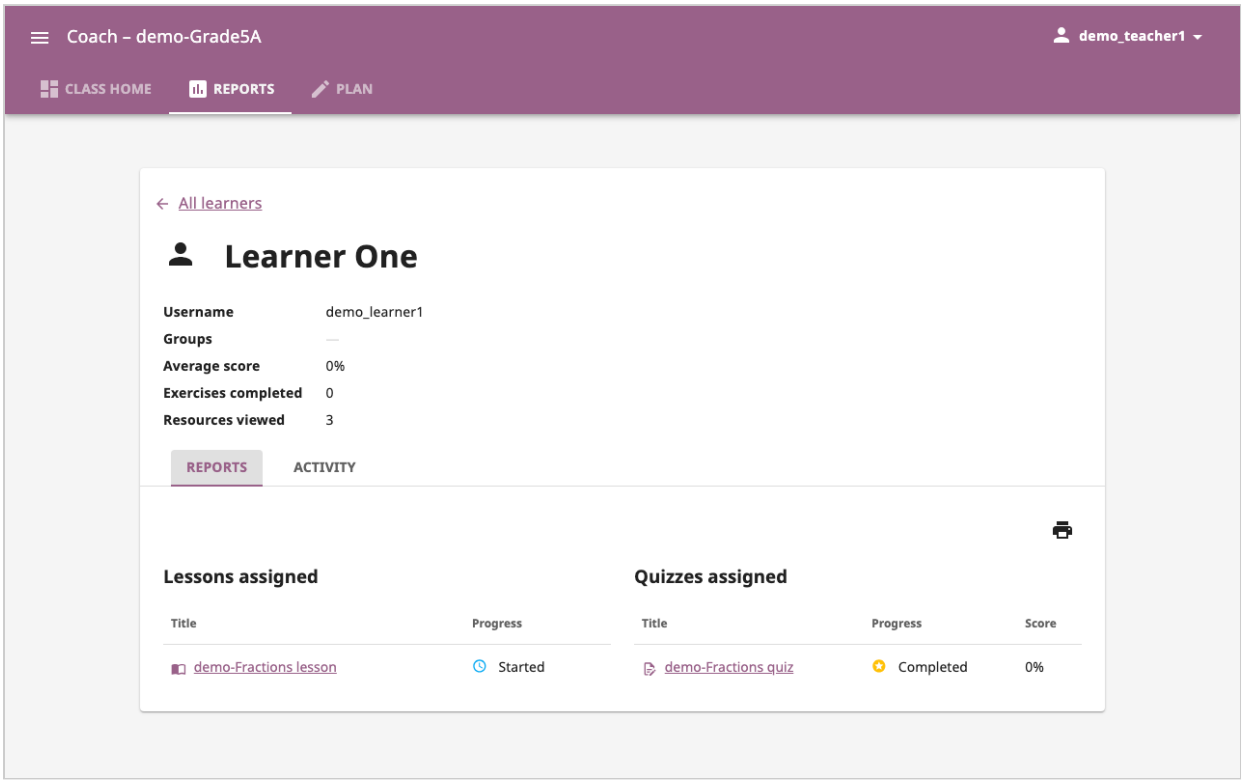


Figure 23. Learner report

Reports can be printed or exported as CSV (icons top right). Tip: do not chase 100%. Look for the question most learners got wrong and teach it again.

4. Kolibri for learners (try it yourself)

Sign in as a learner in a second browser window (private window) to see exactly what your class sees. On the demo school, learners sign in with their **username only, no password** (facility setting; check your school's setting).

> **Note on passwords.** The demo learners sign in without a password. A school can turn learner passwords on: sign in as a Kolibri admin, open **Facility** > **SETTINGS**, tick **Require password for learners**, then click **SAVE CHANGES**. (Setting name from the Kolibri 0.15.5 source code; not yet clicked on IDEA.) Learners who have no password yet then need one set by the admin.

S1. Do a lesson

1. Sign in as `demo_learner1`. **HOME** shows Your classes, Recent lessons and Recent quizzes.

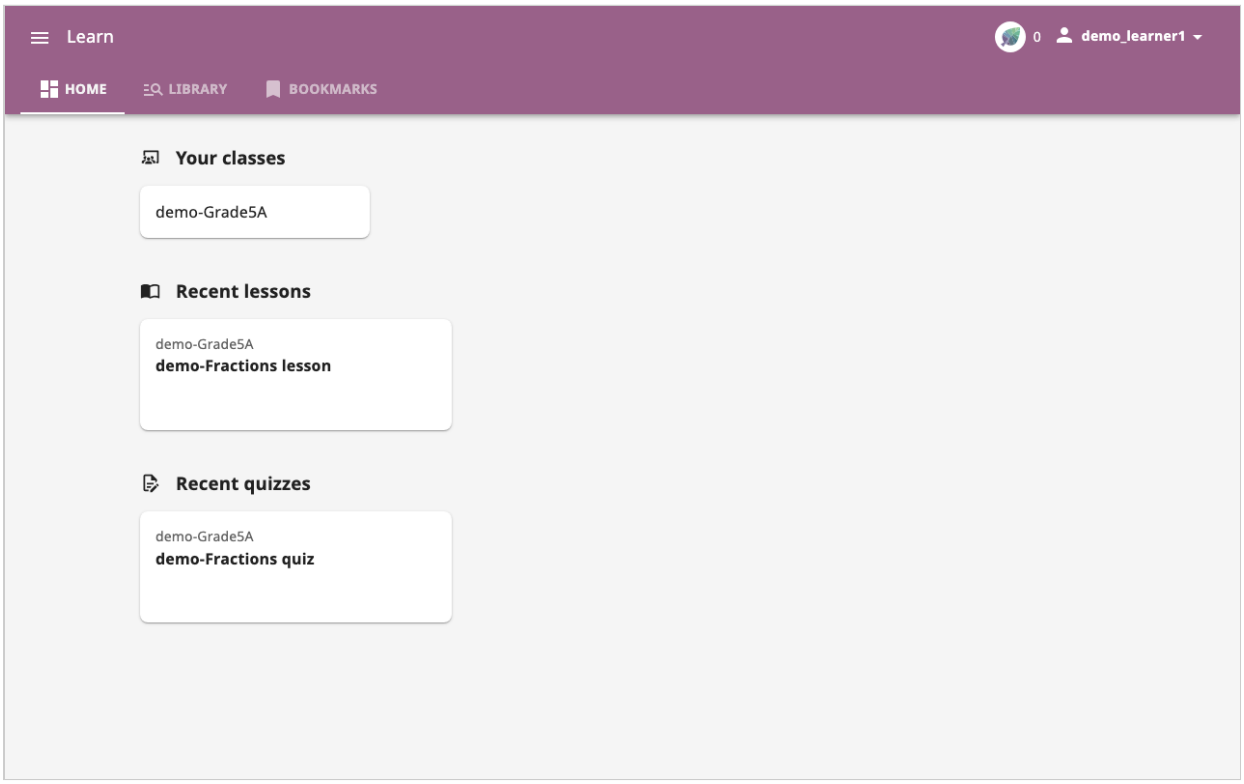


Figure 24. Learner home

2. Click **demo-Fractions lesson**. The 6 items are listed in order.

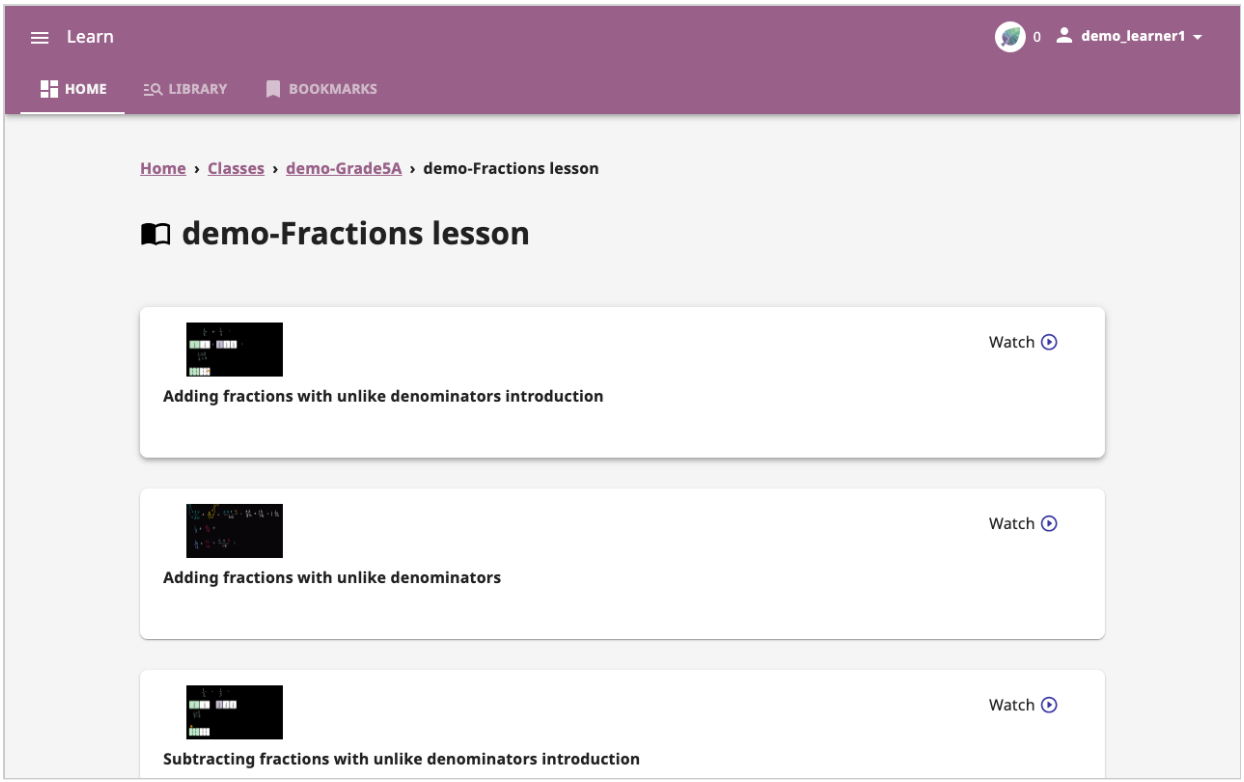


Figure 25. Lesson

3. Watch the first video.

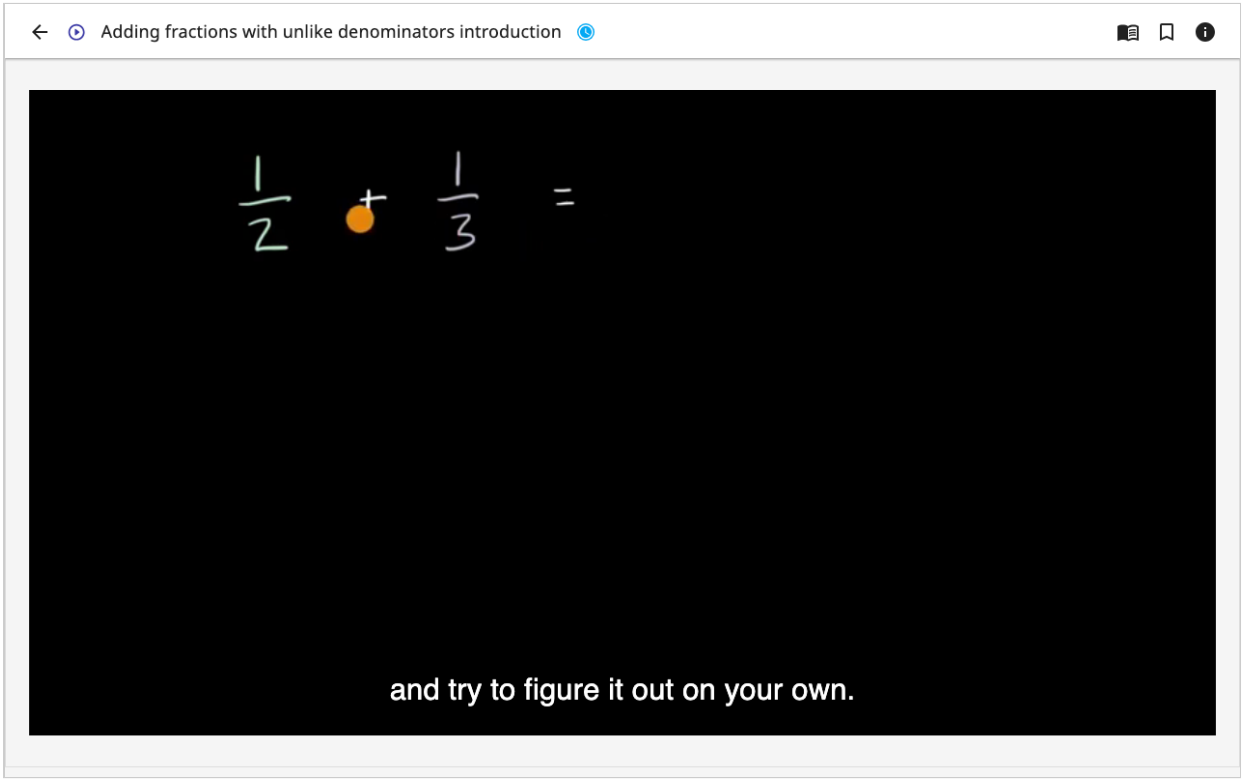


Figure 26. Video

4. Open an exercise, type an answer, click **CHECK**. Keep going until the exercise is complete ("5 correct" goal shown at the top).

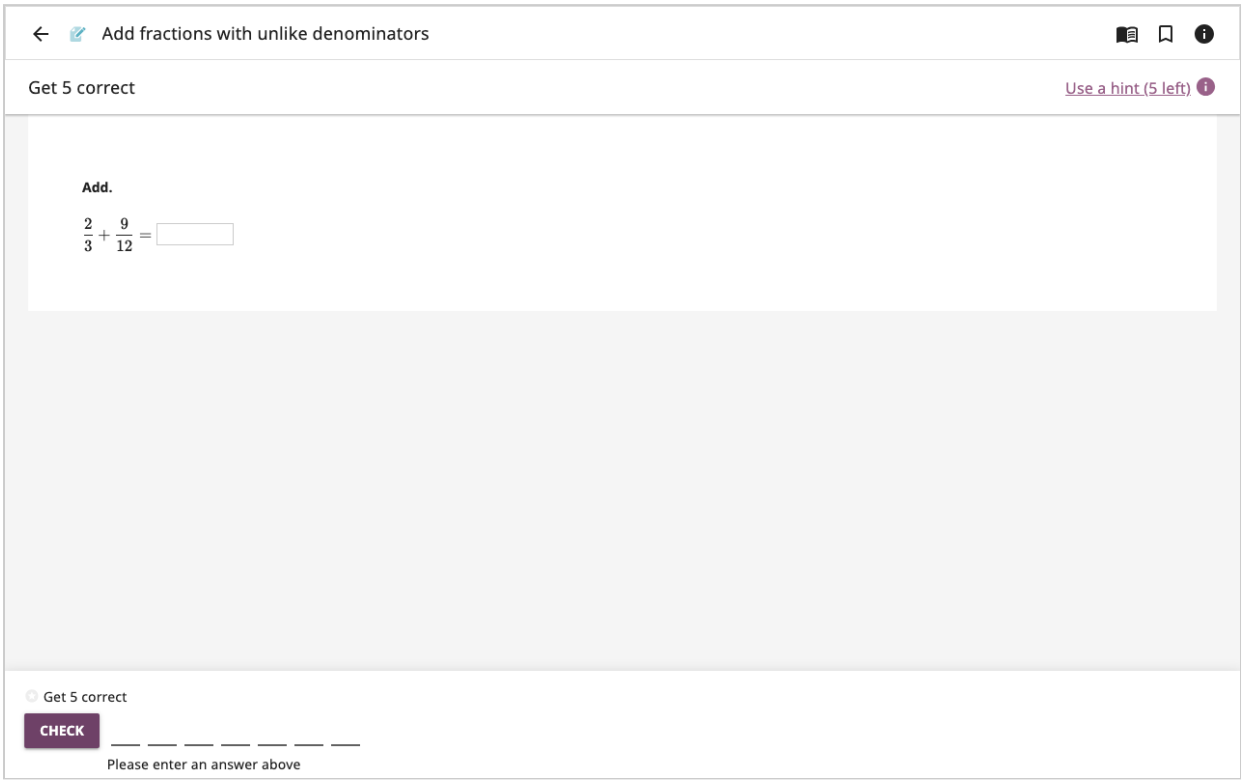


Figure 27. Exercise

5. Sign out. Sign in as `demo_learner2` and open only one video, so the report shows a difference.

S2. Take a quiz

1. Open the quiz from **Recent quizzes** (or the class page).

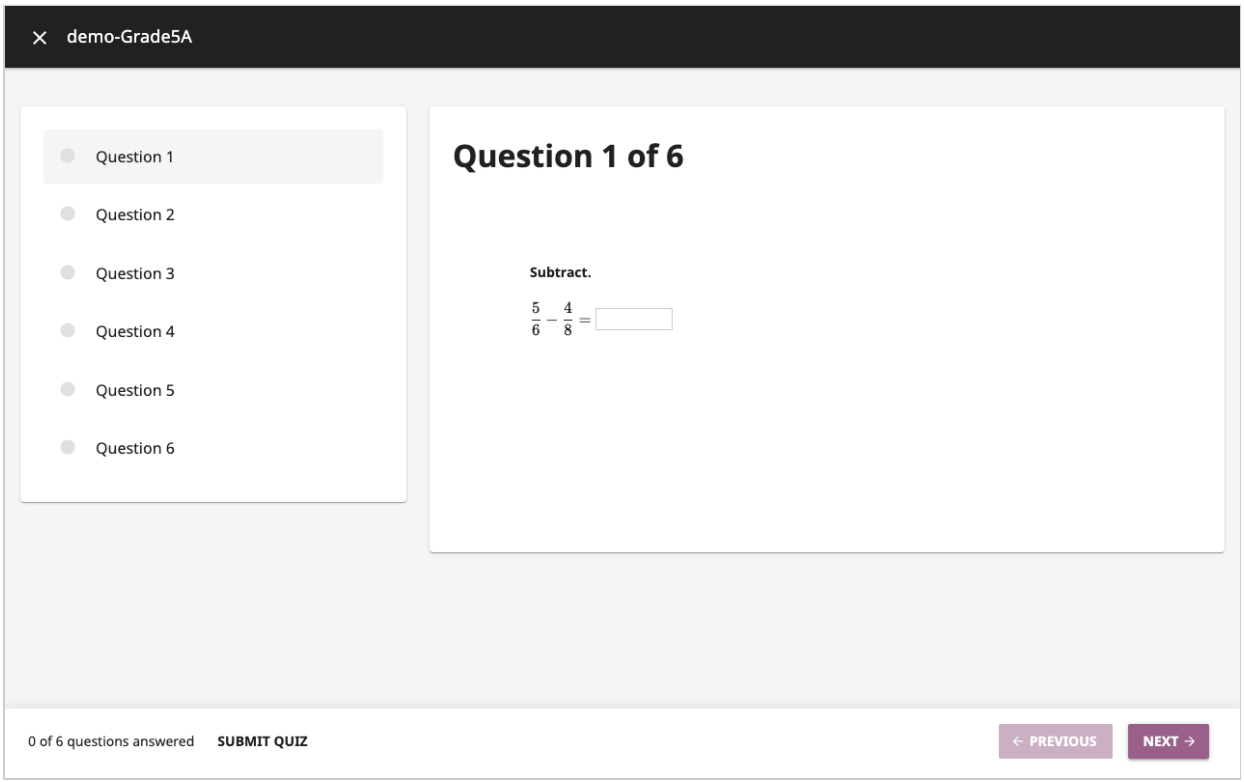


Figure 28. Quiz

2. Answer each question and use **NEXT** / **PREVIOUS**. The list on the left shows answered questions.

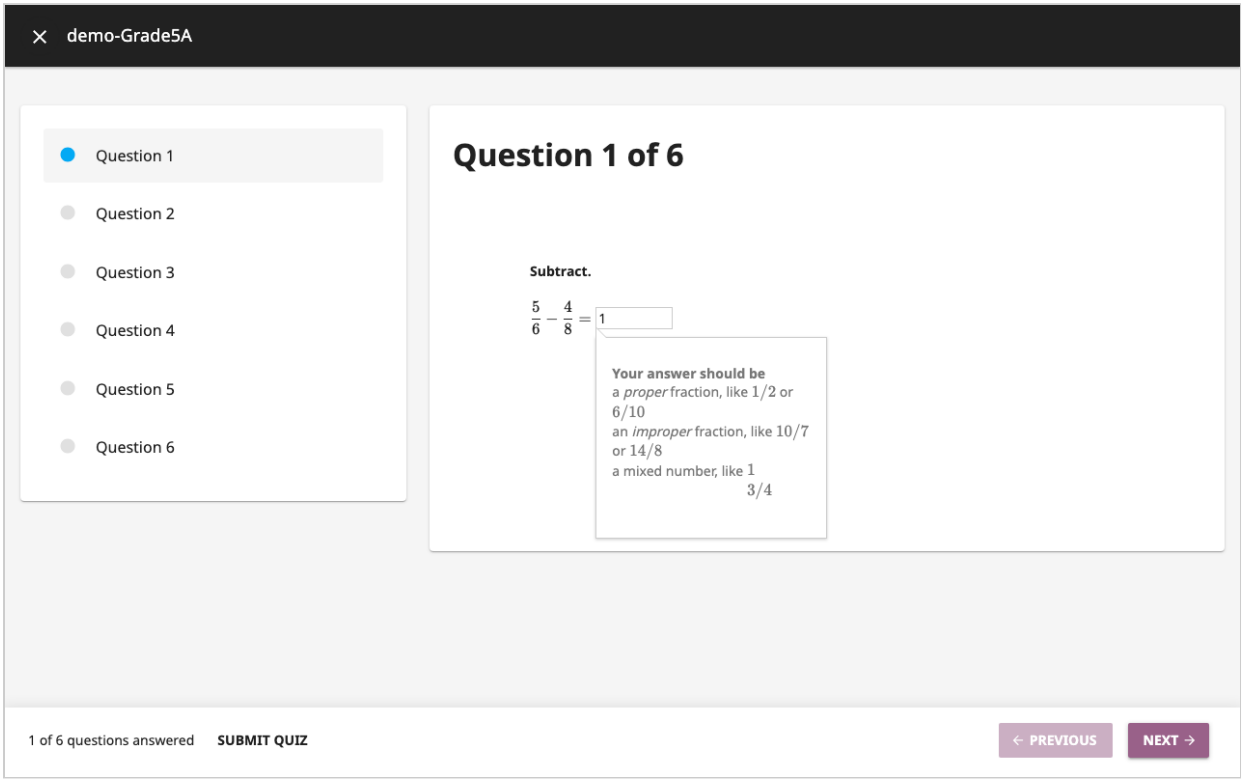


Figure 29. Question

3. Click **SUBMIT QUIZ** and confirm. After submitting, answers cannot be changed.

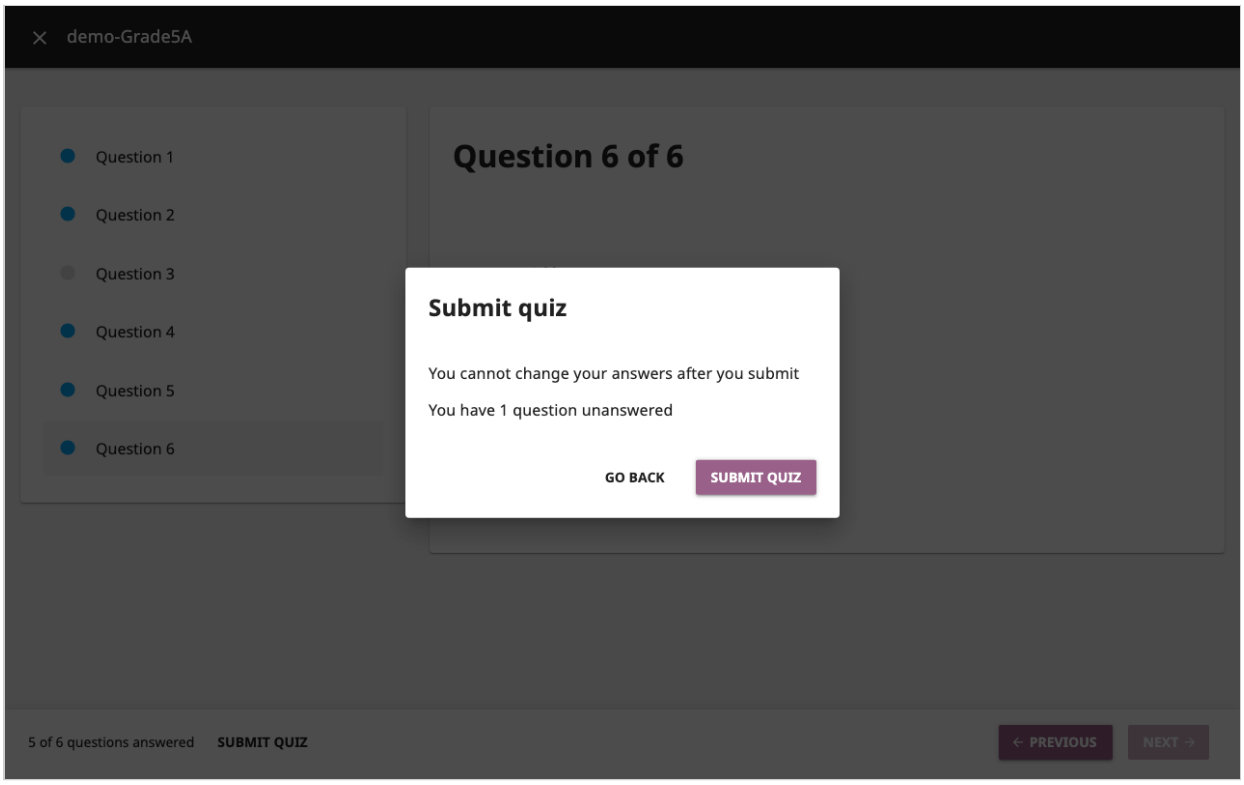


Figure 30. Submit

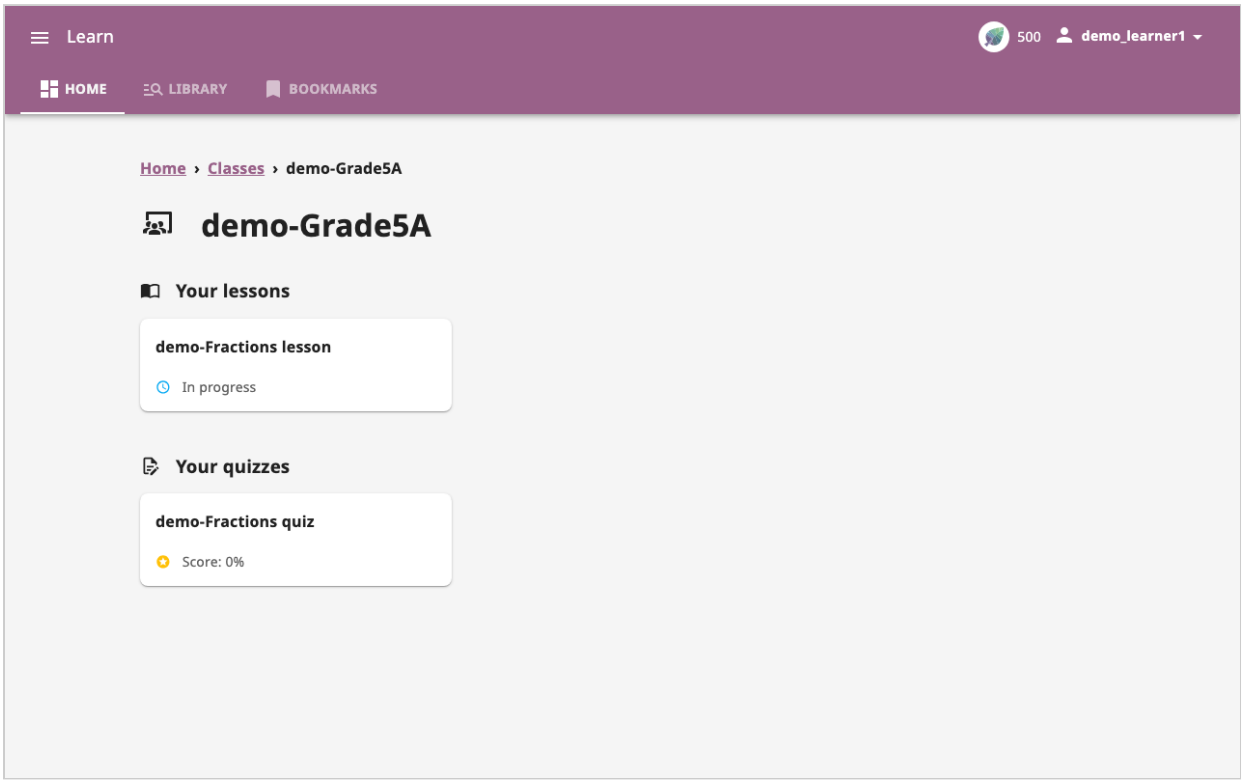


Figure 31. After submit

Now go back to the teacher window and look at Reports (K5).

5. Nextcloud for teachers

Demo: `demo-teacher1`, groups `demo-Grade5A` (class) and `demo-teachers`. Office documents open in **Nextcloud Office (Collabora)**. Open Nextcloud from the Console (chapter 2) and sign in. Click **Files** at the top.

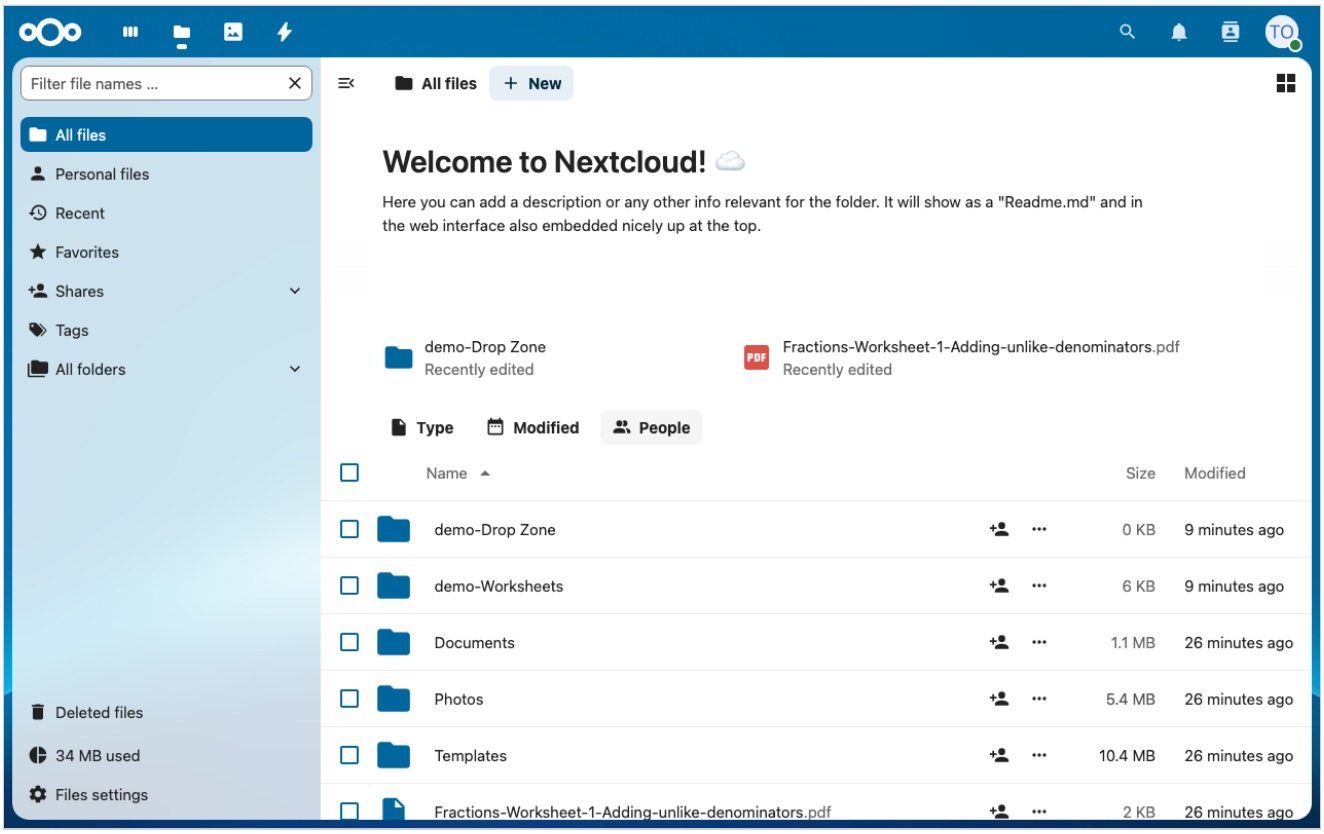


Figure 32. Files

N1. Share worksheets with the class (view only)

1. Next to the folder `demo-Worksheets`, click the share icon (**Show sharing options**). The right panel opens on **Sharing**.

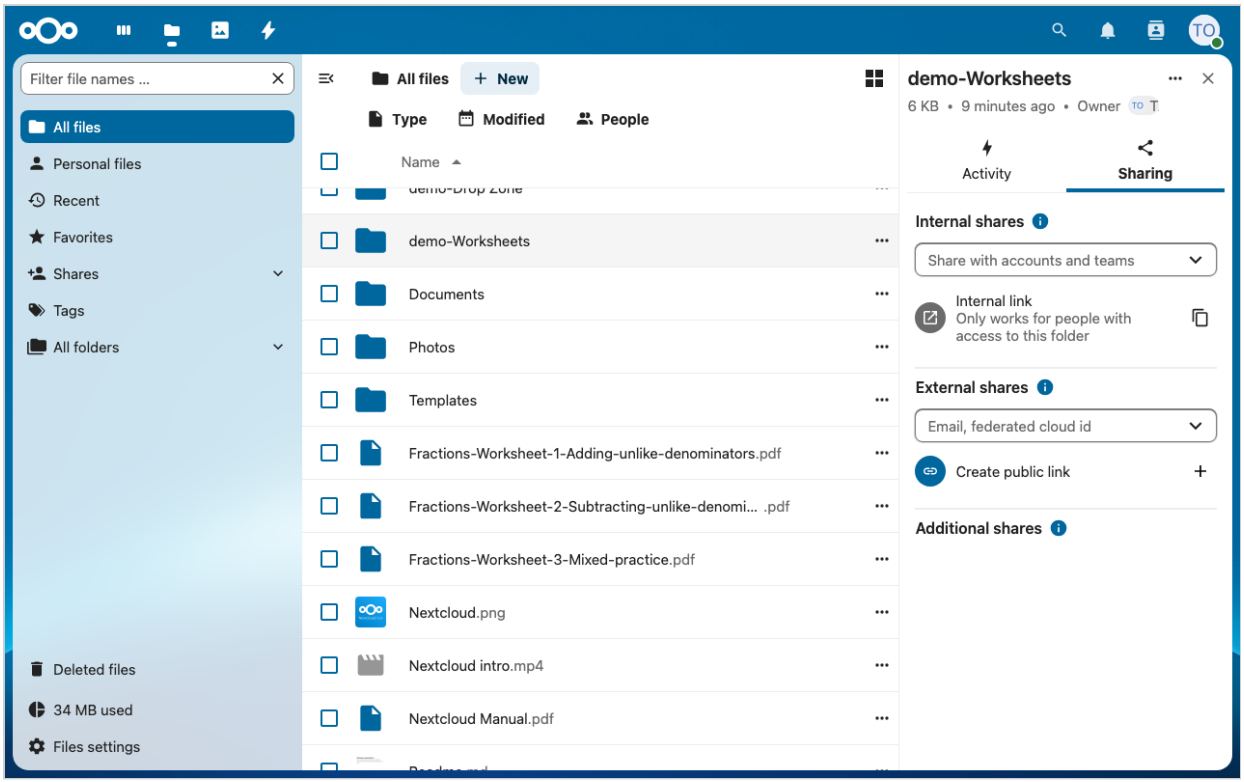


Figure 33. Sharing panel

2. In **Internal shares** ("Share with accounts and teams"), type `demo-Grade5A` and pick the group.
3. Set the permission to **View only**. The group then appears under Internal shares.

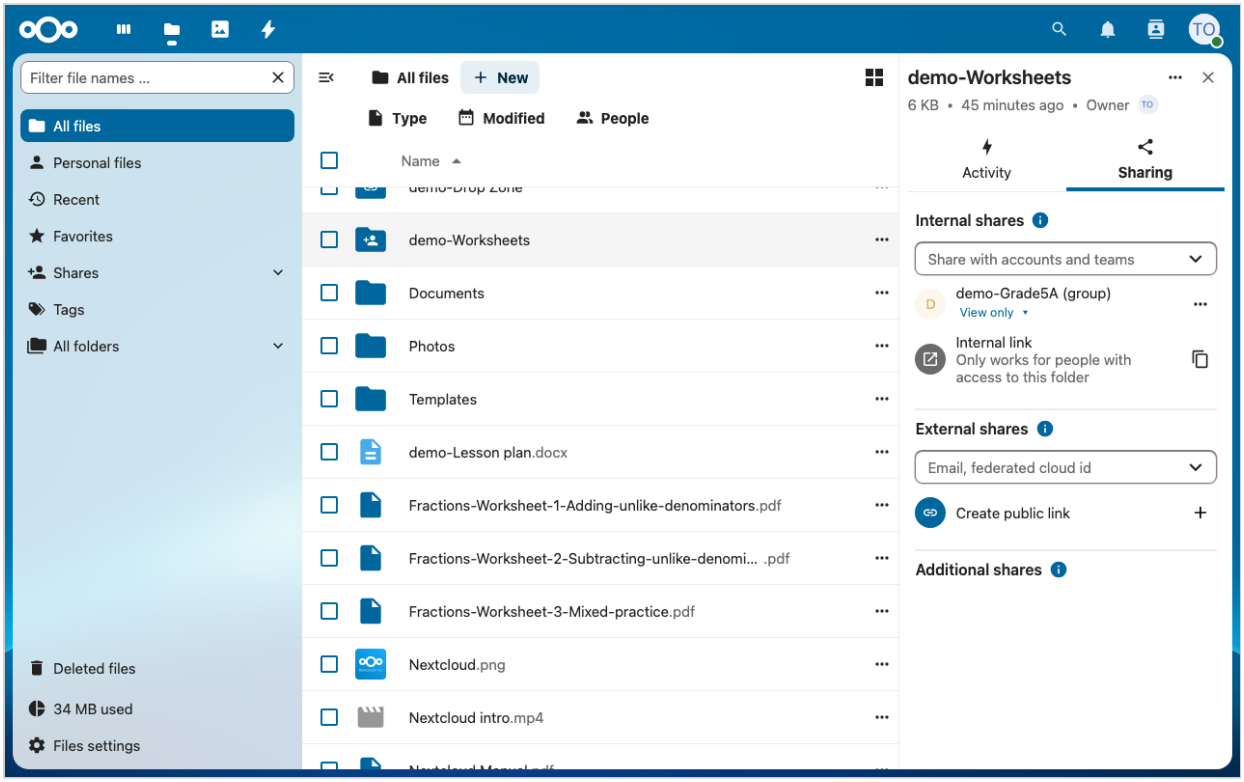


Figure 34. Shared with group

Learners now see the folder but cannot change or delete files.

N2. Write a document together

1. Click **+ New** > **New document**.

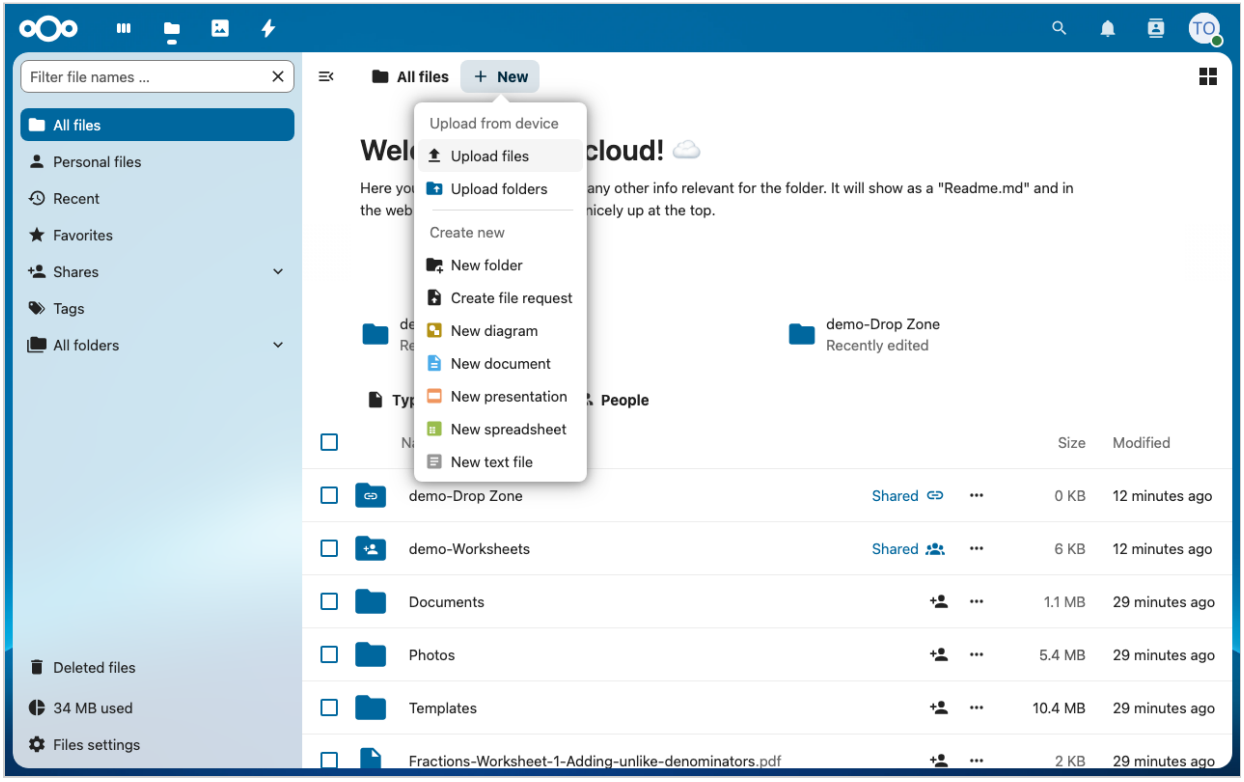


Figure 35. New menu

2. Type a name (`demo-Lesson plan`) and click **Create**.

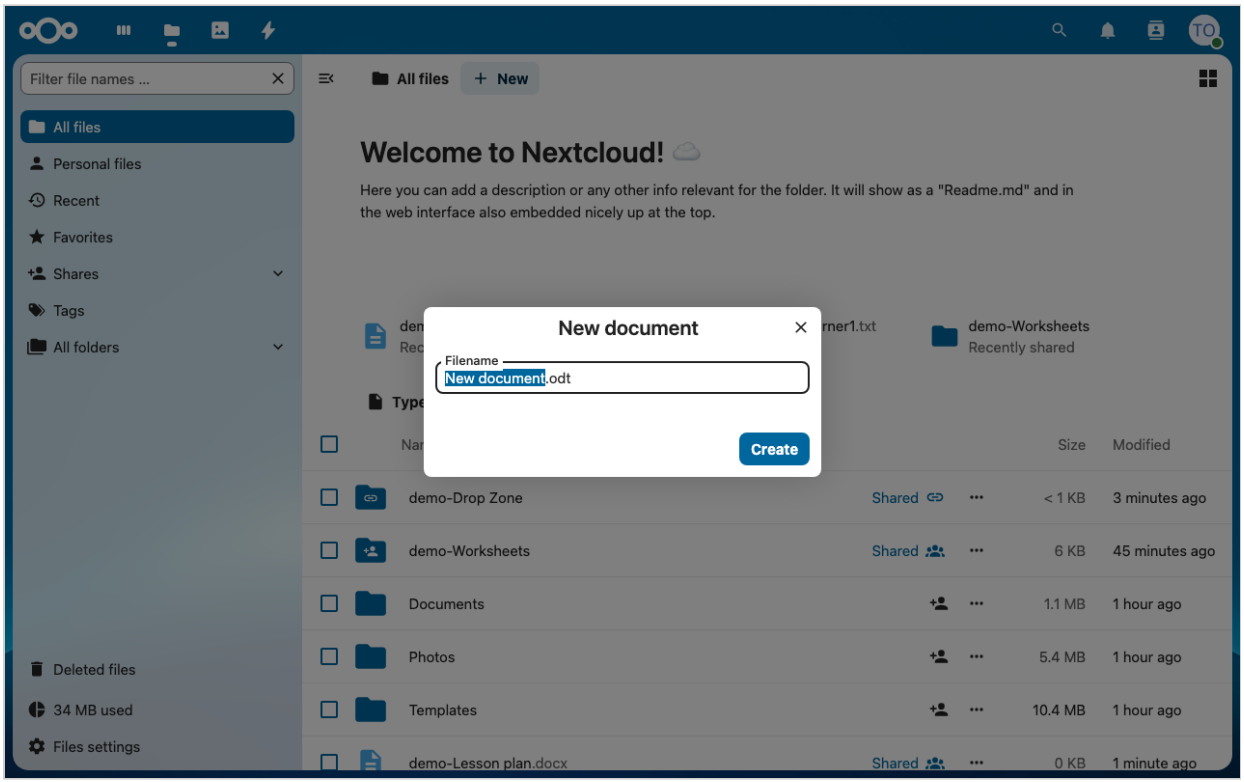


Figure 36. Name

3. Pick a template (Blank is fine) and click **Create**.

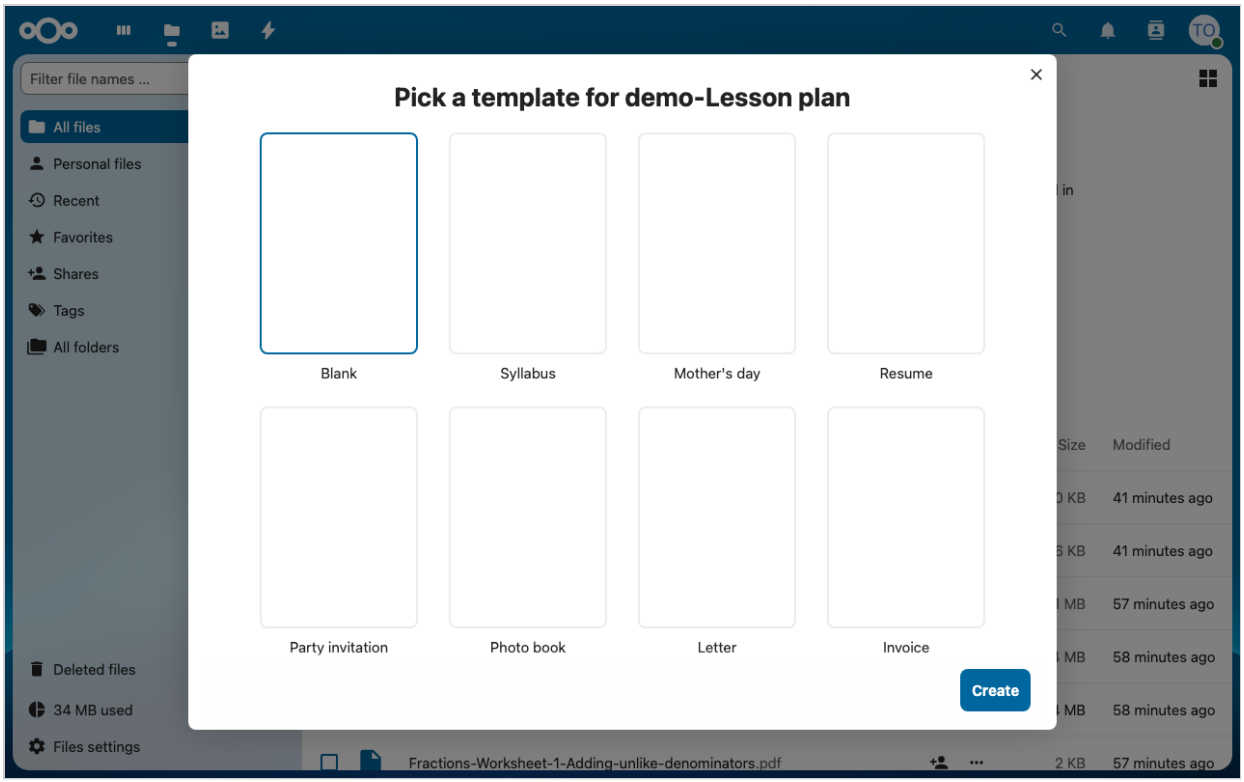


Figure 37. Template

4. Share the file with `demo-teachers` and/or your class group, with editing allowed.

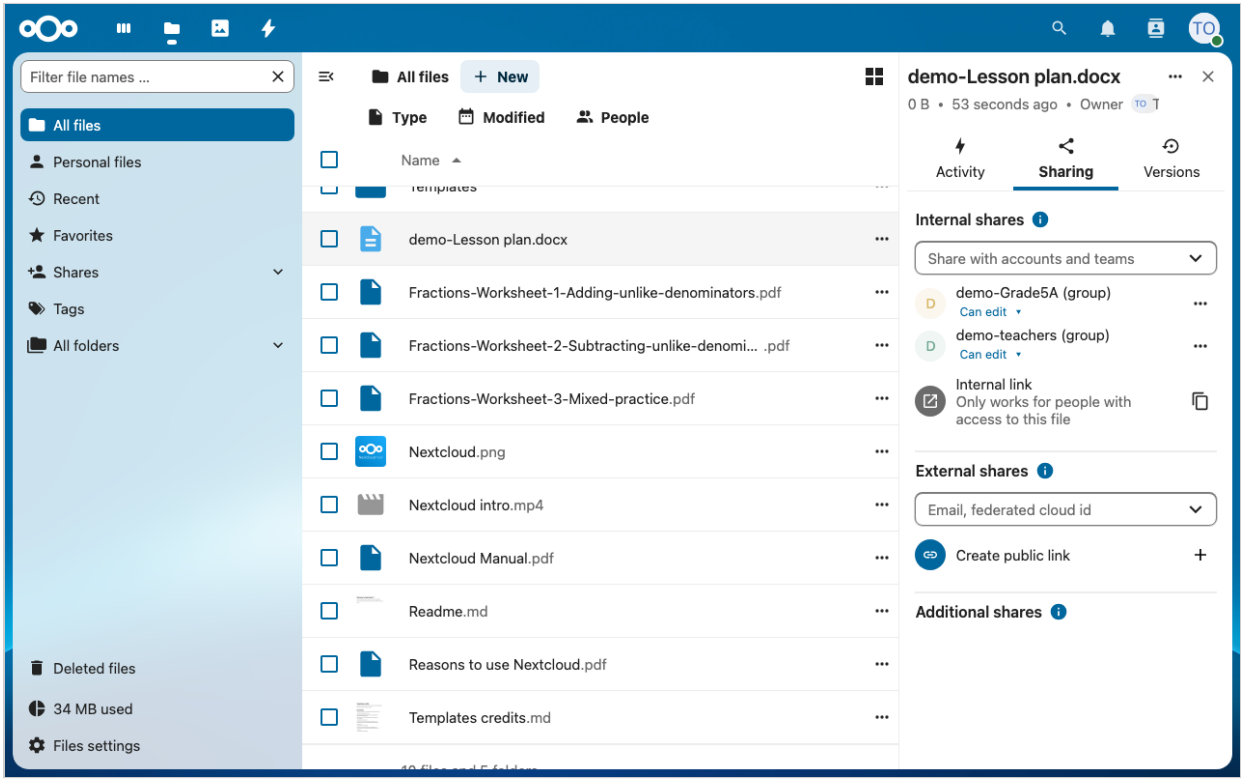


Figure 38. Shared document

5. The document opens in Nextcloud Office. When others open it at the same time, you see their names and cursors.

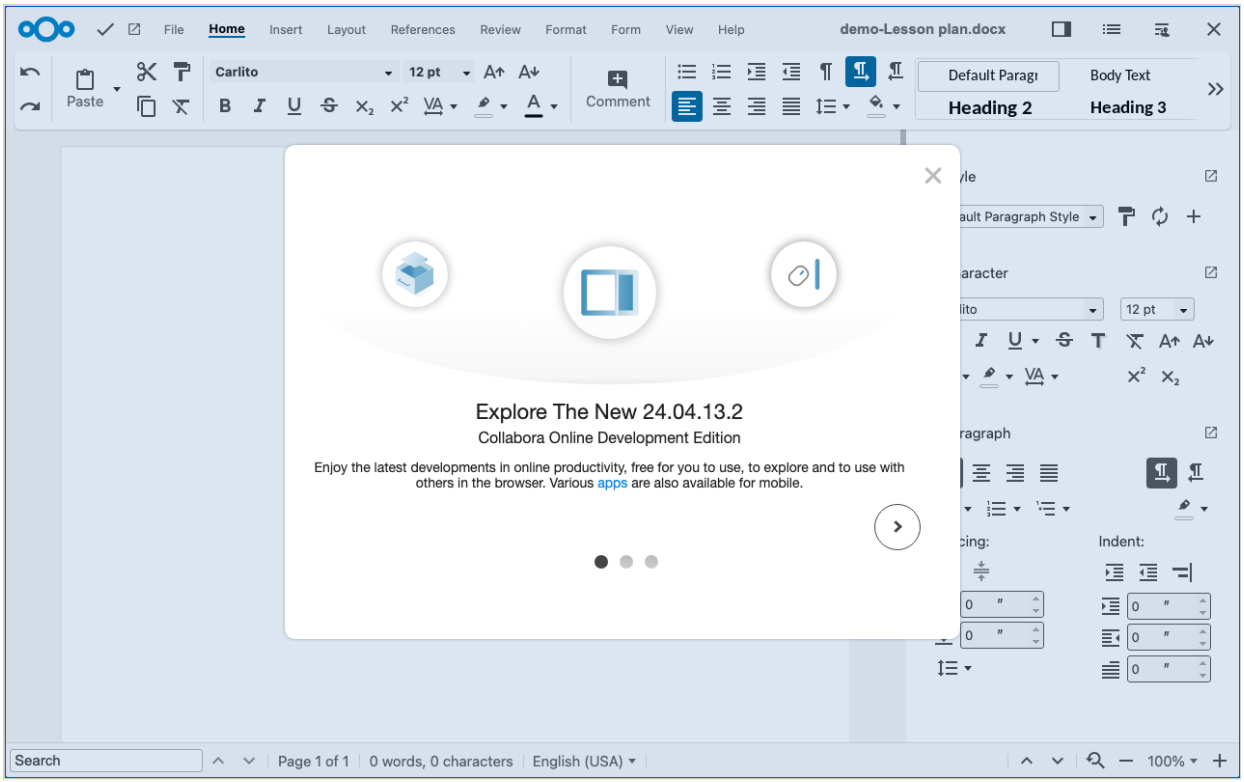


Figure 39. Office editor

N3. Collect homework (File drop)

Learners upload work, but cannot see each other's work.

1. Open sharing for folder `demo-Drop Zone`.

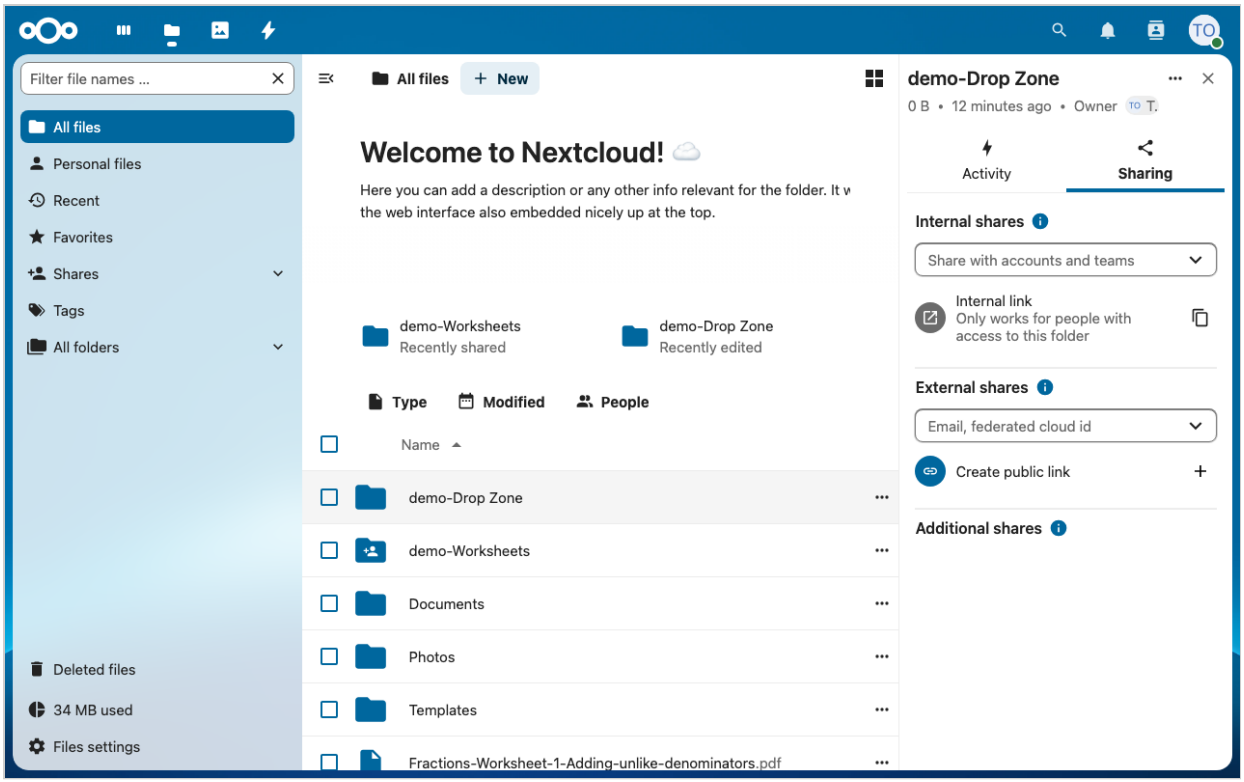


Figure 40. Sharing for Drop Zone

2. Under **External shares**, click **+** next to **Create public link** and choose **File request** (upload only). Copy the link.

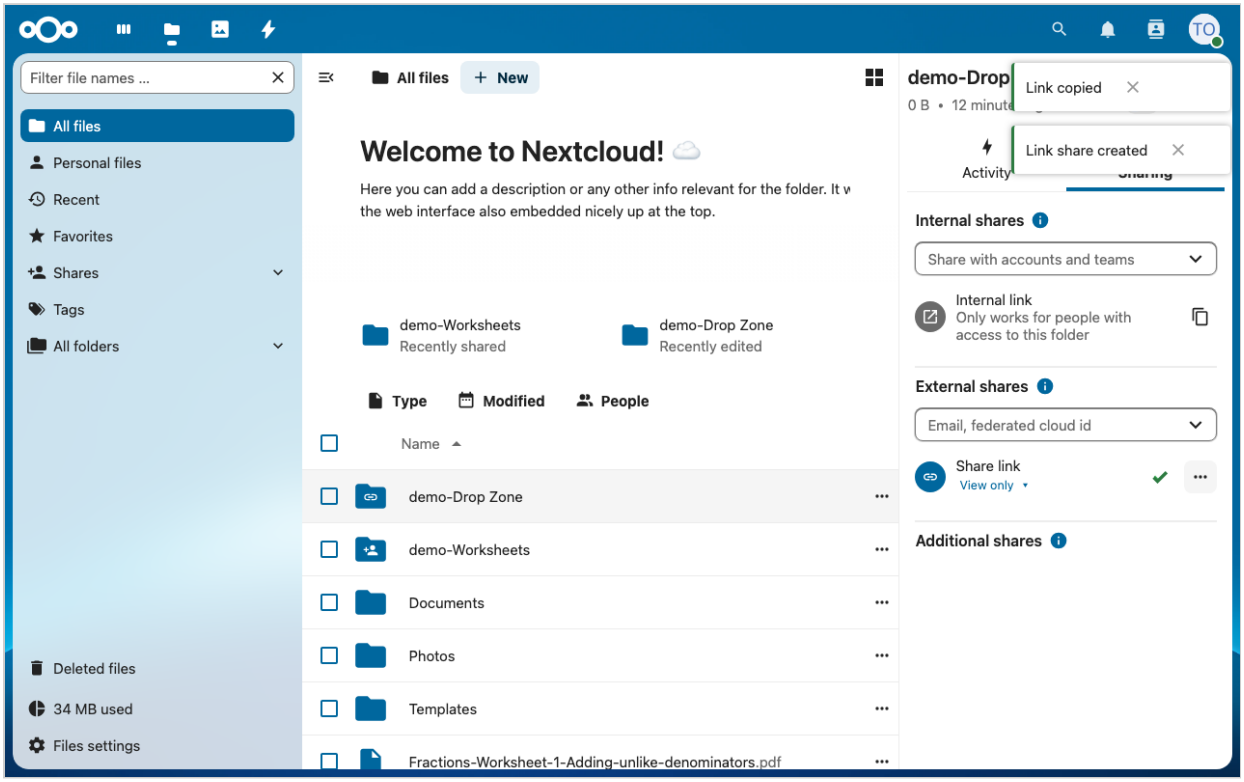


Figure 41. Create link

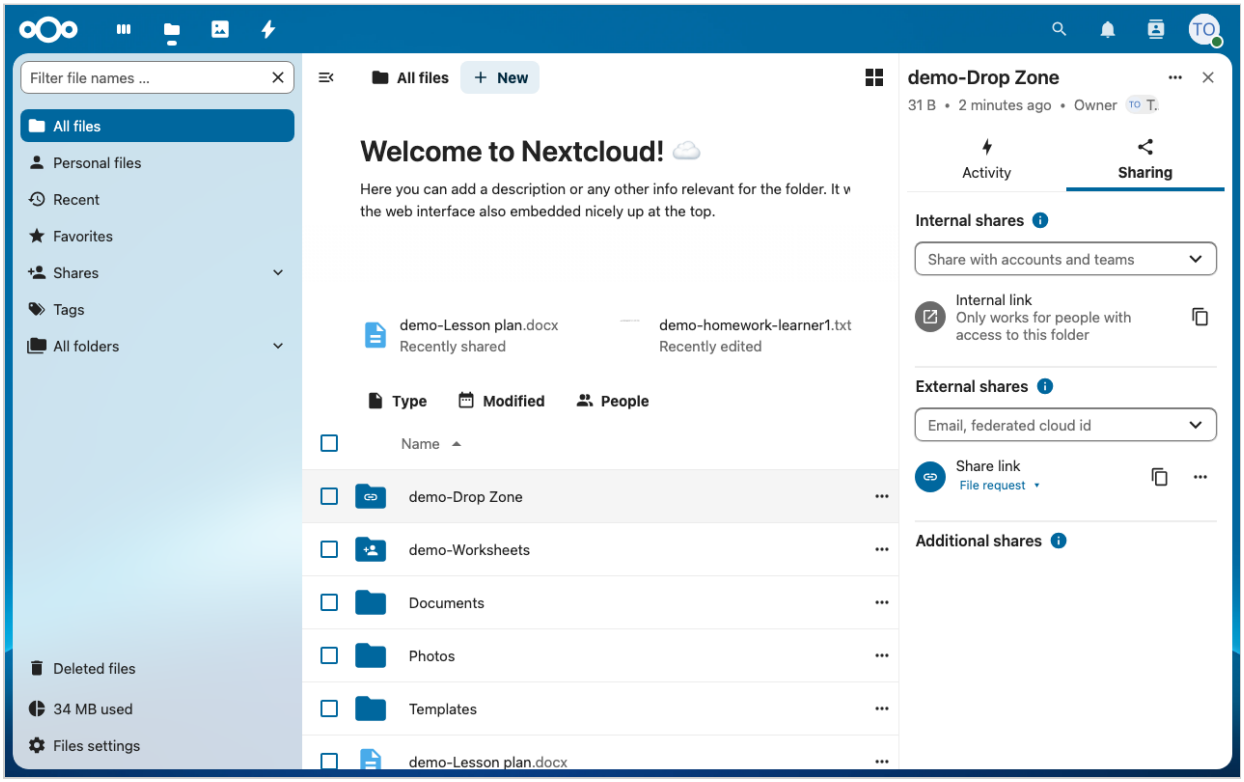


Figure 42. Link created

3. Give learners the link (write it on the board or send it in Talk). Uploaded files appear in your `demo-Drop Zone` folder.

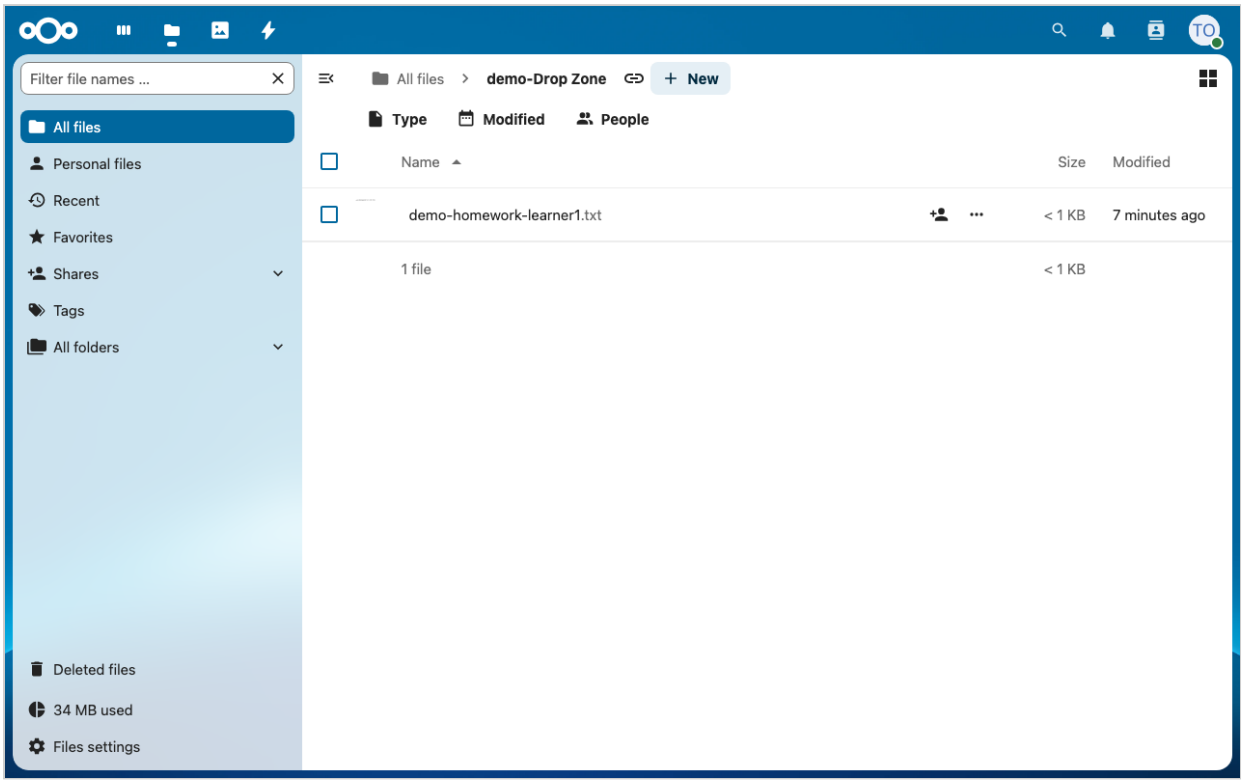


Figure 43. Received homework

6. Nextcloud for learners (try it yourself)

S3. Open class files and co-edit

1. Sign in to Nextcloud as `demo-learner1` (private window). Shared folders and files appear in **Files** with the teacher's name.

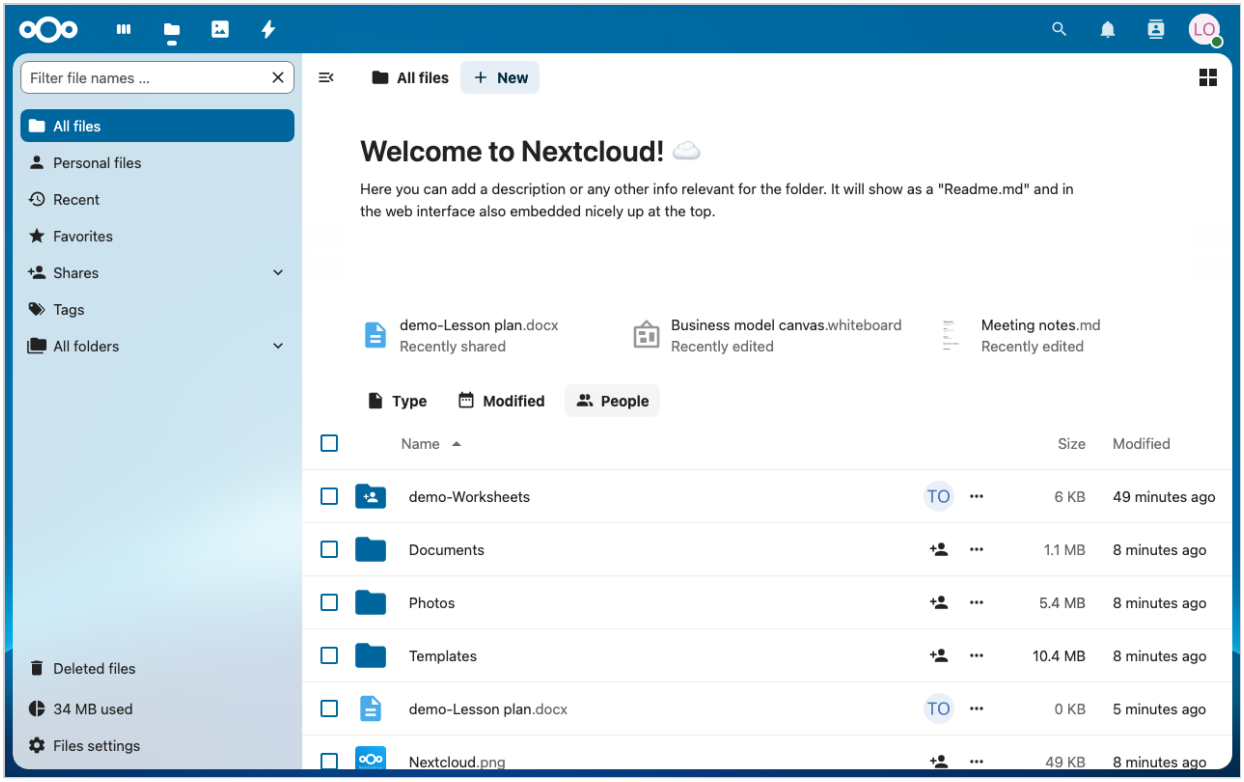


Figure 44. Learner files

2. Open `demo-Worksheets`: the learner can open and download the worksheets, not change them.

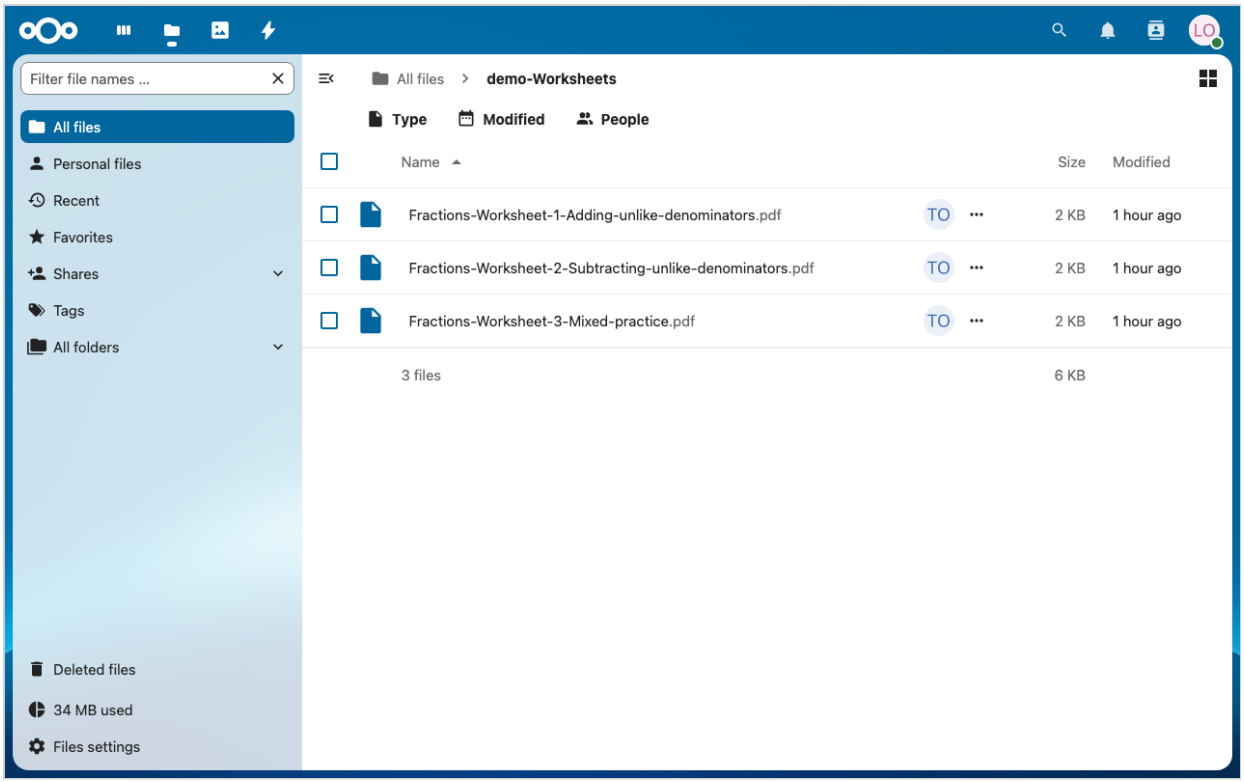


Figure 45. Worksheets

3. Open `demo-Lesson plan.docx`: it opens in Nextcloud Office and the learner can type.

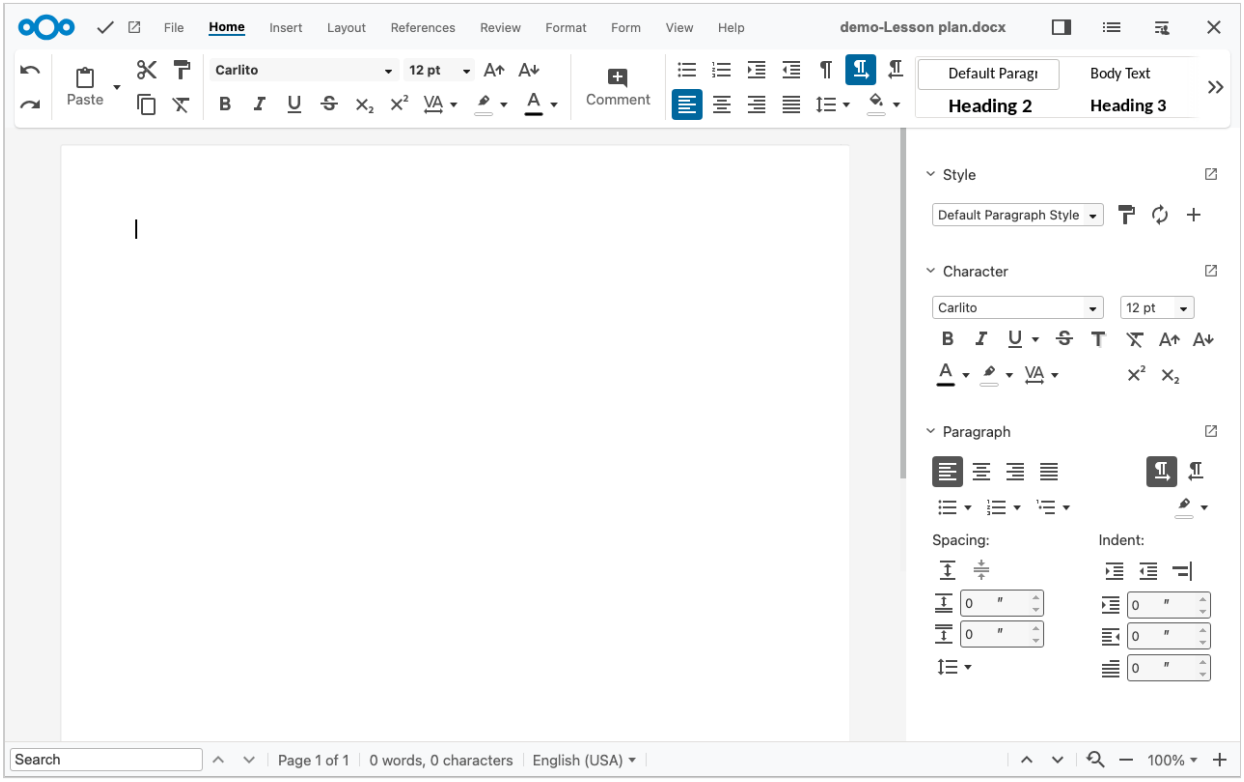


Figure 46. Learner in document

S4. Hand in homework

1. Open the File drop link from N3. The page says **File drop** and shows an **Upload** button; the learner cannot see other files.

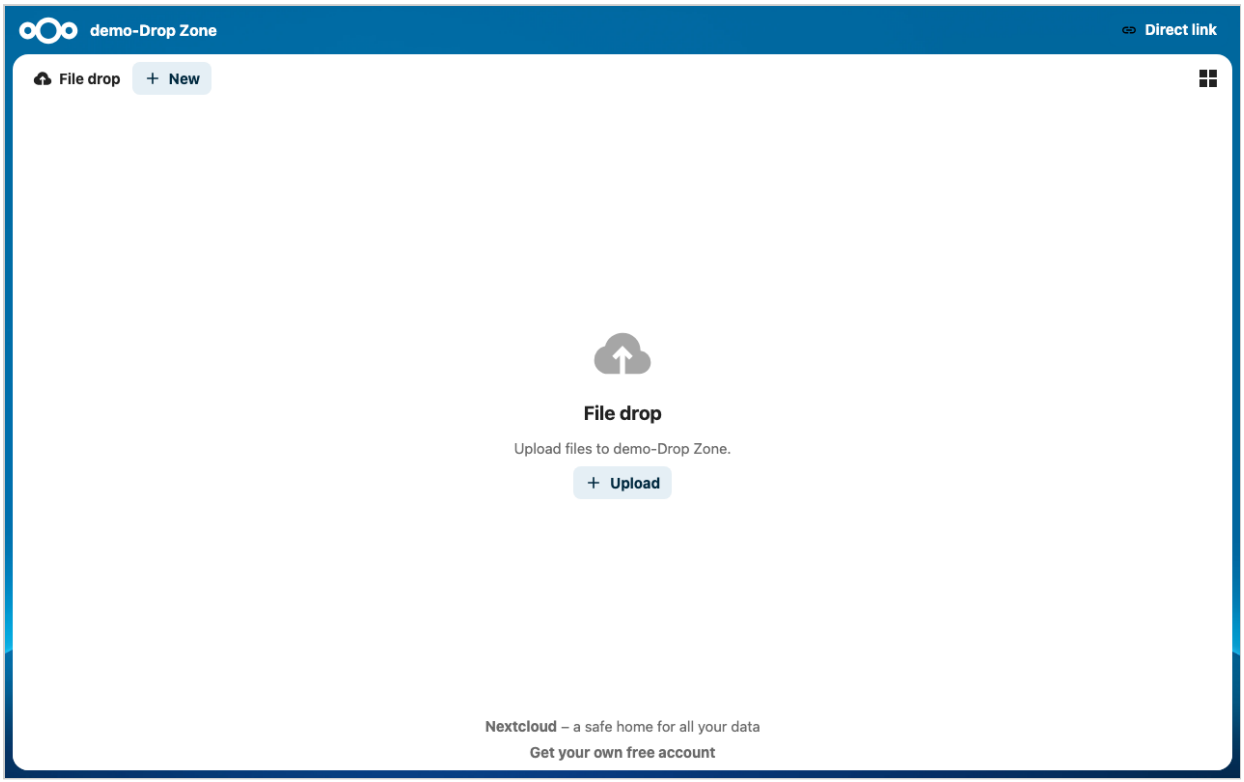


Figure 47. File drop

2. Click **Upload** and choose the file (e.g. a photo of written work). If asked, type your name.

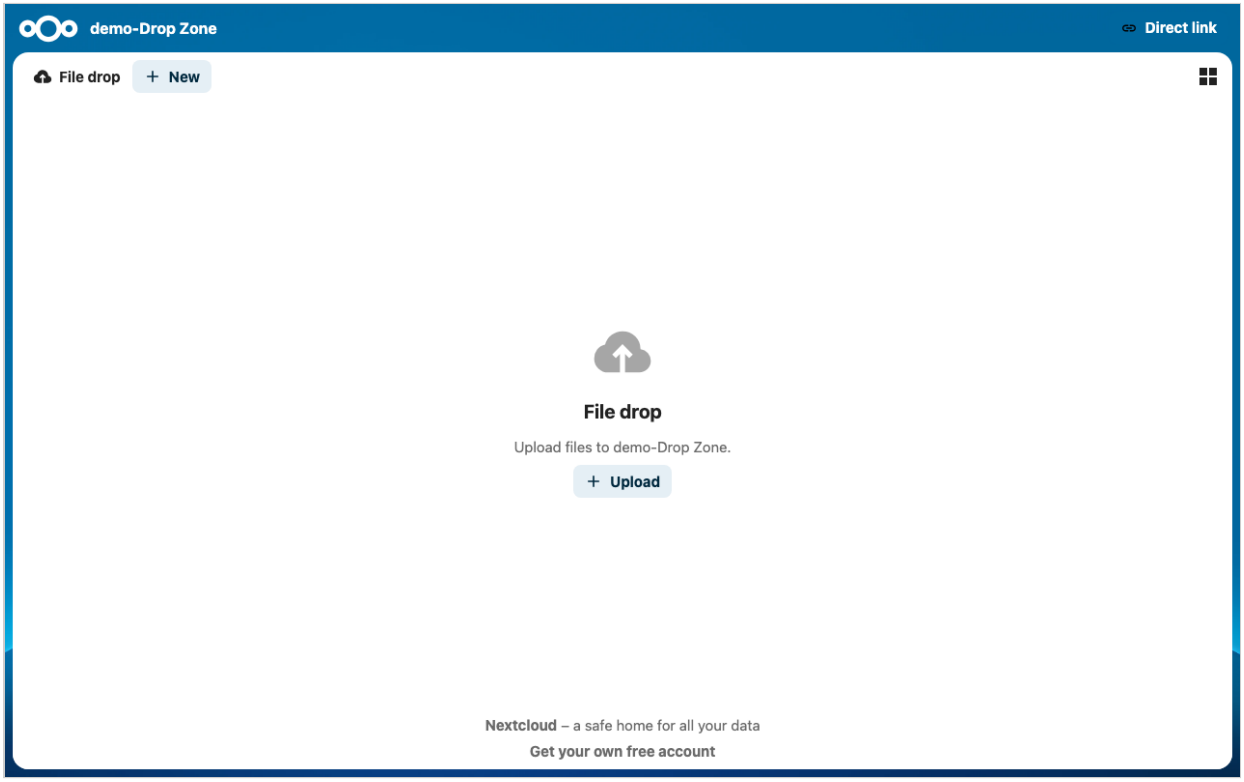


Figure 48. Uploaded

The teacher sees the file in `demo-Drop Zone` (N3).

7. Wikipedia offline (Kiwix)

The demo has **Wikipedia in simple English** (text, no pictures). It works without internet.

W1. Look something up and use it in class

1. Open Kiwix from the Console (chapter 2). The library shows the books available.

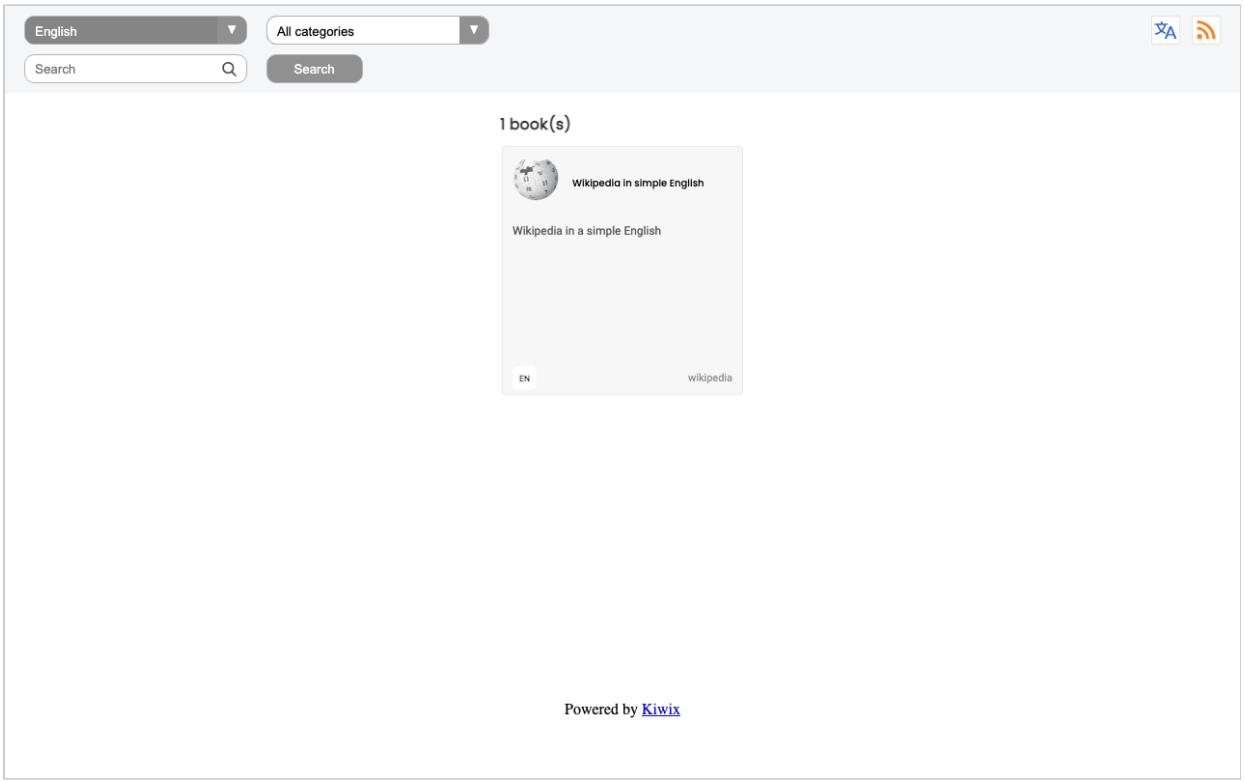


Figure 49. Kiwix library

2. Click **Wikipedia in simple English**. Its welcome page opens.



Figure 50. Wikipedia home

3. Type in the search box at the top, e.g. `Fraction`. Pick a result.

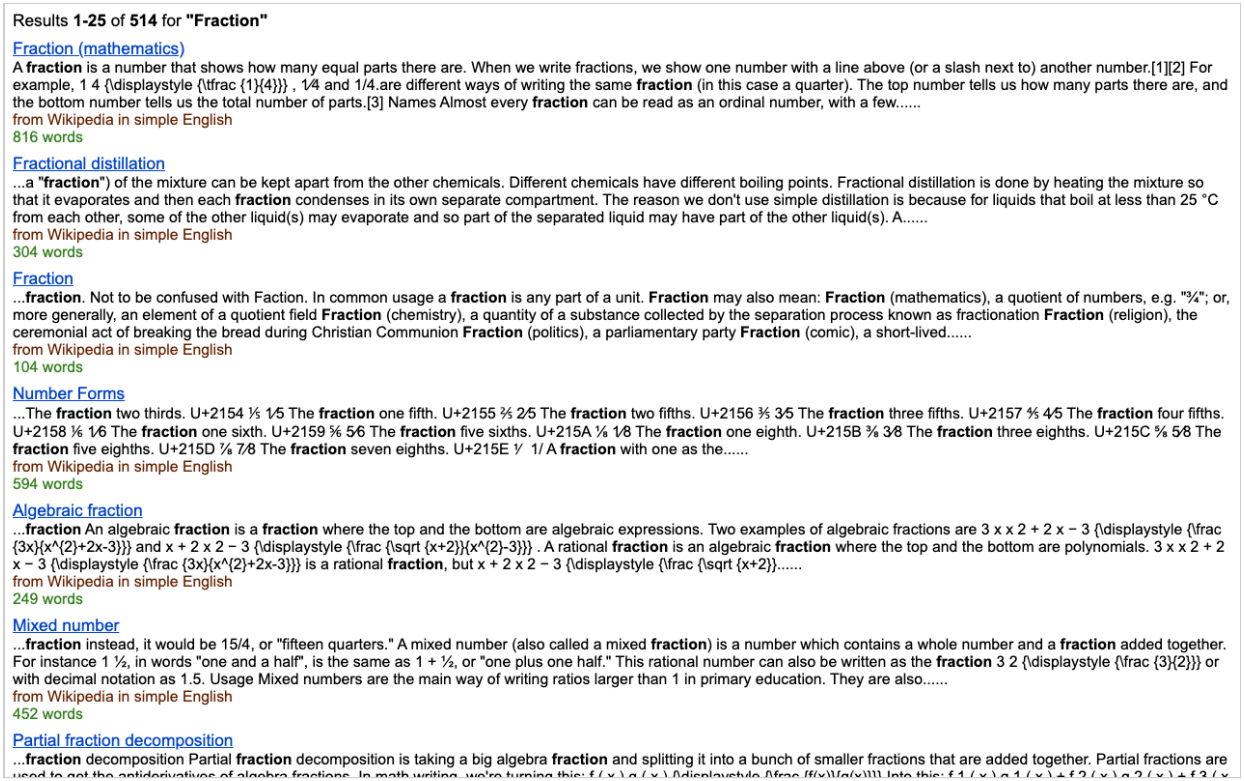
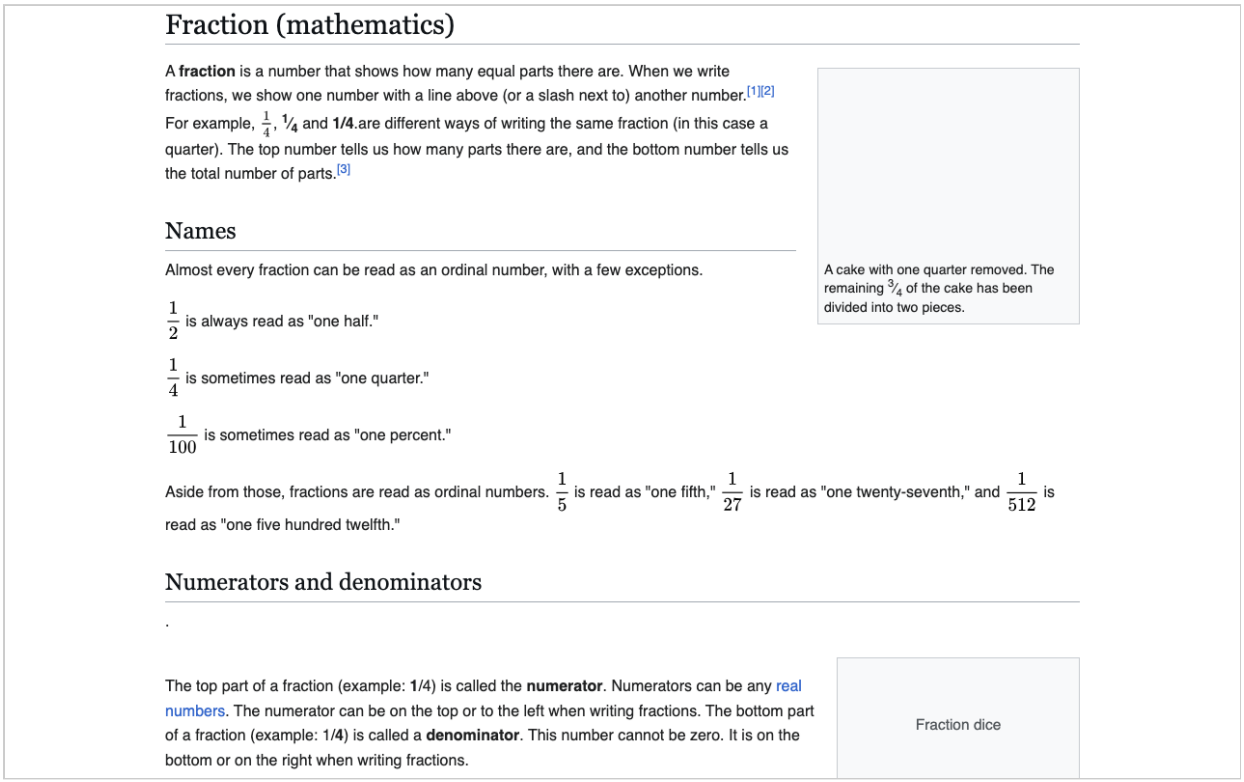


Figure 51. Search results



Fraction (mathematics)

A **fraction** is a number that shows how many equal parts there are. When we write fractions, we show one number with a line above (or a slash next to) another number.[1][2] For example, $\frac{1}{4}$, 1⁄4 and **1/4**.are different ways of writing the same fraction (in this case a quarter). The top number tells us how many parts there are, and the bottom number tells us the total number of parts.[3]

A cake with one quarter removed. The remaining $^3/_4$ of the cake has been divided into two pieces.

Names

Almost every fraction can be read as an ordinal number, with a few exceptions.

$\frac{1}{2}$ is always read as "one half."

$\frac{1}{4}$ is sometimes read as "one quarter."

$\frac{1}{100}$ is sometimes read as "one percent."

Aside from those, fractions are read as ordinal numbers. $\frac{1}{5}$ is read as "one fifth," $\frac{1}{27}$ is read as "one twenty-seventh," and $\frac{1}{512}$ is read as "one five hundred twelfth."

Numerators and denominators

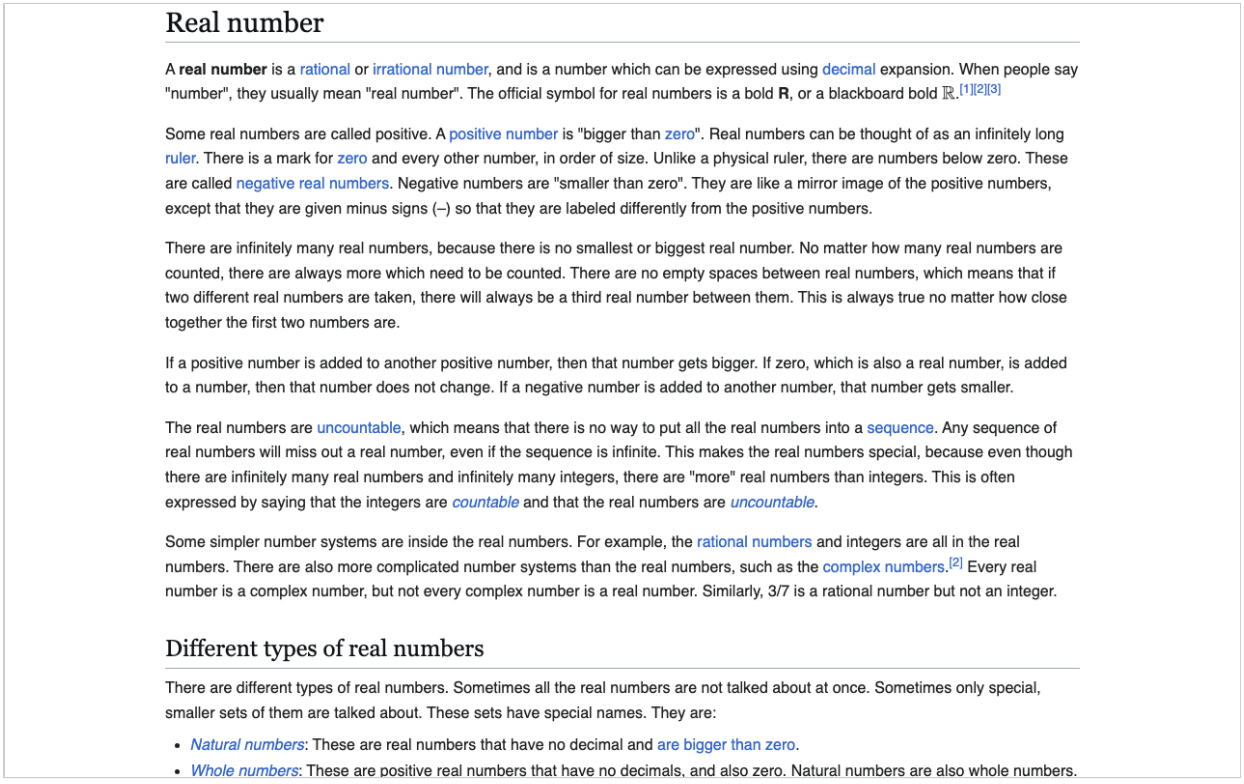
.

The top part of a fraction (example: **1**/4) is called the **numerator**. Numerators can be any real numbers. The numerator can be on the top or to the left when writing fractions. The bottom part of a fraction (example: 1/**4**) is called a **denominator**. This number cannot be zero. It is on the bottom or on the right when writing fractions.

Fraction dice

Figure 52. Fraction article

4. Follow blue links to related articles.



Real number

A **real number** is a rational or irrational number, and is a number which can be expressed using decimal expansion. When people say "number", they usually mean "real number". The official symbol for real numbers is a bold **R**, or a blackboard bold $\mathbb{R}$.[1][2][3]

Some real numbers are called positive. A positive number is "bigger than zero". Real numbers can be thought of as an infinitely long ruler. There is a mark for zero and every other number, in order of size. Unlike a physical ruler, there are numbers below zero. These are called negative real numbers. Negative numbers are "smaller than zero". They are like a mirror image of the positive numbers, except that they are given minus signs (–) so that they are labeled differently from the positive numbers.

There are infinitely many real numbers, because there is no smallest or biggest real number. No matter how many real numbers are counted, there are always more which need to be counted. There are no empty spaces between real numbers, which means that if two different real numbers are taken, there will always be a third real number between them. This is always true no matter how close together the first two numbers are.

If a positive number is added to another positive number, then that number gets bigger. If zero, which is also a real number, is added to a number, then that number does not change. If a negative number is added to another number, that number gets smaller.

The real numbers are uncountable, which means that there is no way to put all the real numbers into a sequence. Any sequence of real numbers will miss out a real number, even if the sequence is infinite. This makes the real numbers special, because even though there are infinitely many real numbers and infinitely many integers, there are "more" real numbers than integers. This is often expressed by saying that the integers are *countable* and that the real numbers are *uncountable*.

Some simpler number systems are inside the real numbers. For example, the rational numbers and integers are all in the real numbers. There are also more complicated number systems than the real numbers, such as the complex numbers.[2] Every real number is a complex number, but not every complex number is a real number. Similarly, 3/7 is a rational number but not an integer.

Different types of real numbers

There are different types of real numbers. Sometimes all the real numbers are not talked about at once. Sometimes only special, smaller sets of them are talked about. These sets have special names. They are:

- *Natural numbers*: These are real numbers that have no decimal and are bigger than zero.
- *Whole numbers*: These are positive real numbers that have no decimals, and also zero. Natural numbers are also whole numbers.

Figure 53. Linked article

Classroom idea: give learners 3 questions to answer from one article; they read on their own device.

8. Bringing your own content (Files Disk)

An operator prepares a Files Disk in the Console (F1); teachers then use it in Nextcloud (F2).

What a Files Disk is

- A USB stick or disk that the operator turns into a shared file store in the Console. It shows the **files** badge.
- Its folder (for example **School Files**) appears inside **Nextcloud**. In version 1 it is available **only in Nextcloud**, not in Kolibri.
- **Backups do not cover Files Disk content** in version 1. Keep your own copy of important files.

F3. Can I put my own files into Kolibri?

Not directly. Kolibri only imports ready-made **Kolibri channels** (from Kolibri Studio online, another Kolibri server, or a drive with a channel exported from another Kolibri). Your own videos/PDFs must first be built into a channel on Kolibri Studio, which needs internet and an account. On our IDEA setup, Kolibri does not see USB disks or the Files Disk, so "Import from attached drive" is not available today. Use Nextcloud for your own files, and Kolibri for the prepared lessons.

Note (not yet tested on IDEA): Kolibri can also copy channels from **another Kolibri server on the same local network**, without internet (Device > Channels > Import > "Local network or internet", enter the other server's address). Both servers need Kolibri 0.11 or later. This only moves ready-made Kolibri channels, not your own files. Source: Kolibri docs, "Import from a local network or internet".

F1. Making a Files Disk (operator, captured 2026-10-12, run did not finish)

1. Select the new disk (UNFORMATTED badge), click **Make this a Files Disk**. The panel **MAKE THIS A FILES DISK** asks for a **SHARE NAME** and offers **Erase the disk first** ("Removes everything on it, then makes it a Files Disk").
2. The confirmation **Erase ?** shows model, size and contents. Type the disk name exactly, then click **Erase and make a Files Disk**.

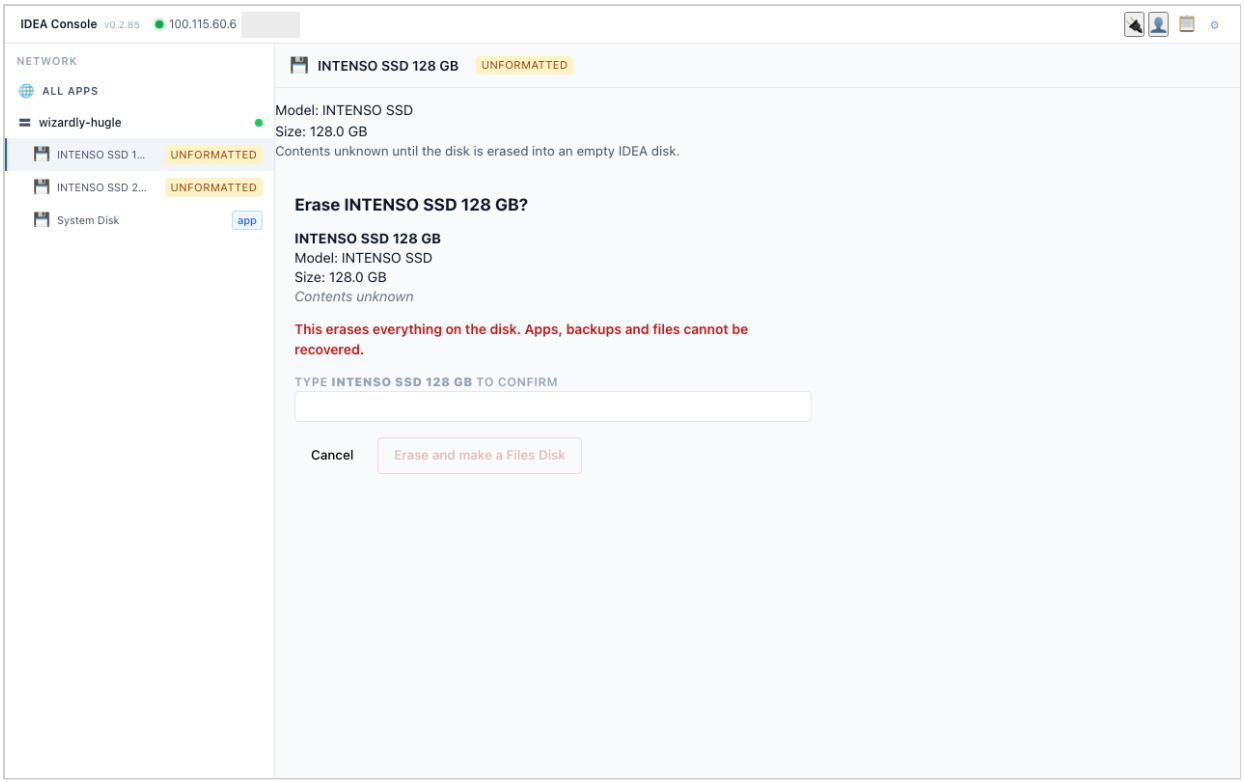


Figure 54. Erase confirmation

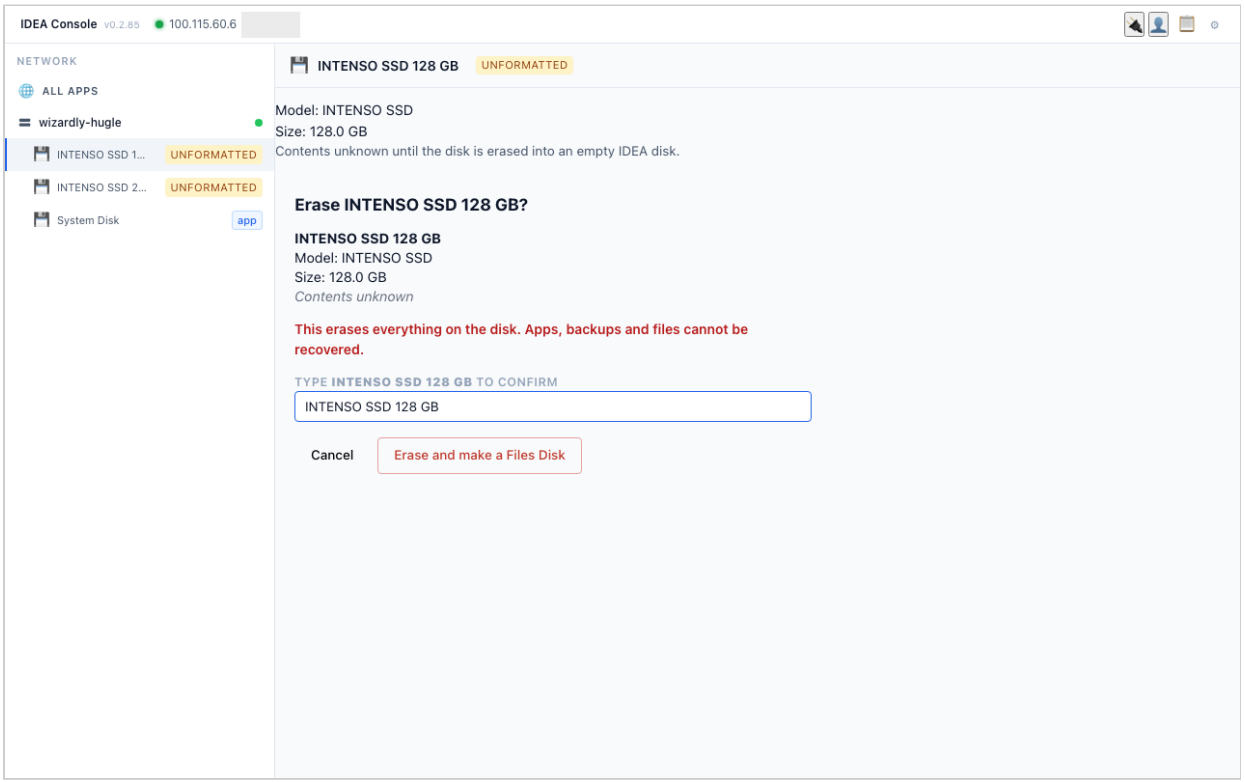


Figure 55. Name typed

3. **Known issue (being fixed):** in Console v0.2.85 the erase can finish correctly while the Console says "Erase finished but the disk did not come back as empty" and the disk disappears from the list. The erase did work; an Engine bug hides the disk afterwards, and it reappears as an empty **IDEA Disk** once the IDEA team has restarted detection. If you see this message: do **not** erase again and do not unplug the disk. Call your IDEA contact.

F1 on a disk that is already an empty IDEA Disk (captured 2026-10-12, completed)

1. Select the **IDEA Disk** marked **empty**. The view asks "What would you like to do with this disk?": **Make this a Files Disk**, **Make this a Backup Disk**, **Install App**, and **Erase this disk…** at the bottom.

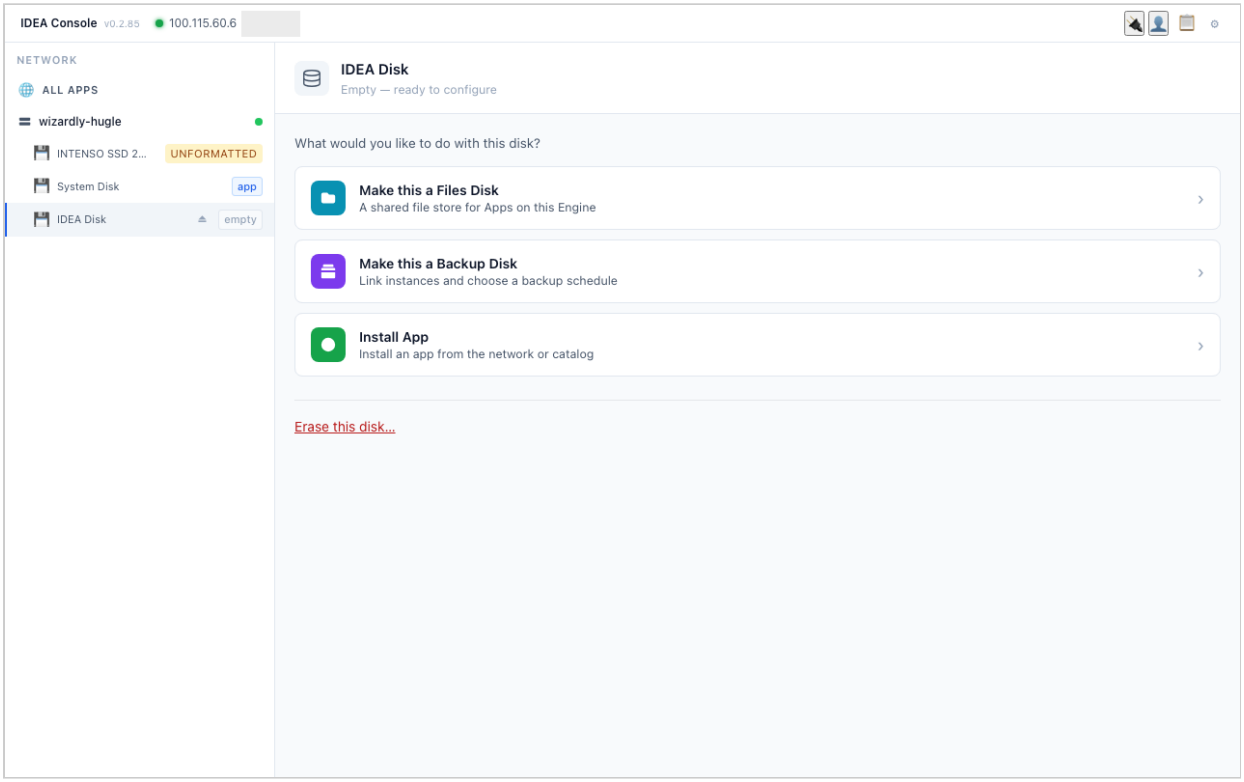


Figure 56. Empty IDEA Disk

2. Click **Make this a Files Disk**. The panel explains "Nothing on the disk is erased", with **SHARE NAME** pre-filled as **School Files**. Click **Make this a Files Disk**. (The red **Erase the disk first** is only for disks with other content.)

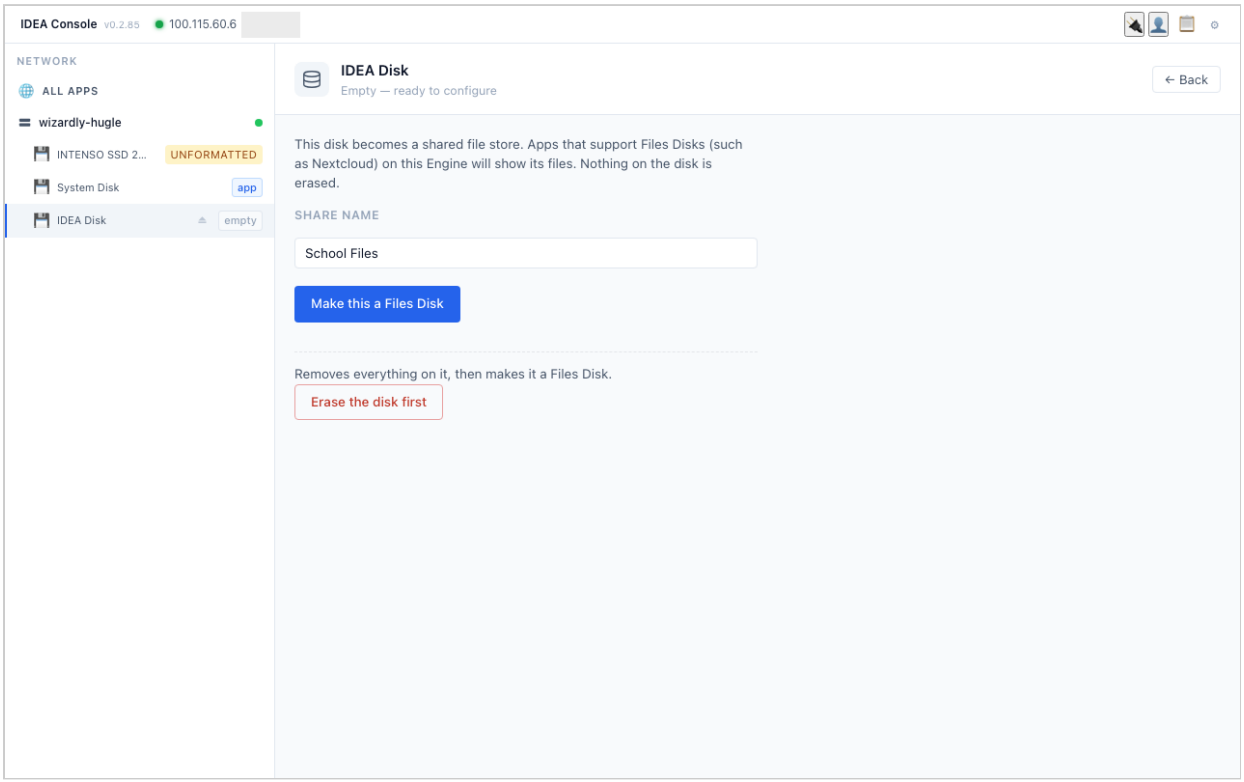


Figure 57. Make this a Files Disk

3. After a short wait the disk shows the **files** badge, "Share name: School Files", free space (here 119.0 GB free of 125.4 GB), an **Eject** button and **Erase this disk…**.

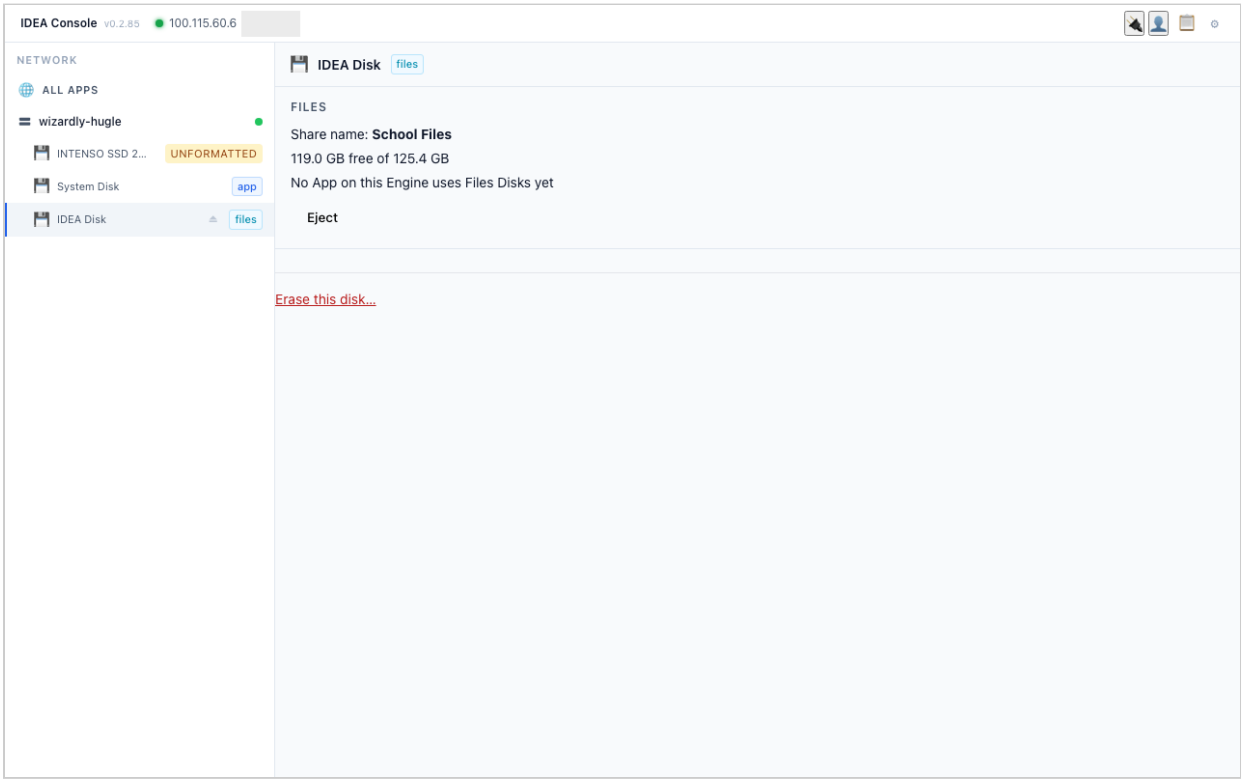


Figure 58. Files Disk

F2. Using the Files Disk in Nextcloud (captured 2026-10-13)

1. In the Console, the Files Disk view now says **"Available in: Nextcloud (…)"**, with the share name, free space, **Eject** and **Erase this disk…**.

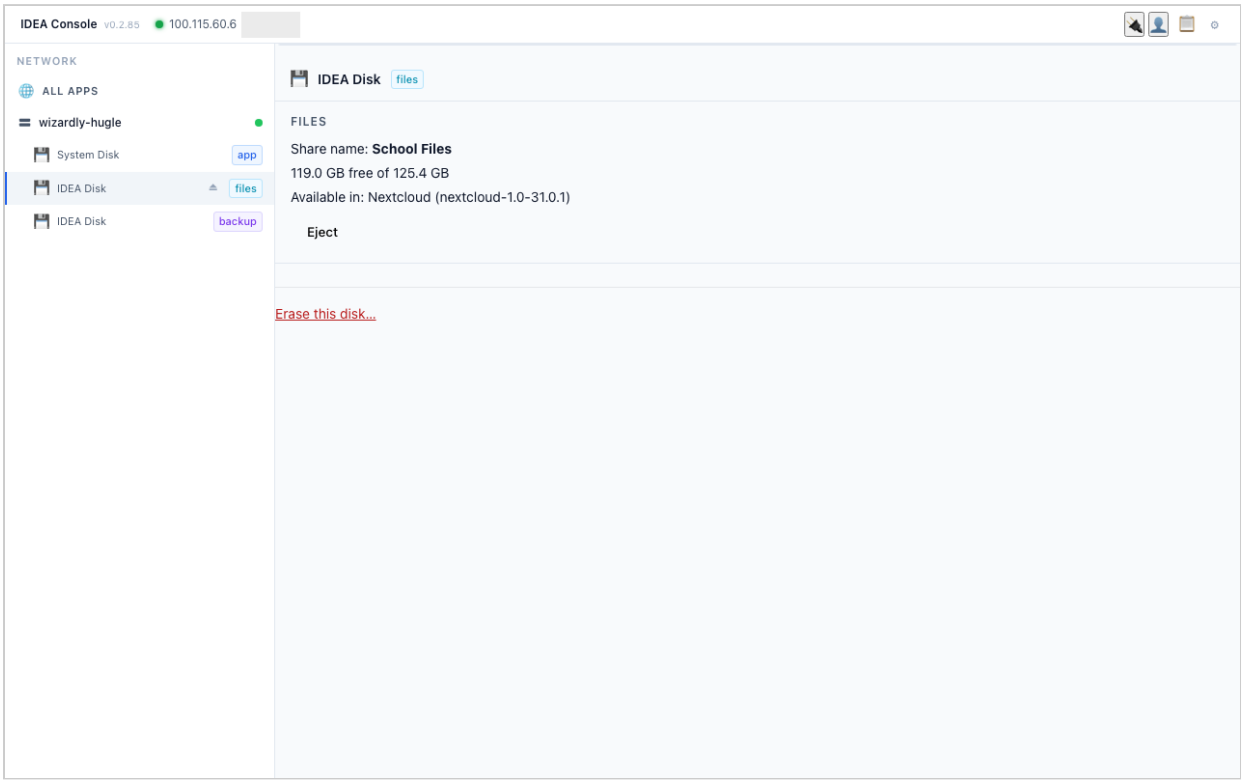


Figure 59. Files Disk available in Nextcloud

2. In Nextcloud (as a teacher), **Files** shows a folder **School Files** (marked as External storage in the left menu).

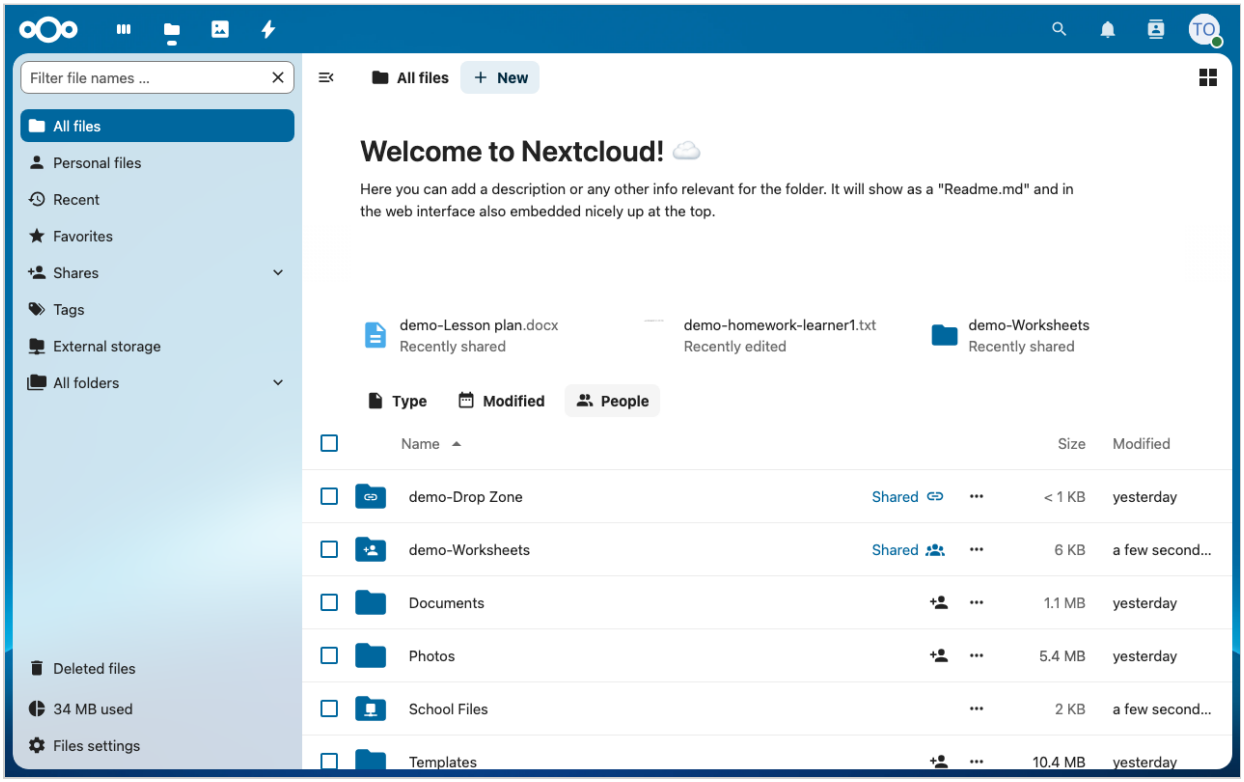


Figure 60. School Files in Nextcloud

3. Open **School Files** to see what is on the disk. Files copied onto the disk (here `demo-Fractions-Worksheet-1.pdf`) appear here.

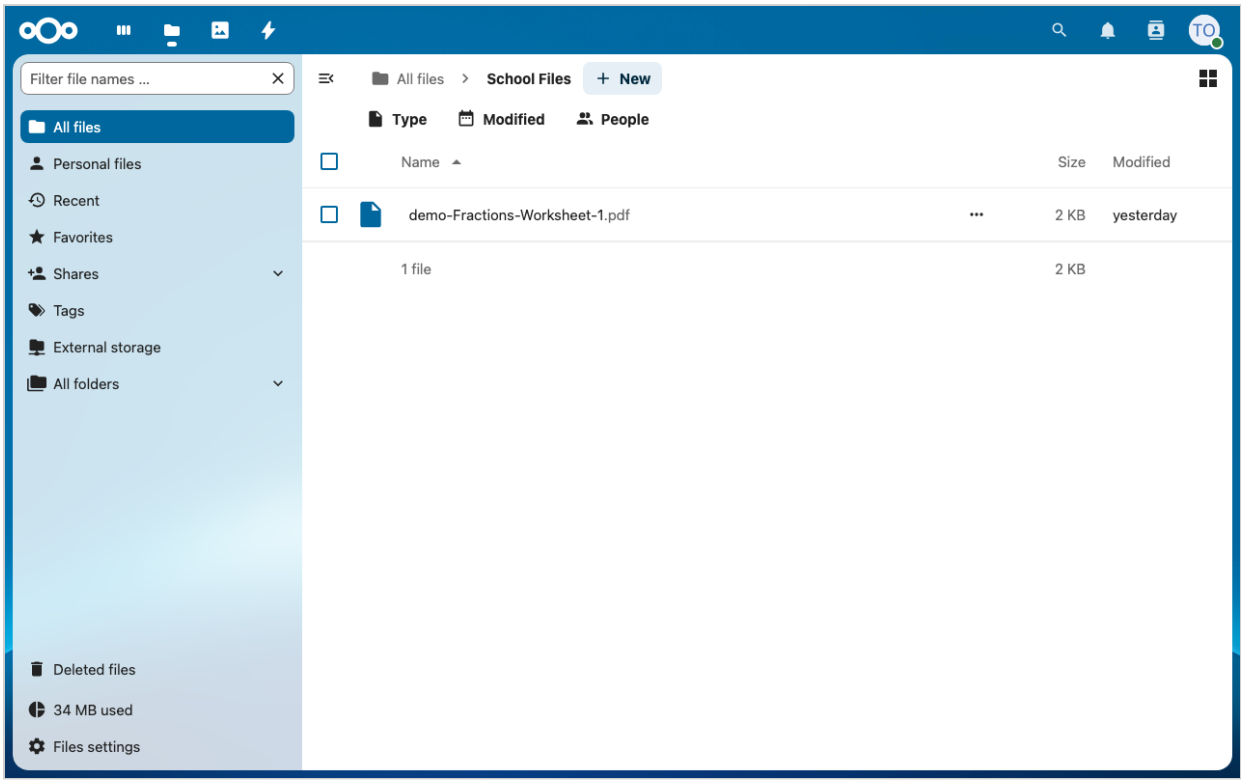


Figure 61. School Files content

4. To share a file with a class: on the file's row click ··· (Actions) > **Show sharing options** (or the share icon). In **Internal shares**, type the class group name (here `demo-Grade5A`), choose it from the list, keep **View only** and save. The group then appears under Internal shares and the row shows "Shared with group demo-Grade5A".

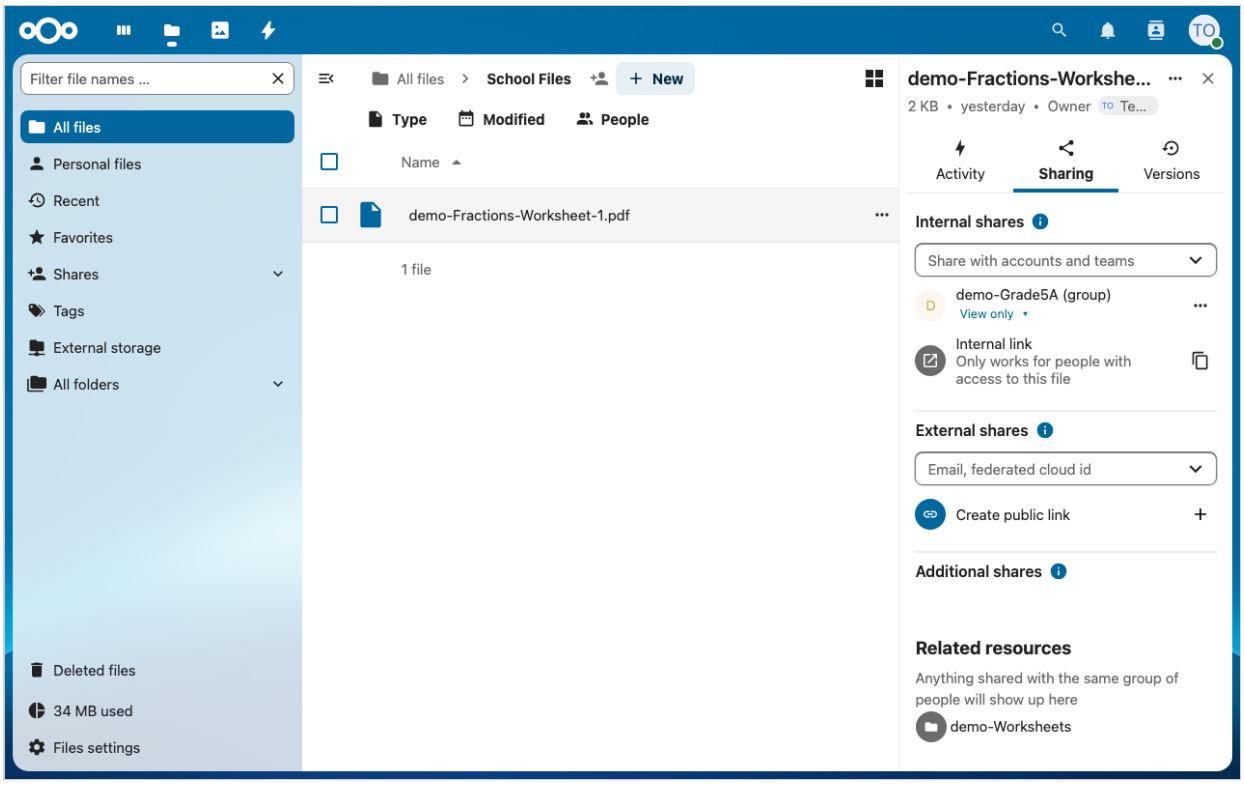


Figure 62. File in School Files shared with demo-Grade5A

5. A learner in that group (here `demo-learner1`) signs in and finds the file in **All files** (and under Recently shared).

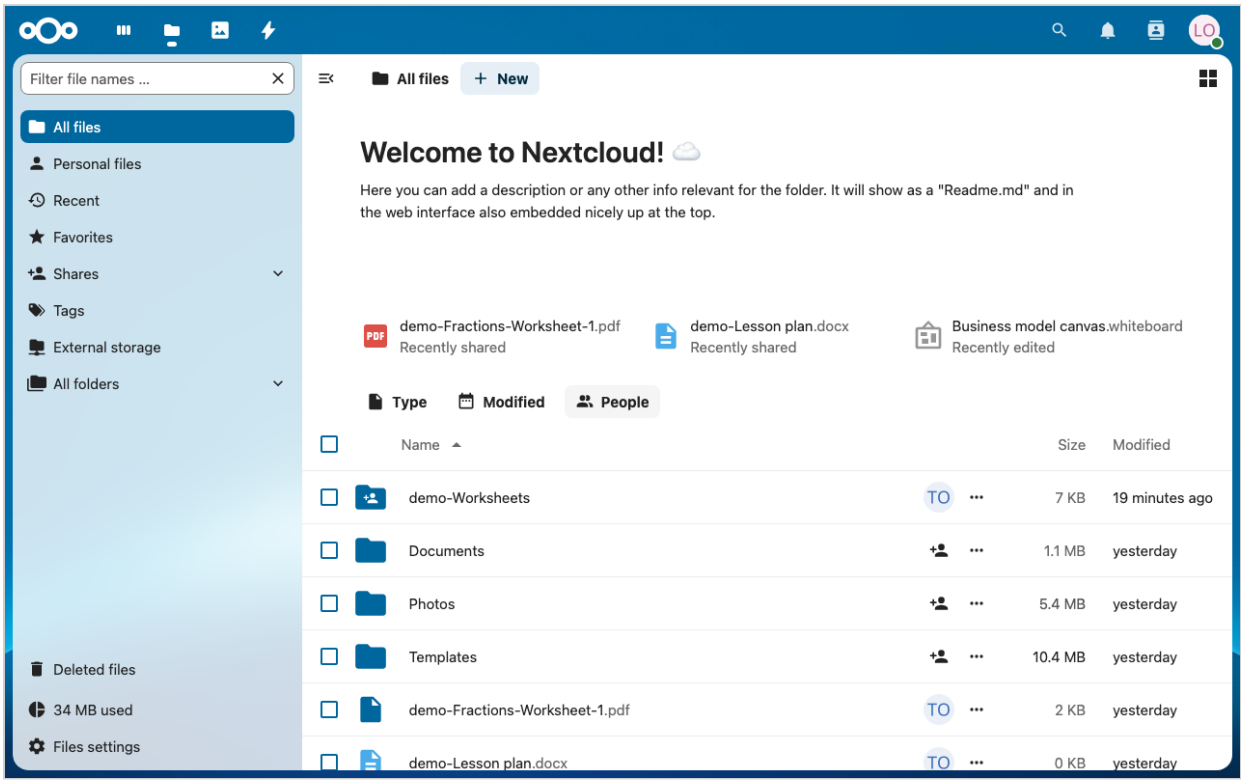


Figure 63. Learner sees the shared file

Only teachers see the **School Files** folder; learners get only the files that are shared with them.

Reminder: in version 1 a Files Disk is for **Nextcloud only** (not Kolibri), and **backups do not cover Files Disk content** (the Backup Disk links app instances only). Keep your own copy of important files.

9. Keeping data safe: the Backup Disk (concepts)

Making and restoring backups is done by the operator and practised on a test Pi. Teachers should know:

- A **Backup Disk** is a separate disk with the **backup** badge in the Console.
- It saves the **apps' data**: Kolibri (classes, learners, progress) and Nextcloud (accounts, files inside Nextcloud).
- It does **not** save Files Disk content (version 1).
- During a backup, the app stops for a short time and then starts again. Plan backups outside lesson time.
- If a Pi breaks, the operator restores the latest backup onto a new disk. You lose only what was done after the last backup.
- Never unplug a disk while a backup is running. If it happens, the backup restarts next time the disk is connected. How often the school backs up (each time the disk is plugged in, or on request): TO CONFIRM per school.

Hands-on (operator, captured on the demo box 2026-10-12/13)

B2. Make a Backup Disk

1. Plug in a spare disk. If it is new, it first has to become an empty **IDEA Disk** (chapter 8, Erase).
2. Select the **IDEA Disk** marked **empty** and click **Make this a Backup Disk** ("Link instances and choose a backup schedule").

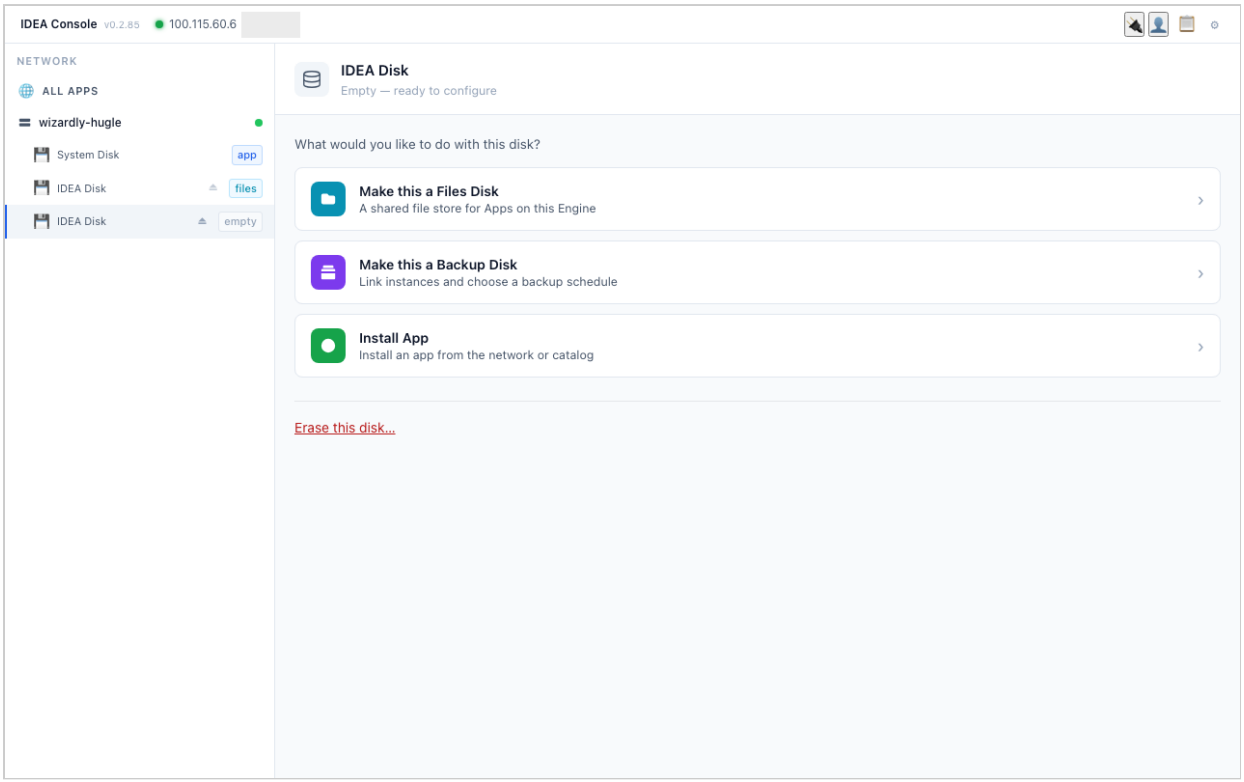


Figure 64. Empty disk options

3. Choose when backups run: **On demand** (only when you press **Back up**; used in the demo), **Immediate** (as soon as the disk is docked) or **Scheduled**.

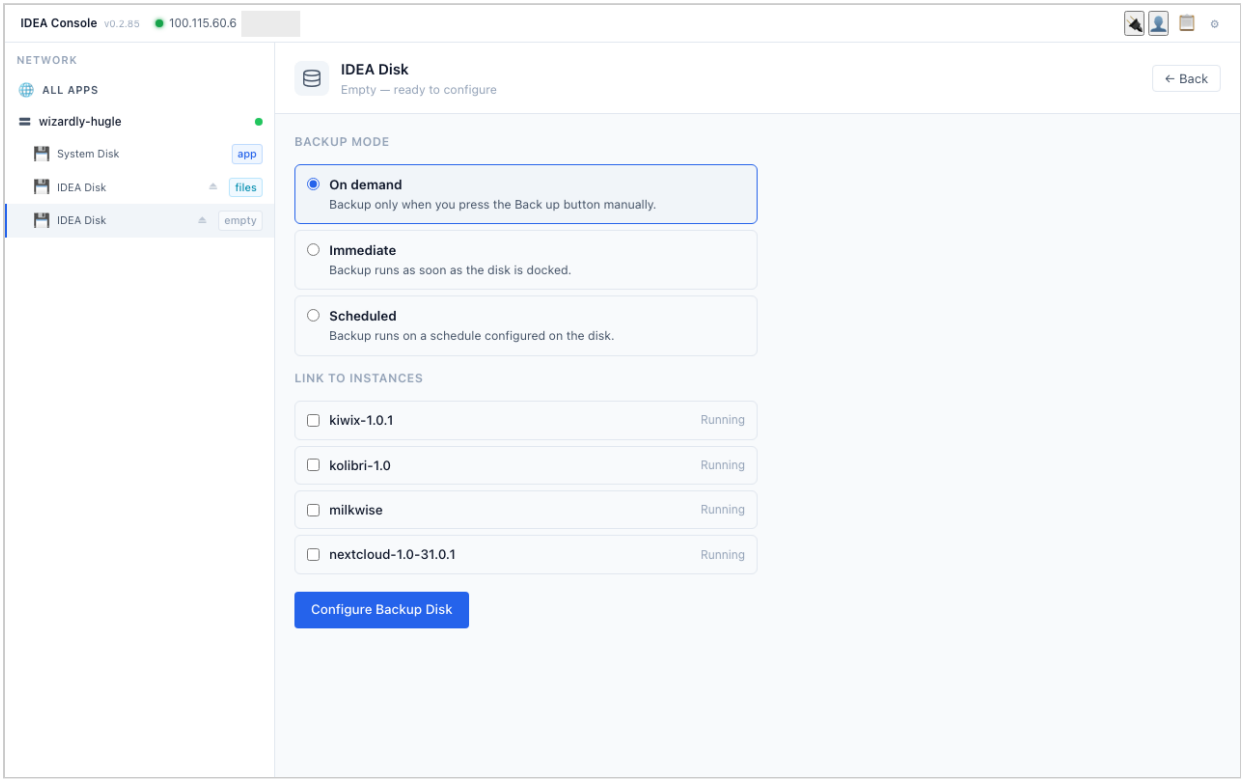


Figure 65. Backup mode

4. Under **LINK TO INSTANCES**, tick only the school apps to protect (demo: **kolibri-1.0** and **nextcloud-1.0-31.0.1**). Leave other apps unticked.

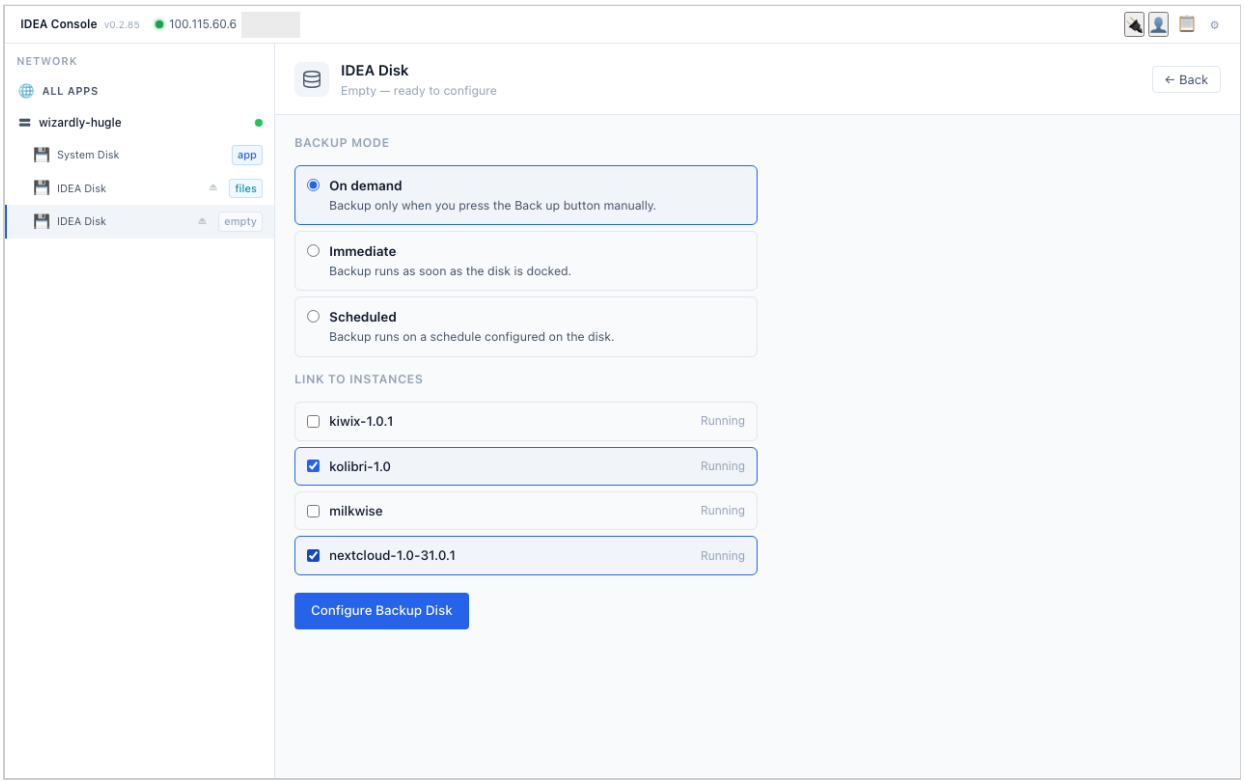


Figure 66. Linked apps

5. Click **Configure Backup Disk**. The disk gets the **backup** badge and a **Backups** list: each linked app, "Never backed up", and **Restore**.

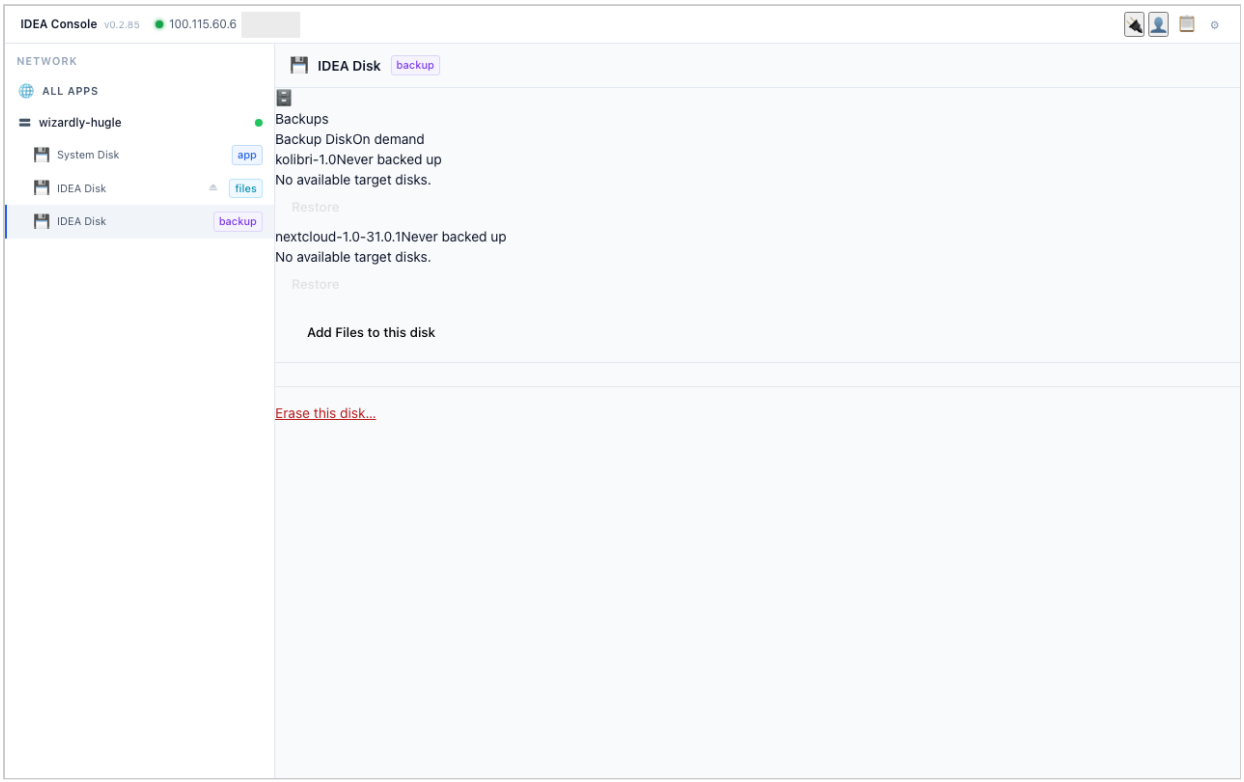


Figure 67. Backup Disk configured

B3. Run a backup

1. In **ALL APPS**, each linked app row has a **Back up** button next to Start / Stop / Open. Click it for Kolibri.

   > **Screenshot B3-10a.png pending retake** (Back up clicked).

2. The app stops for a short time while it is copied, then starts again. On the Backup Disk the app shows **Operation in progress**. Do this outside lesson time.

   > **Screenshot B3-15.png pending retake** (Operation in progress).

3. When it is done, the Backups list shows **Last backup:** (demo: Kolibri 8:46, Nextcloud 8:49, about 3 minutes each). Repeat for Nextcloud.

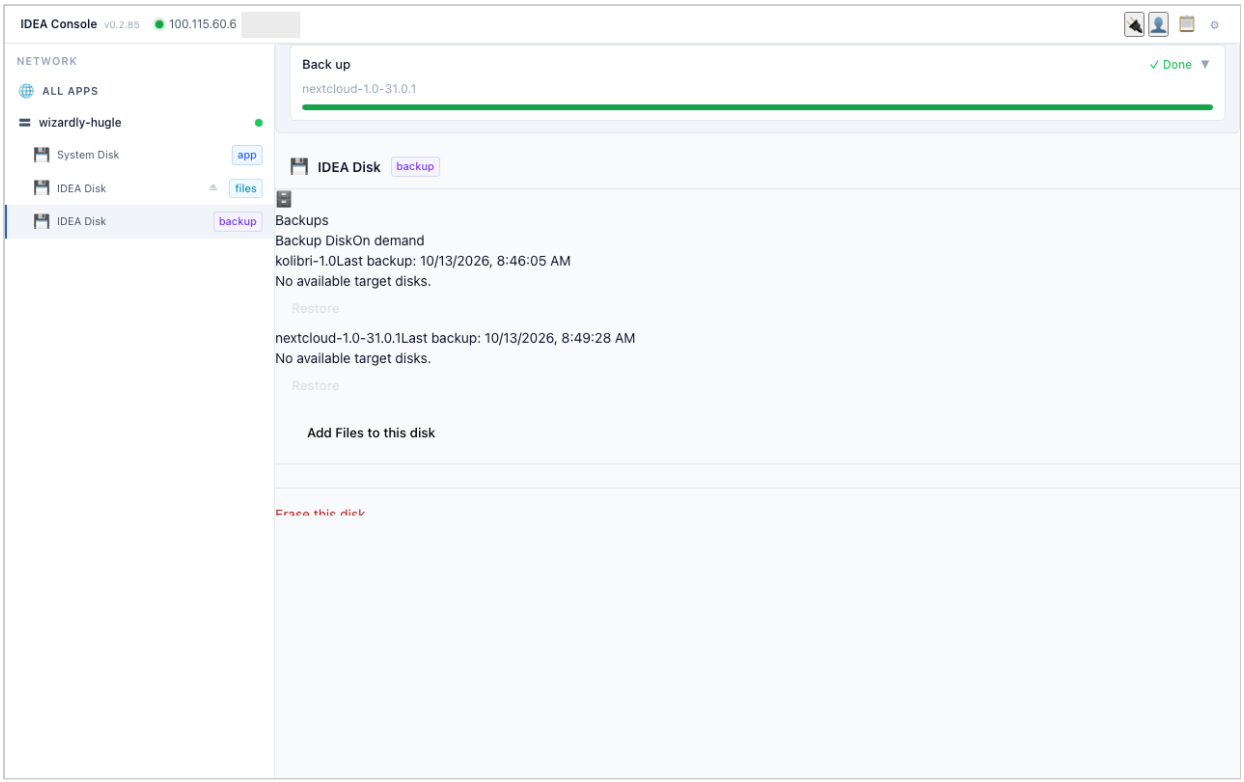


Figure 68. Last backup for both apps

4. The **OPERATIONS** box at the top keeps old messages (in the demo: failed attempts from the day before, when the backup tool was not yet installed). Check the **Last backup** time, not old messages. Close old messages with ×.

B4. Restore after a crash

1. On the Backup Disk, each app in the **Backups** list has a **Restore** button.

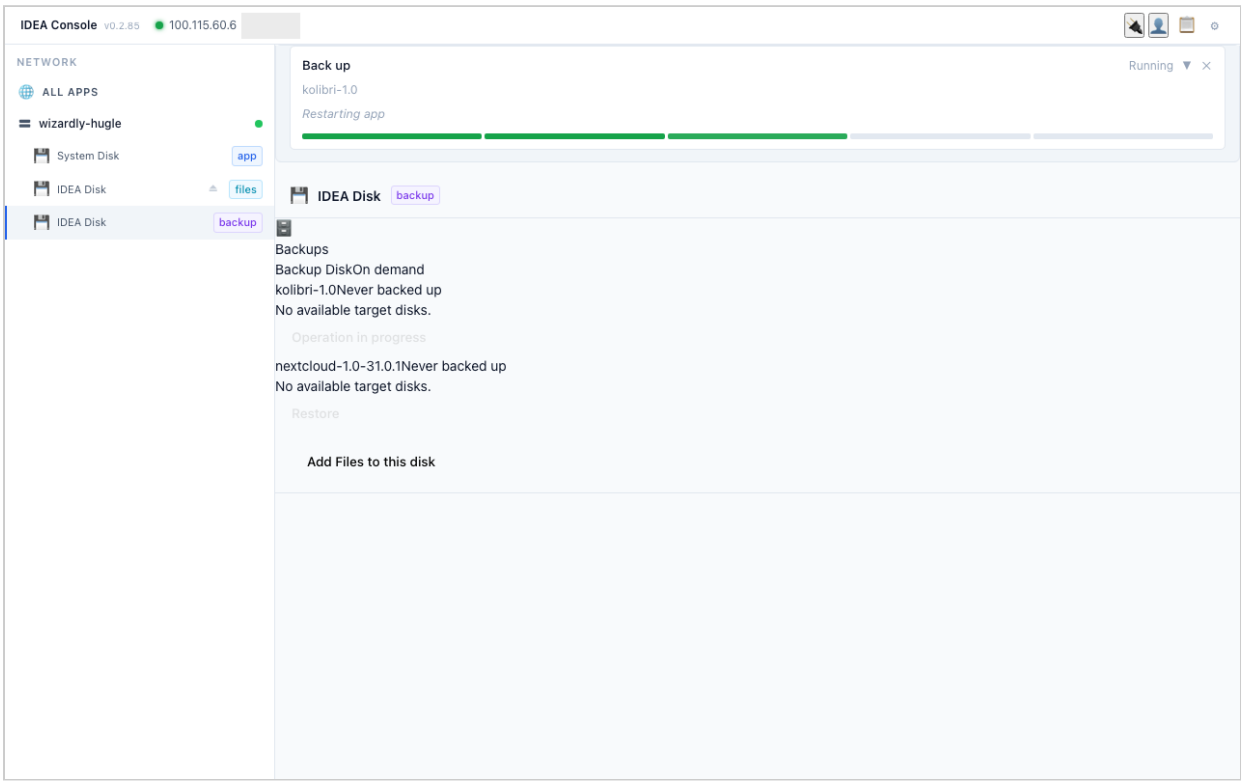


Figure 69. Restore and no target disks

2. Restore needs a **target disk**: a spare, empty IDEA Disk to put the app on. On the demo box it says **"No available target disks."** because the only other disks are the Files Disk and the System Disk, and the Console does not offer those as targets.
3. With a spare empty IDEA Disk plugged in, choose it as the target and confirm. The full steps, tried on the demo box, are below. Never restore onto a disk that holds other apps you need.

> **Erasing a Files Disk that an app is using.** If Nextcloud is using the Files Disk (School Files), the erase cannot run because the disk is busy, and the Console gives no clear error. Stop Nextcloud first (with your IDEA contact), then erase.

B4 on the demo box (2026-10-13): restoring Kolibri onto a spare disk

Before you start you need a spare, **empty IDEA Disk** (here the 128 GB disk, erased first; see the note above about stopping Nextcloud).

1. Open the **Backup Disk**. In the **Backups** list each app shows **Last backup:** , a **Select target disk…** list and a **Restore** button (grey until a target is chosen).

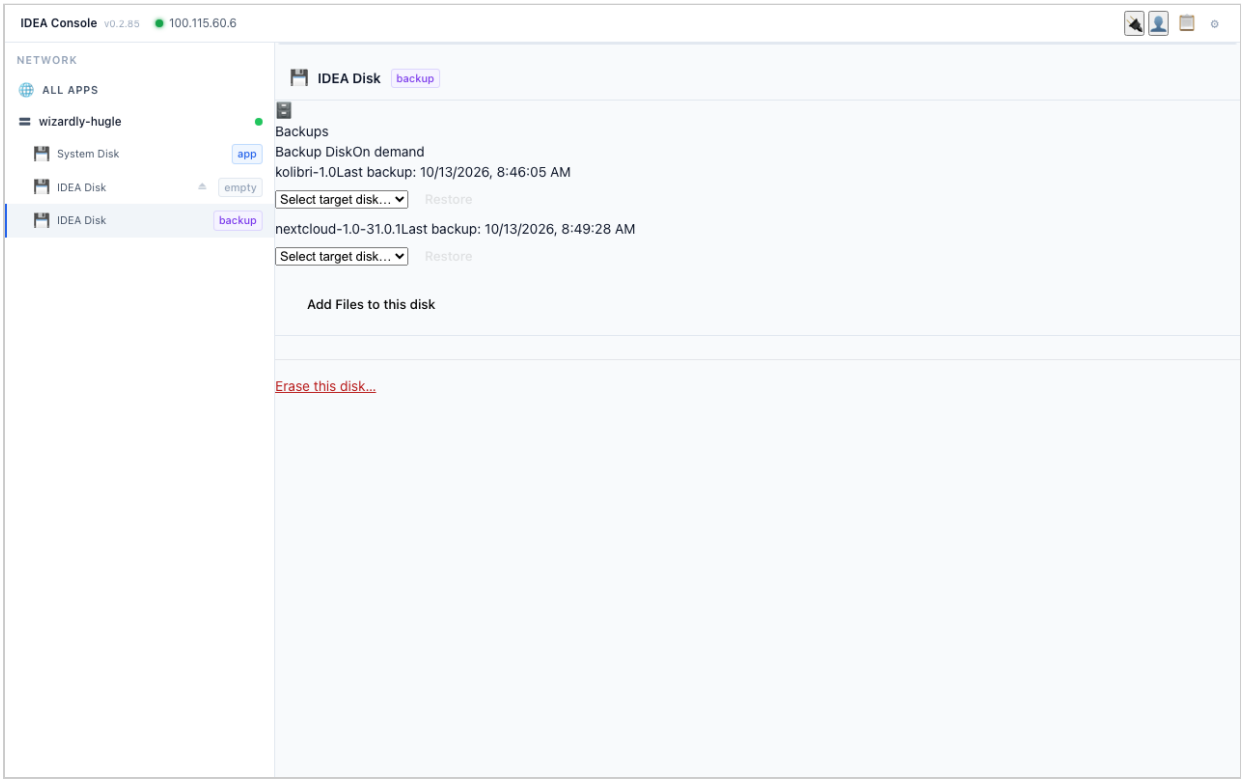


Figure 70. Backups list

2. In **Select target disk…** choose the spare disk. Only spare, non-system disks are offered (here one **IDEA Disk**). Then click **Restore**.

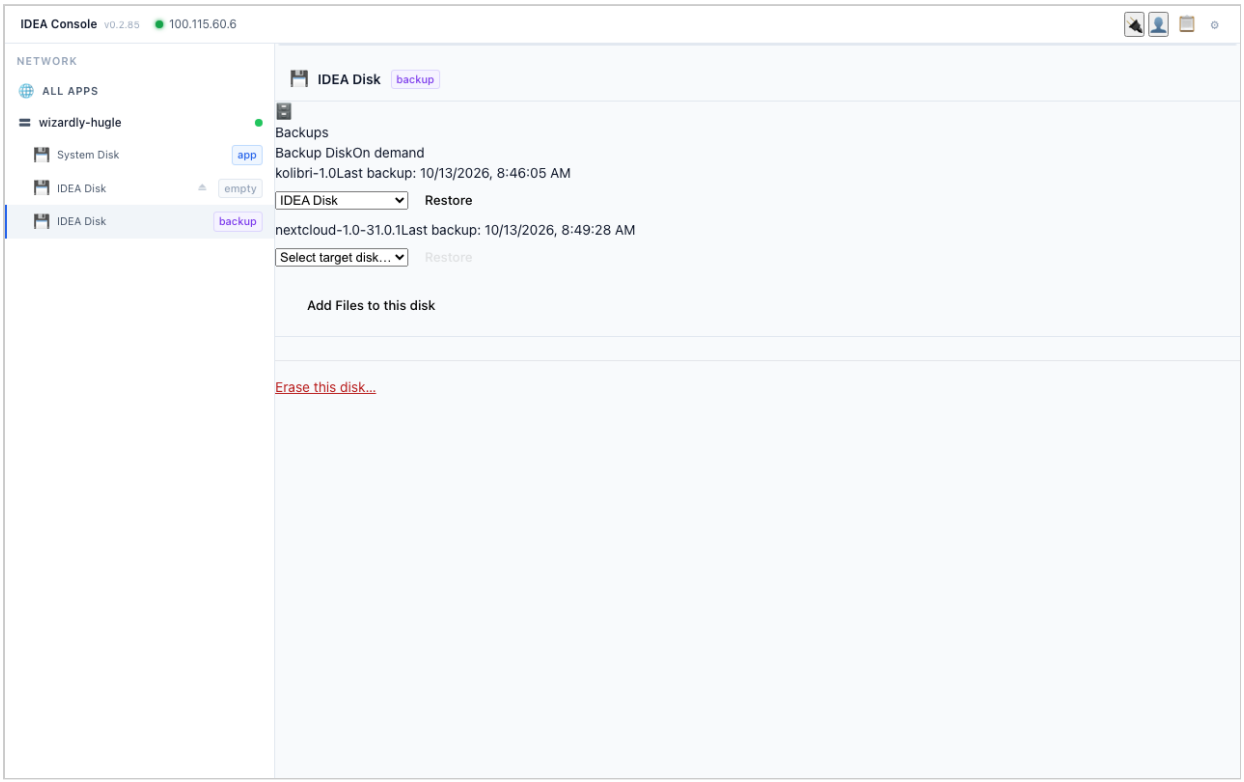


Figure 71. Target chosen

3. The Console asks: "Are you sure? This will overwrite **kolibri-1.0** on **IDEA Disk**." Click **Confirm Restore** (or **Cancel**).

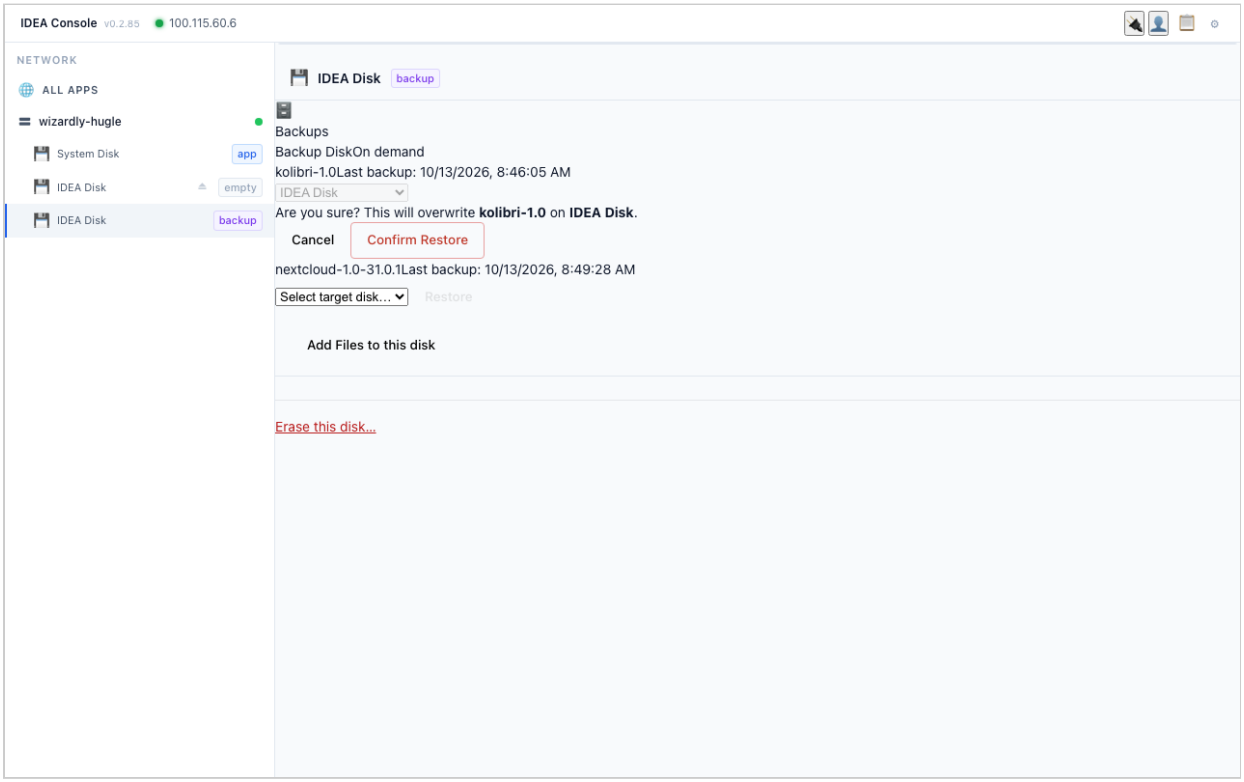


Figure 72. Confirm Restore

4. After about a minute the spare disk shows the **app** badge and lists **kolibri-1.0**.

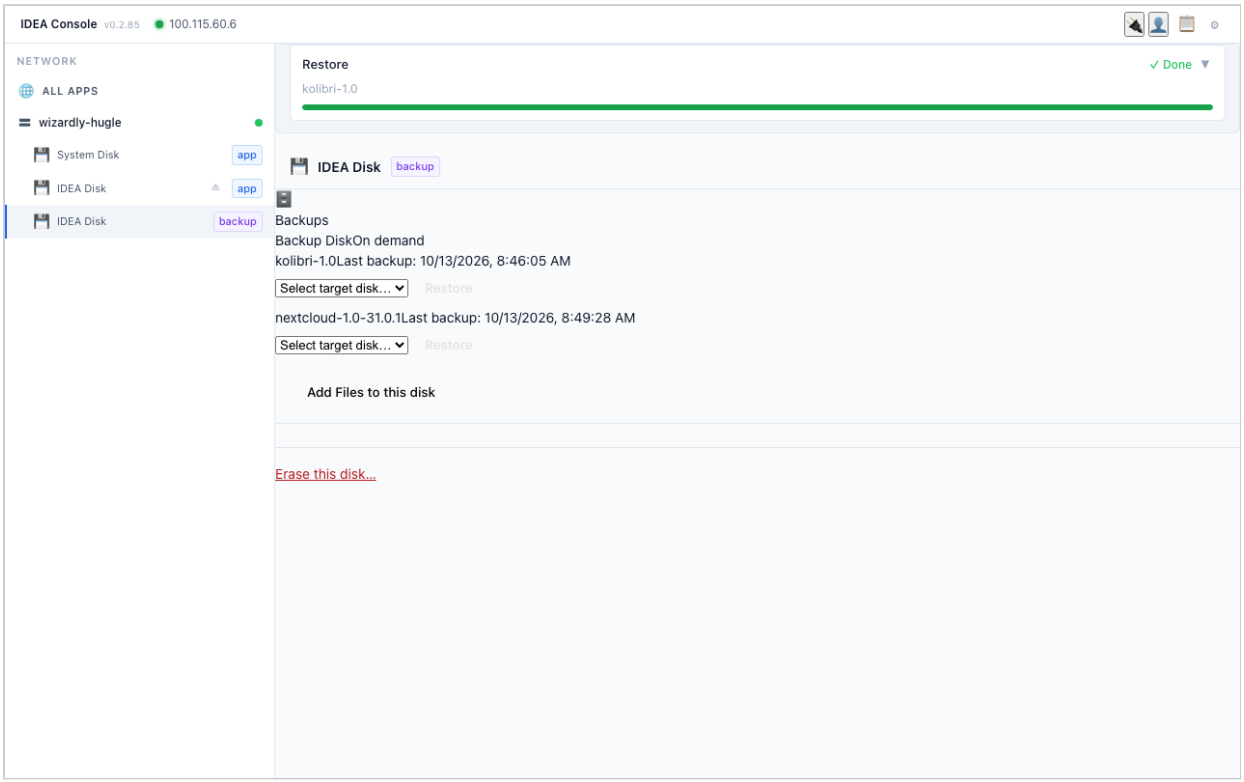


Figure 73. Restore running

> **Screenshot B4-25.png pending retake** (Restored Kolibri on the spare disk).

5. **Important (seen on the demo box):** the restore **moves** the app. Kolibri is no longer listed on the System Disk (it now shows only the other apps), and the restored Kolibri is **stopped** (orange dot, **Start** button). Until you start it, Kolibri is not reachable. So a restore is for a real breakdown, not for testing on a working school box.

> **Screenshot B4-27.png pending retake** (System Disk after restore).

6. On the spare disk, click **Start** on the restored Kolibri row. After about 30 seconds it is running again (demo: started 11:15) and the row shows **Open ↗**. The Console may still show an old red message; check **Open**, not the message.

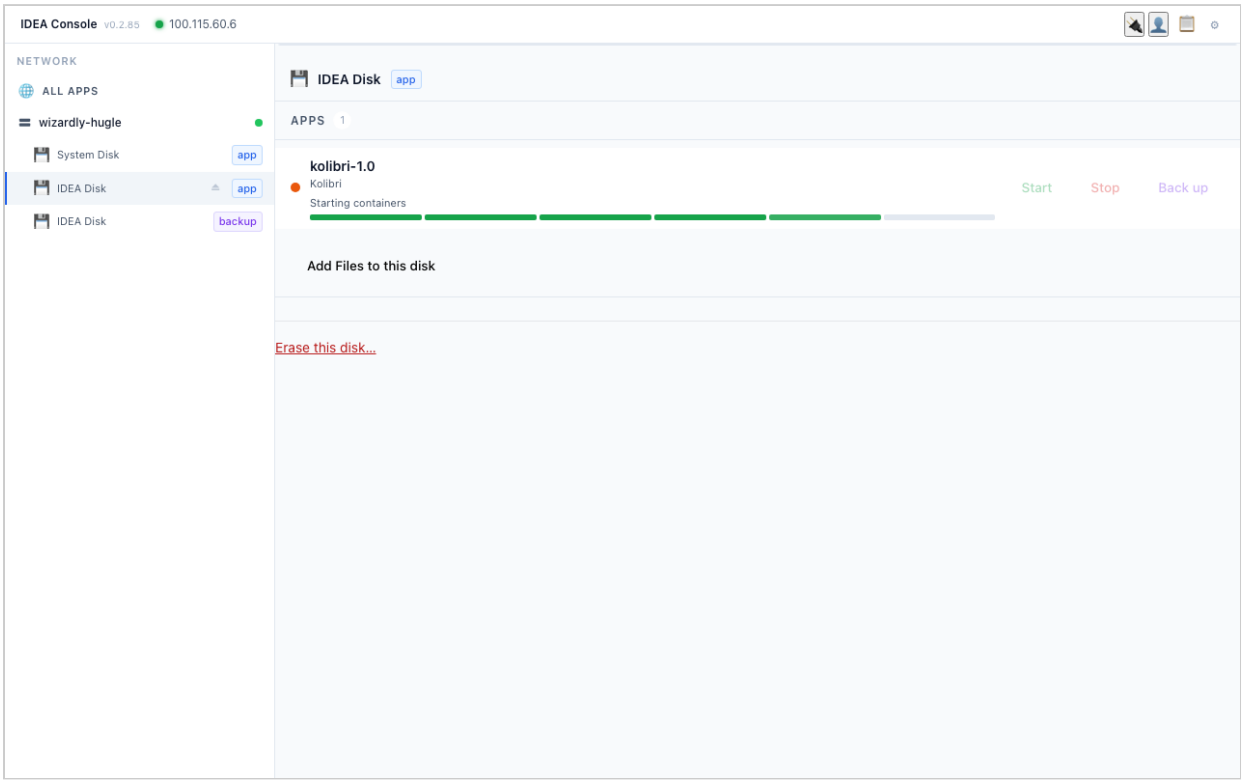


Figure 74. Start clicked

> **Screenshot B4-29.png pending retake** (Restored Kolibri running).

7. Open Kolibri from its **Apps** card as usual (chapter 2) and sign in as a coach.

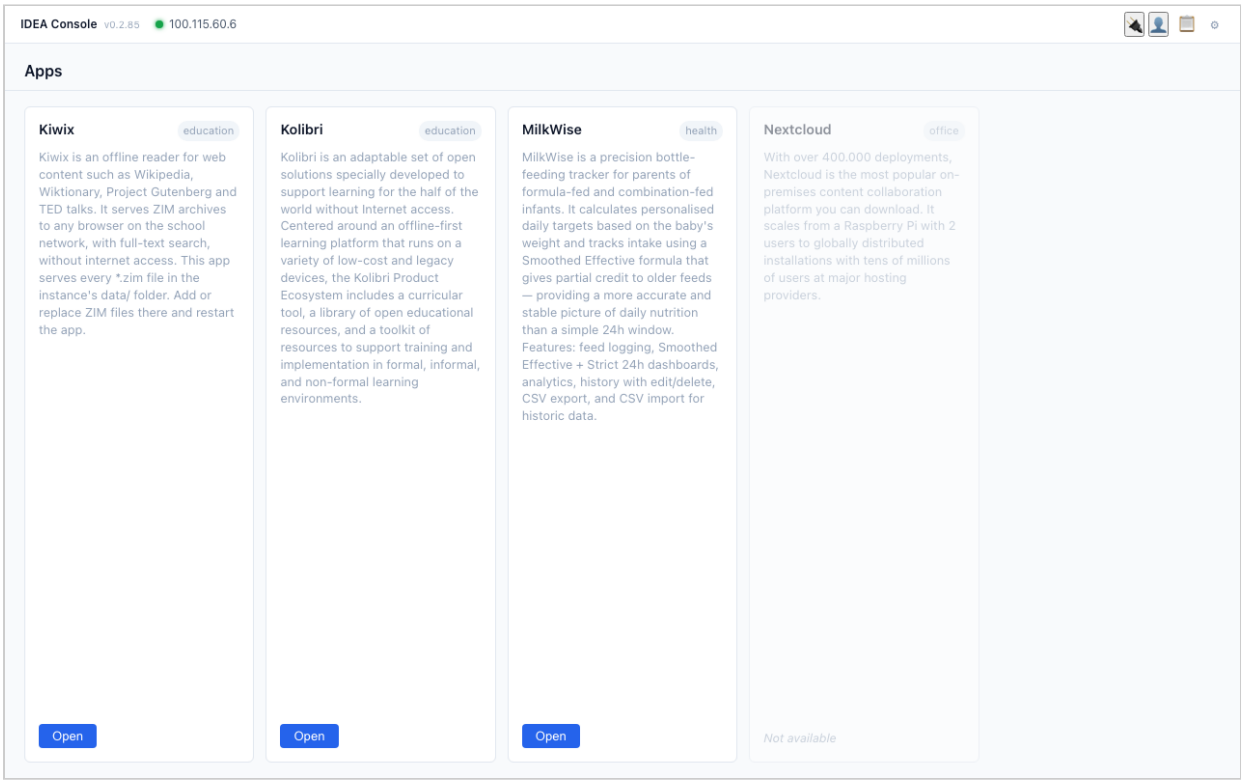


Figure 75. Apps page

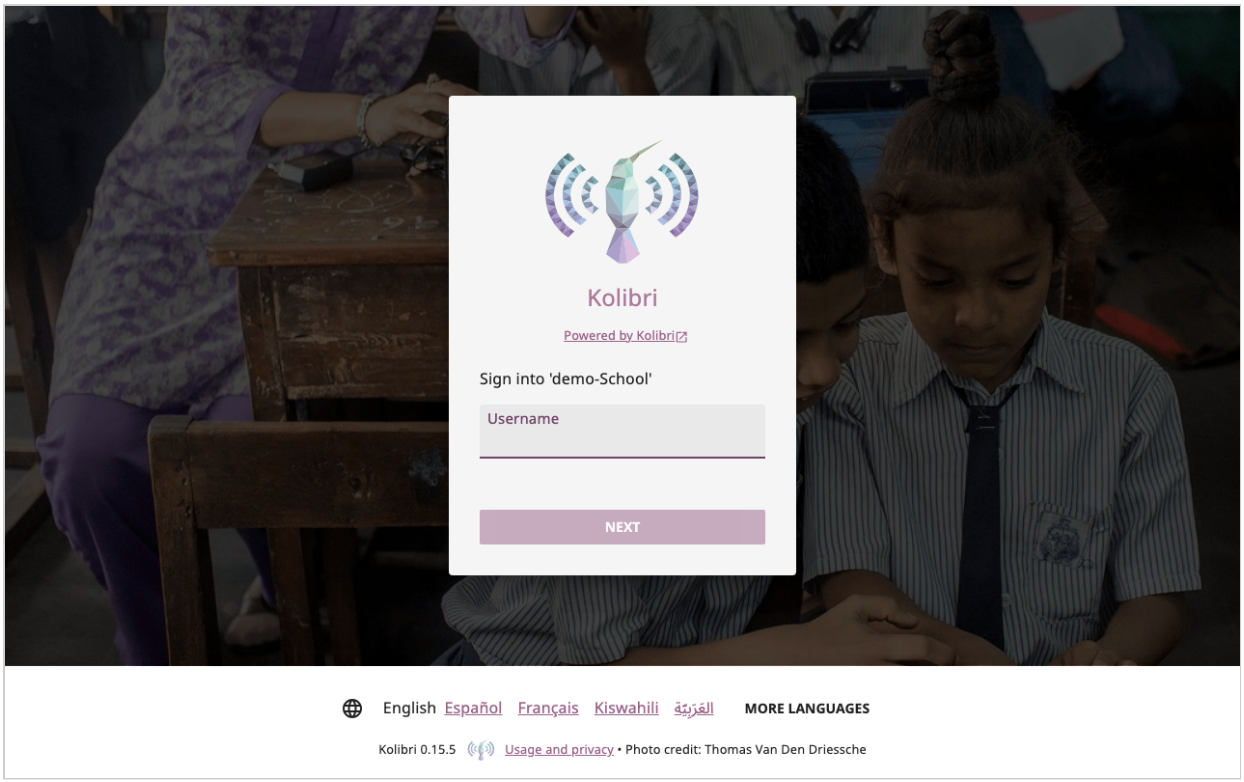


Figure 76. Kolibri sign-in

8. Check that your data is back. On the demo box the class **demo-Grade5A** had its 3 learners, the **demo-Fractions quiz** (completed by 1 of 3), the **demo-Fractions lesson** (2 of 3 started) and the class activity, exactly as at the last backup (8:46).

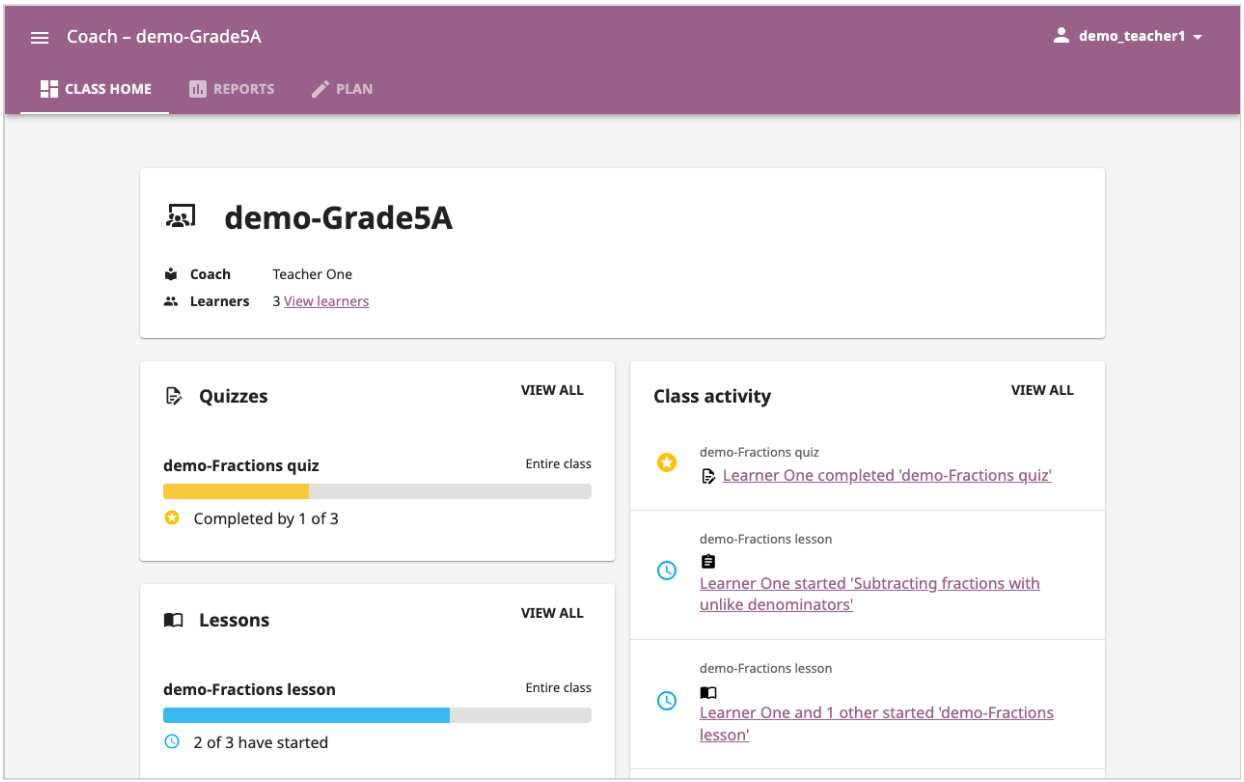


Figure 77. Class after restore

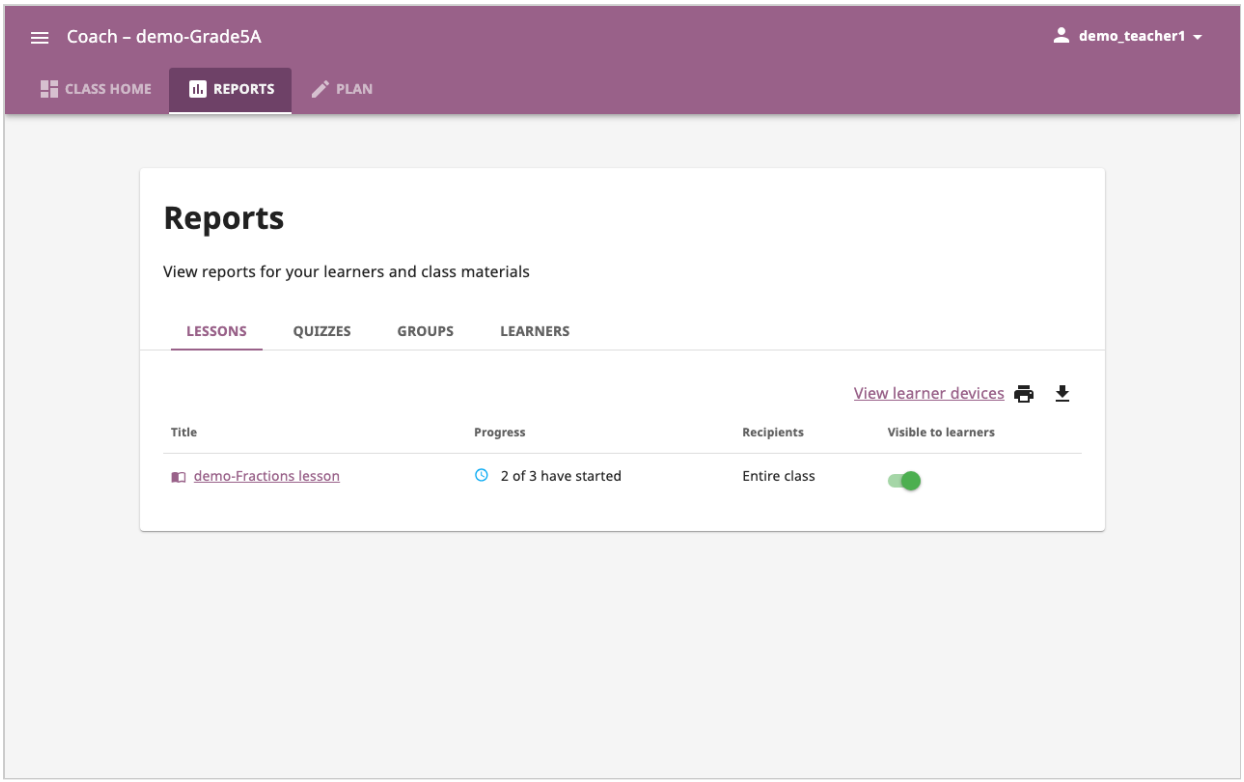


Figure 78. Reports after restore

Anything done after the last backup is lost. Make a new backup of the restored app as soon as it runs.

Known Console quirks (v0.2.85)

- The **green dot** next to an app in ALL APPS is not always the real app state: on the demo box Nextcloud showed green while it was stopped. Check whether **Open** works.
- An app's version label can be out of date (Nextcloud showed **1.0-31.0.1** while 1.1-31.0.1 was installed).
- After an erase the disk can disappear from the list for a while (Engine lock bug). Do not erase again; call your IDEA contact.
- Old red **Back up Failed** messages stay in OPERATIONS until closed with ×.

10. Console basics

C1. Read the Console

1. The **Apps** page shows one card per app with a short description and a category (education, office, …).

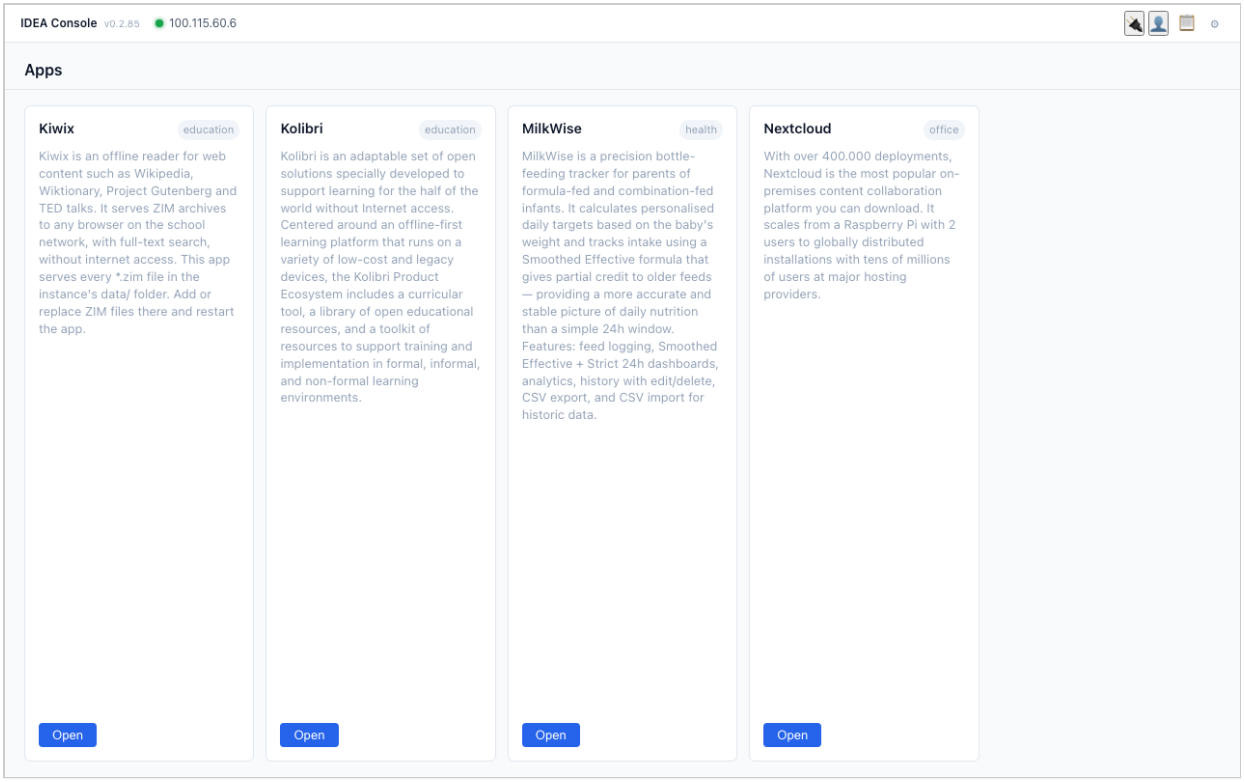


Figure 79. Console

2. A card with a blue **Open** button is running. A greyed-out card with **Not available** is not.
3. The green dot and address at the top show the Console is connected to the IDEA box. Orange "Connecting…" means wait a moment.

C2. App not running?

Do not try to start it. Note the app name and tell your operator.

C3–C4. Operator area

The icons top right (plug: Connection Management, person: Operator Login, clipboard: Command History, gear: Settings) are for operators. The disk view (Network, disks, Eject, Erase, Install) is only visible after **Operator Login** and is **not** covered in this teacher guide (TO CONFIRM with Steve whether to show it).

11. Operator basics

For the school's IDEA operator, not for every teacher. Console v0.2.85 on the demo box. Everything here is **look only**: do not press Start, Stop or other action buttons unless your IDEA contact asks you to.

O1. Operator Login

1. In the Console, click the person icon (top right). The **Operator Login** panel opens with a username and a password field (the eye icon shows/hides the password).

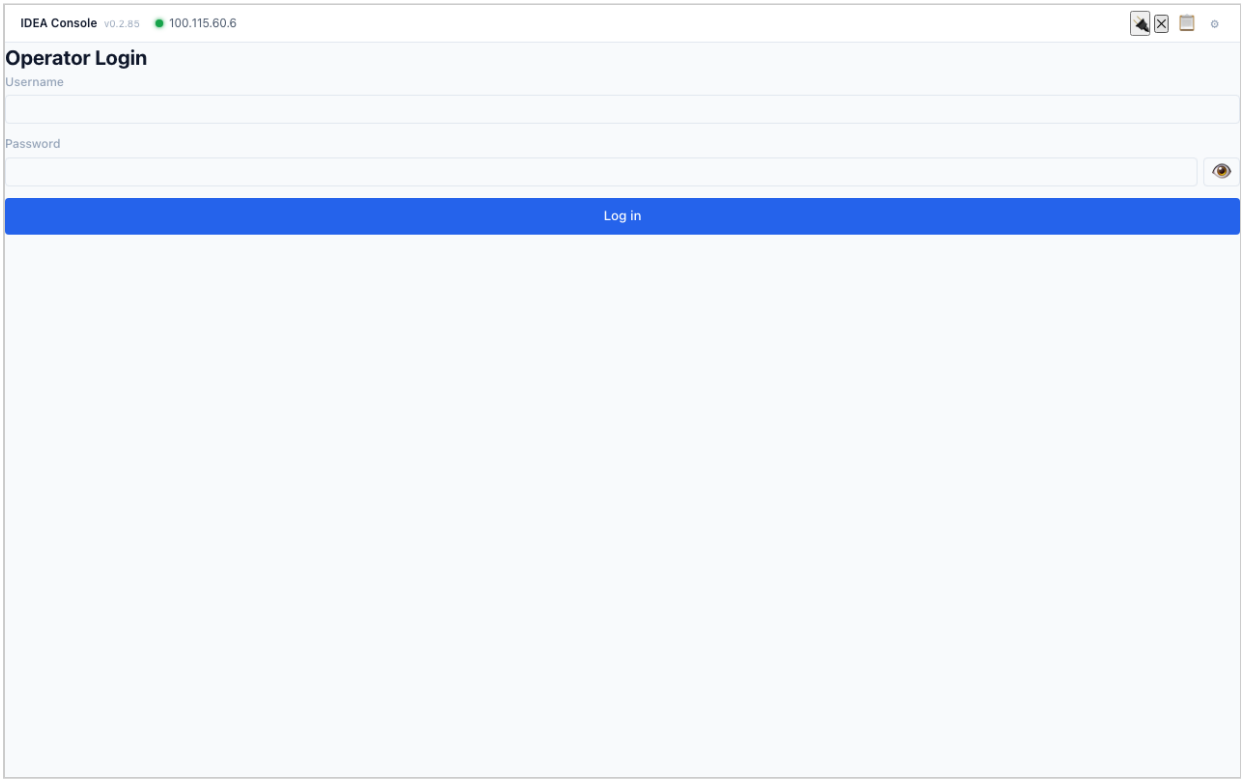


Figure 80. Operator Login

2. Type the operator username and password, click **Log in**. The **Account** panel then shows your name, role **Operator**, **Change Password**, **Manage Operators** and **Log out**.

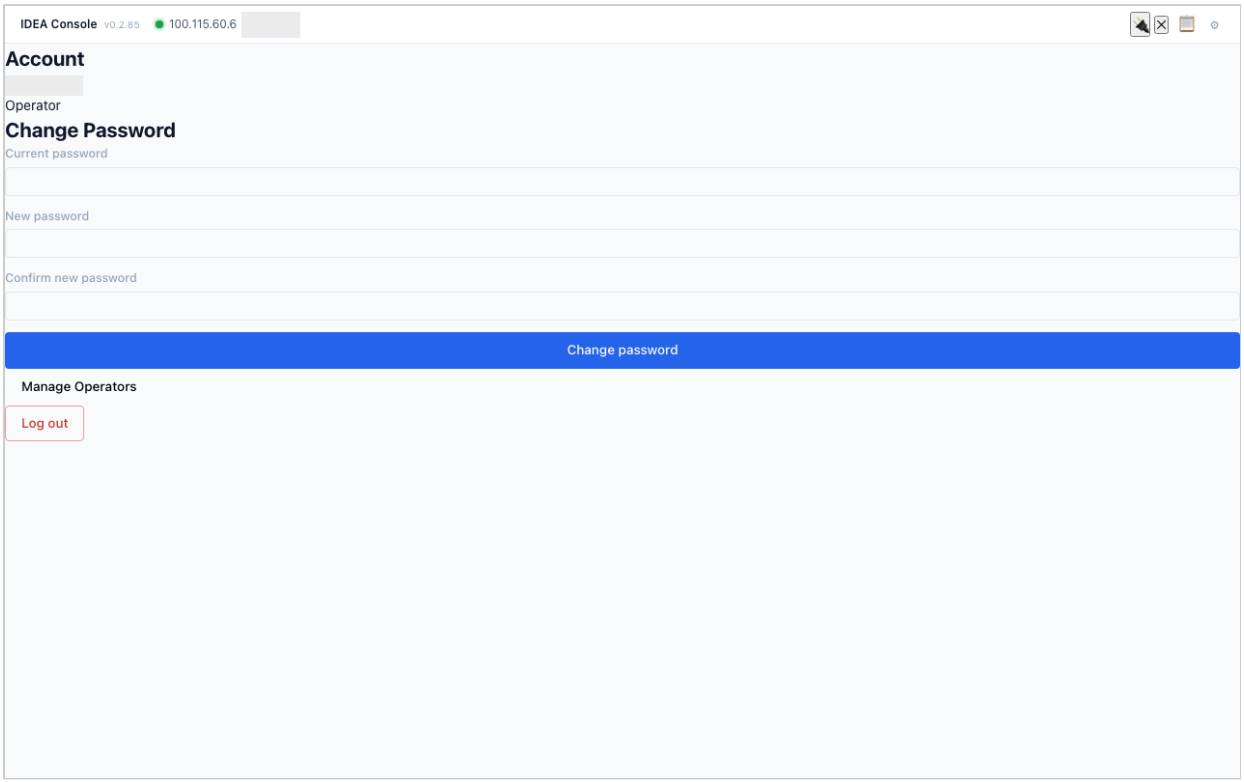


Figure 81. Account panel

3. **On a real school deployment, change the default operator password straight away** (Account > Change Password: current password, new password twice, **Change password**). Keep the new password with the school head, not on a sticker on the box.
4. Close the panel with ×. Use **Log out** in the same panel when you are done.

O2. The Network view: Engine and disks

1. After login the left side shows **NETWORK**: **ALL APPS**, then the IDEA box (Engine, here `wizardly-hugle`, with an **Online** status) and its disks.

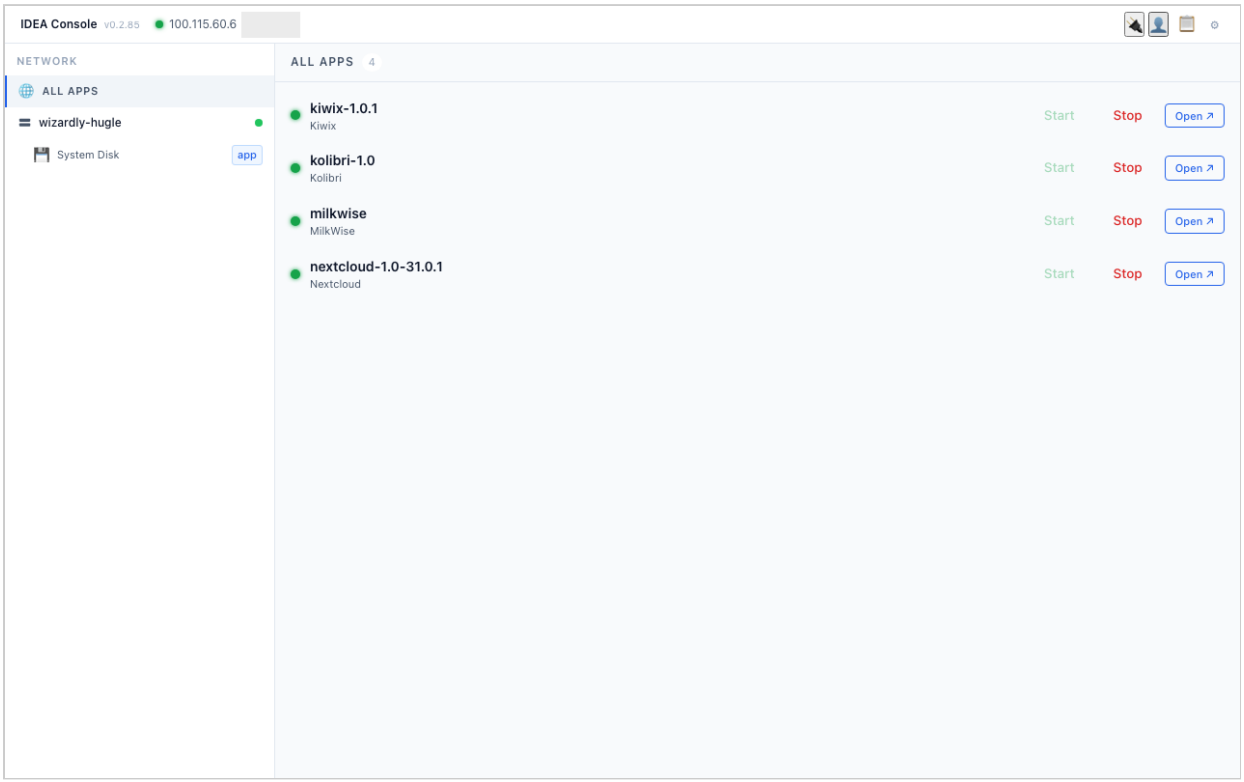


Figure 82. All apps

2. Click the Engine name to select it.

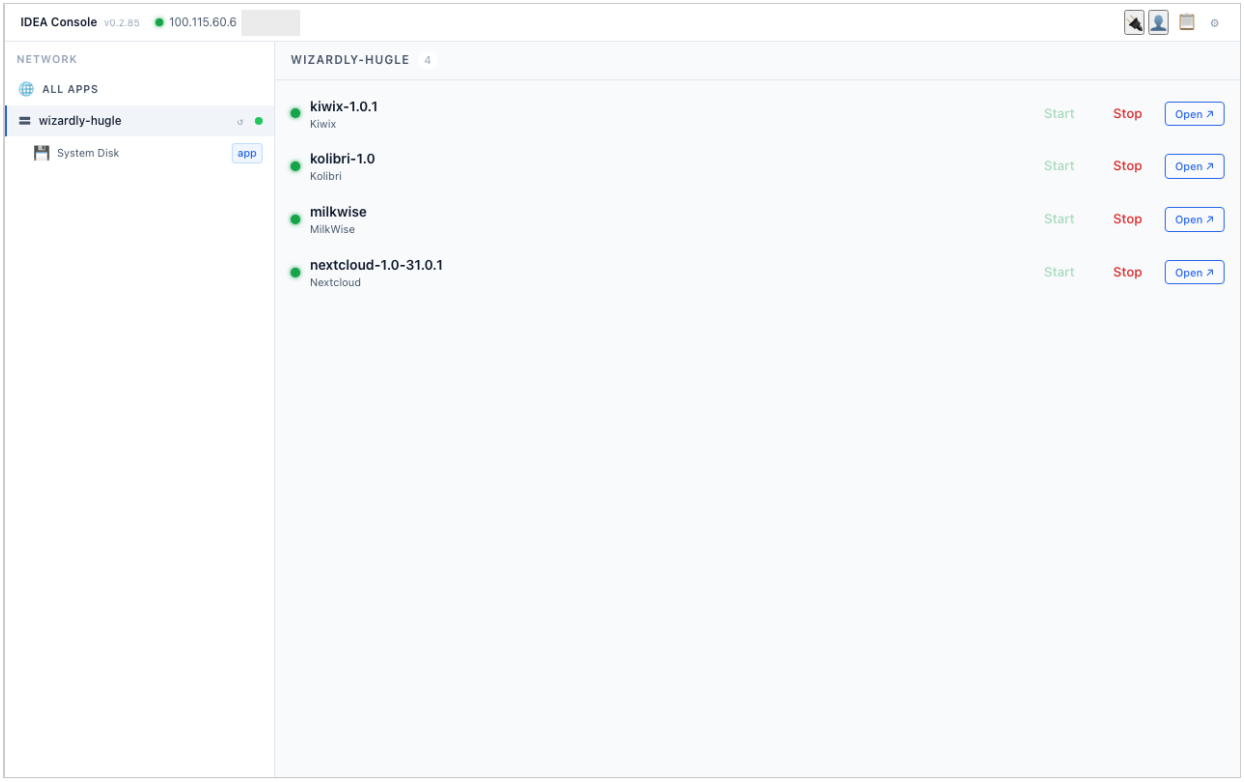


Figure 83. Engine selected

3. Click a disk (here **System Disk**). Each disk shows role badges: **app**, **backup**, **files**, **empty**. The right side lists the **Apps** on that disk with a count.

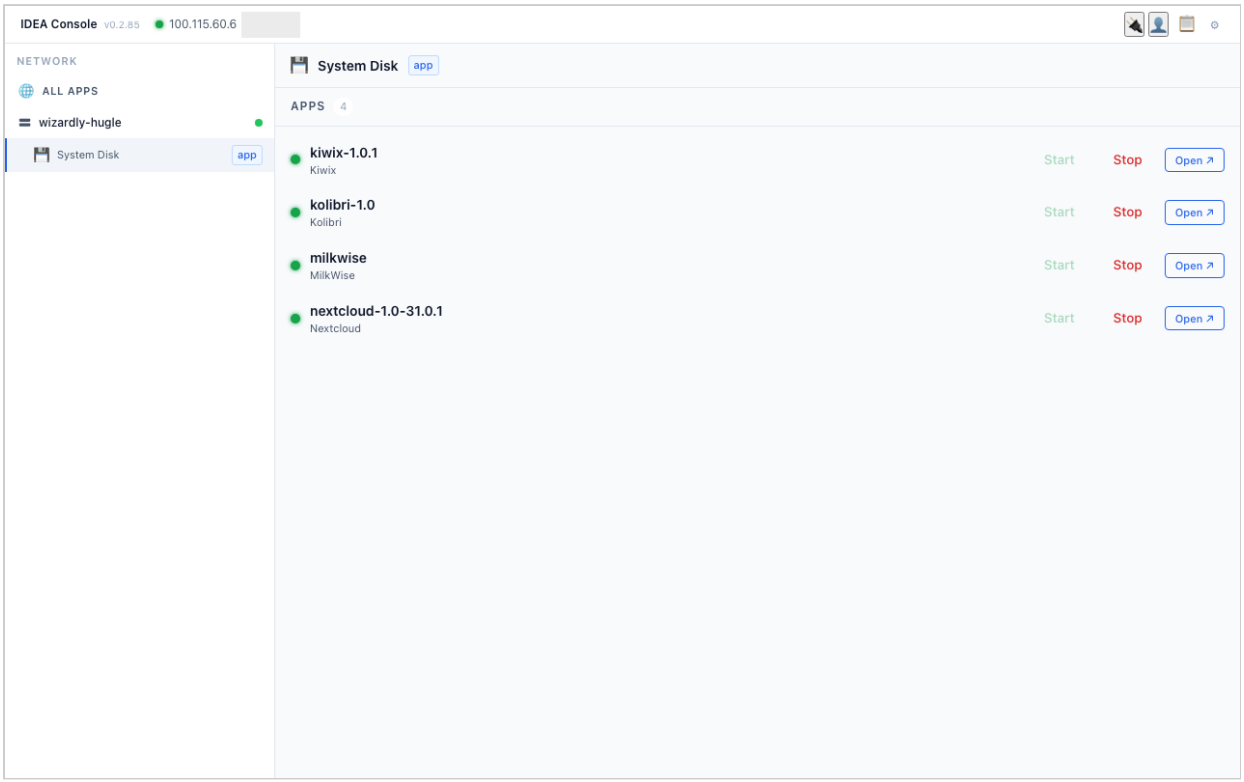


Figure 84. Disk view

O3. App status

1. Each app row shows the app name and version (e.g. `kolibri-1.0`), a coloured status dot, and buttons **Start**, **Stop** and **Open ↗**.
2. Click the status dot (**Show details**) to see: STATUS (e.g. **Running**), VERSION, ENGINE, PORT, CREATED, LAST STARTED, **LAST BACKUP** (e.g. "Never"), the description, a short log, and CONTAINER METRICS (CPU, memory, network, disk).

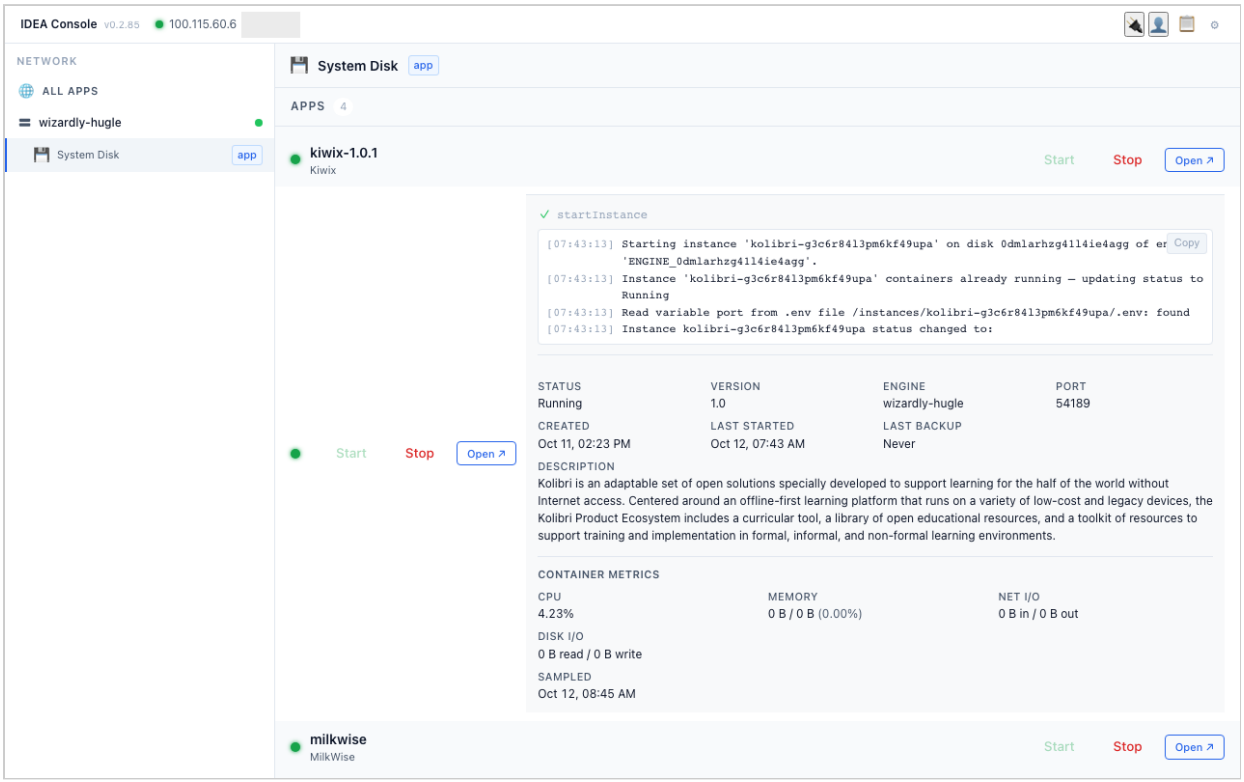


Figure 85. App details

3. Only Start or Stop an app when your IDEA contact asks you to. Never touch **MilkWise** or other non-school apps.

O4. Command History

1. The clipboard icon (top right) lists recent Engine commands (for example `startInstance`, `installApp`) with ✓ (done) or ✗ (failed) and how long ago. Use it when you report a problem.

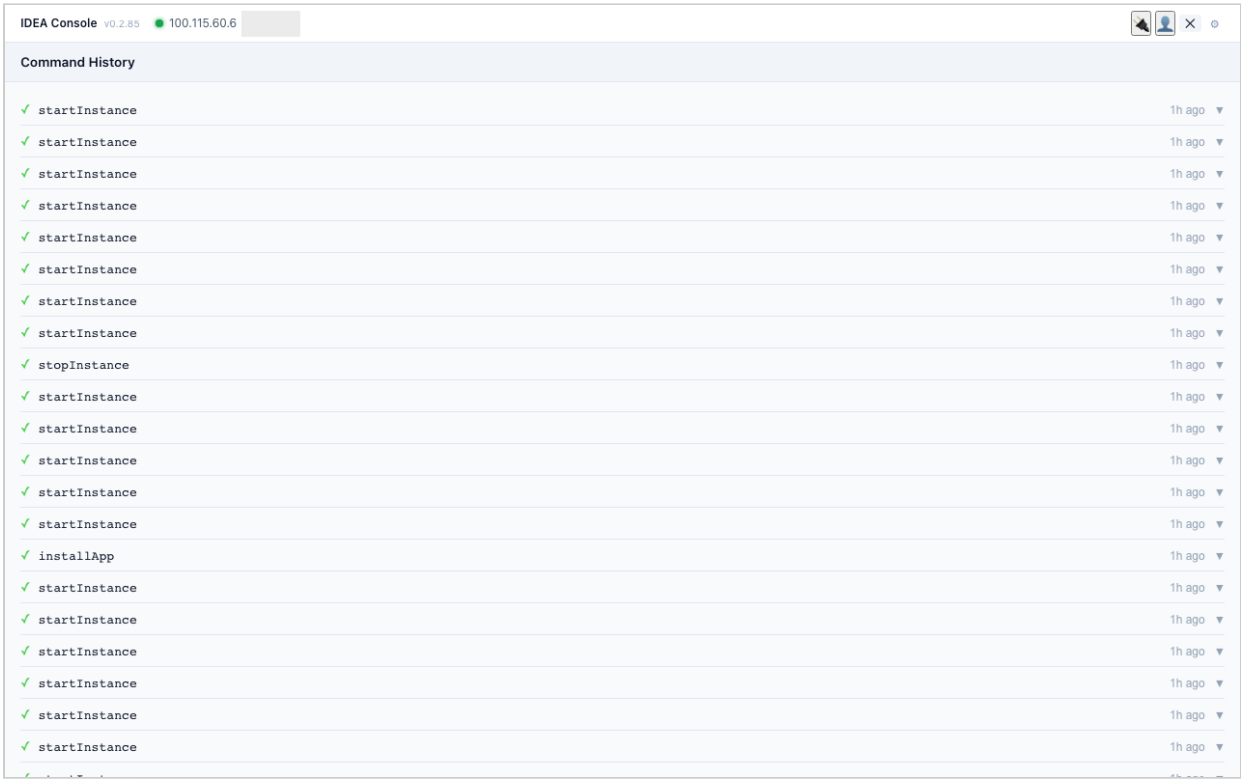


Figure 86. Command History

O4b. The ↺ button is REBOOT

The small **↺** next to the Engine name is **Reboot engine** (it restarts the whole IDEA box). Never press it during lessons.

O4c. A new (unformatted) USB disk

1. A new disk appears under the Engine with its model and size and an **UNFORMATTED** badge (e.g. "INTENSO SSD 256 GB — 256.1 GB").

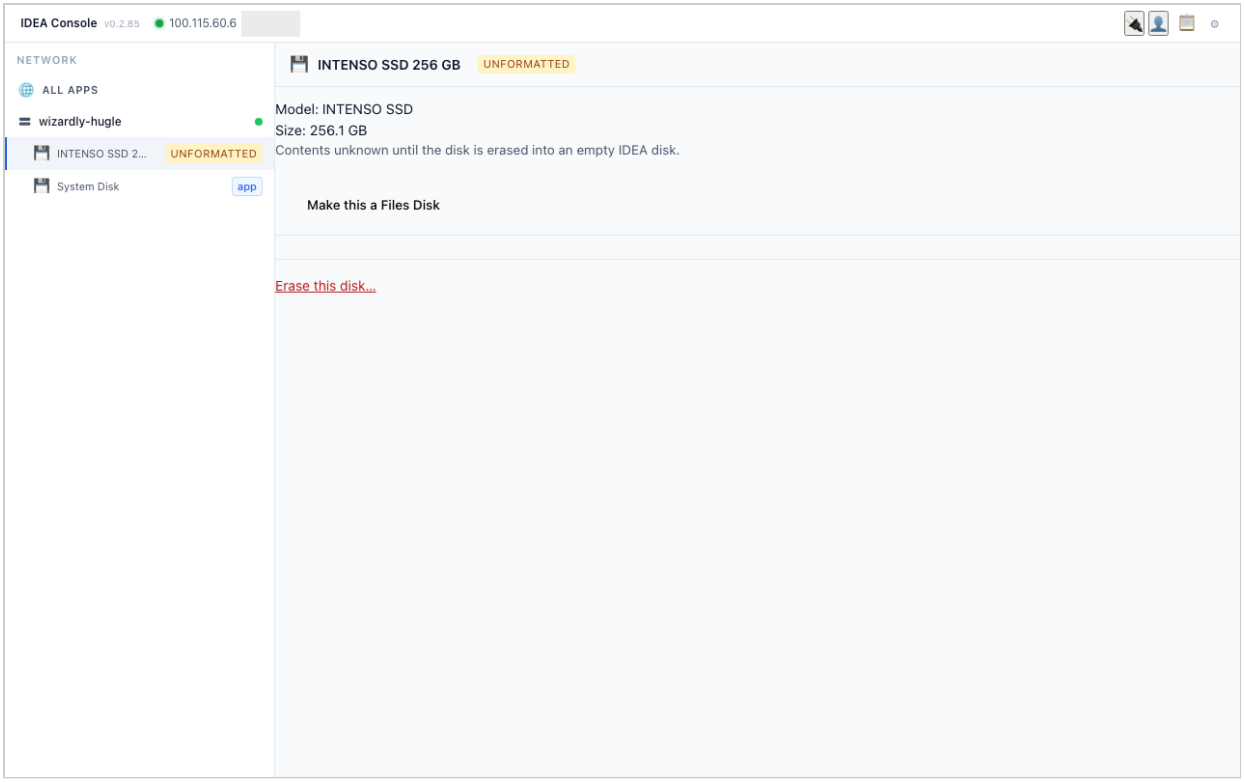


Figure 87. Unformatted disk

2. Its view shows Model, Size, "Contents unknown until the disk is erased into an empty IDEA disk", and two buttons: **Make this a Files Disk** and **Erase this disk…**.
3. **Erase this disk…** opens a confirmation with the disk name, model, size and contents, the red warning "This erases everything on the disk…", and a box **TYPE TO CONFIRM**. The **Erase this disk** button stays grey until the name is typed exactly. Press **Cancel** if anything looks wrong.

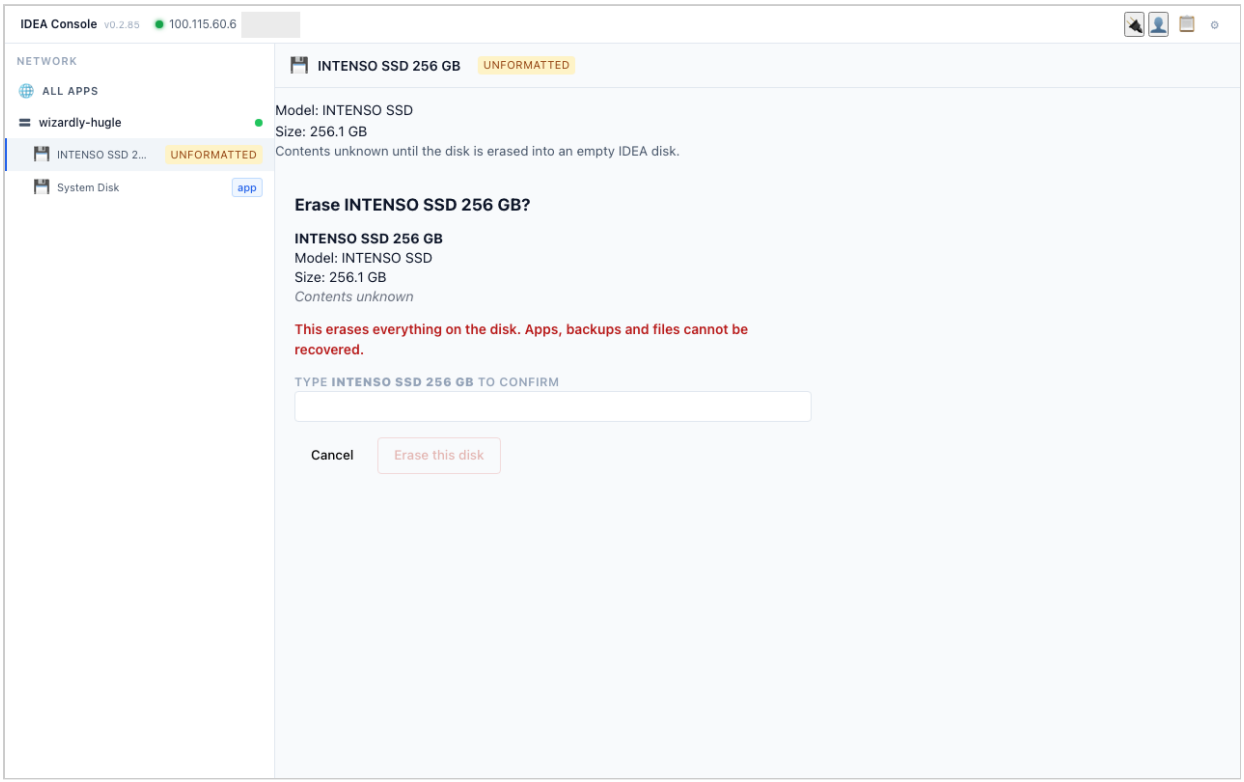


Figure 88. Erase confirmation

Note: the dialog shows model and size but **not the serial number**. If two identical disks are plugged in, unplug one before erasing.

O5. Safely ejecting a disk

The **System Disk** (the box's own disk) has no Eject button. A USB IDEA Disk has a small ⏏ icon next to its name in the Network list, and an **Eject** button in its disk view (below the disk details, above **Erase this disk…**). Eject lists what will stop before you confirm (TO CONFIRM: not clicked on the demo box). Never eject or unplug during a backup.

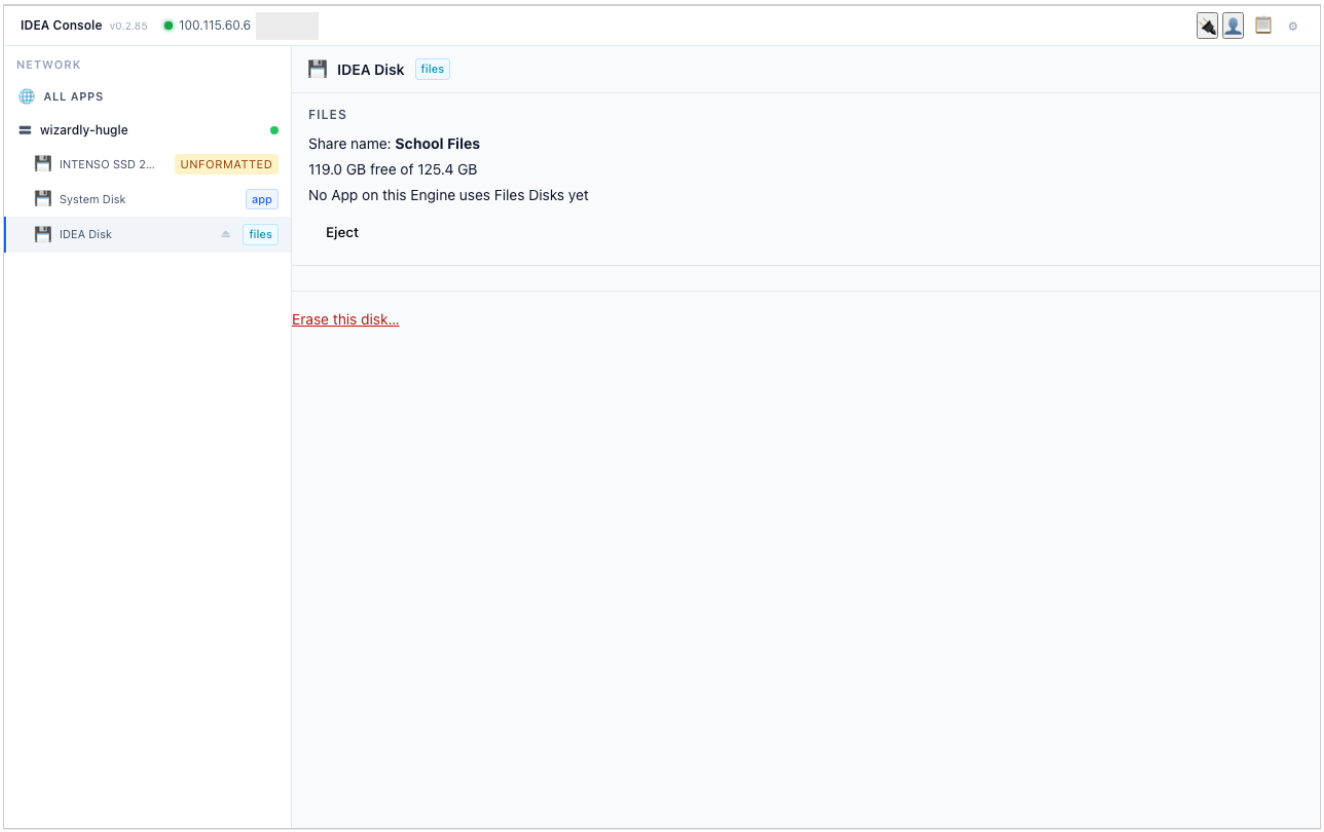


Figure 89. Eject on a Files Disk

O6. Dangerous buttons

Erase this disk…, **Install App**, **Make this a Files/Backup Disk**, **Stop**: only with your IDEA contact on the phone. (Erase and Make this a Files Disk appear only on non-system disks; confirmed on a spare disk. The System Disk has neither.)

O7. Backup and restore (later)

See chapter 9 (hands-on): Make this a Backup Disk, On demand / Immediate / Scheduled, LINK TO INSTANCES, **Back up** button on each linked app row, **Restore** on the Backup Disk (needs a spare target disk). The **LAST BACKUP** line in app details (O3) shows when an app was last backed up.

O8. Known Console quirks

- The coloured status dot can be wrong (Nextcloud showed green while stopped). Check that **Open** works.
- Version labels can be stale (Nextcloud showed 1.0 while 1.1 was installed).
- A restore moves the app to the target disk and leaves it stopped (chapter 9).

Appendix: demo accounts

> Real accounts on wh (2026-10-12): Kolibri uses UNDERSCORES (demo_teacher1, demo_learner1-3, facility demo-School, learners sign in without password); Nextcloud uses hyphens. Created by **Atlas** before capture, removed by Atlas after. All carry the `demo-` prefix. Passwords: set by Atlas, kept out of this file and out of screenshots. Facility/site names on wizardly-hugle: **TO CONFIRM** (Atlas).

Username	App	Role	Class / group	Used in
demo_teacher1	Kolibri	Facility admin + coach	demo-Grade5A	K1–K5, W1
demo_learner1	Kolibri	Learner	demo-Grade5A	S1, S2
demo_learner2	Kolibri	Learner	demo-Grade5A	S1, S2 (2nd data point in reports)
demo_learner3	Kolibri	Learner	demo-Grade5A (enrolled live in K2)	K2
demo-teacher1	Nextcloud	User (no admin)	groups: demo-Grade5A, demo-teachers	N1–N3
demo-teacher2	Nextcloud	User (no admin)	group: demo-teachers	N2 co-editing
demo-learner1	Nextcloud	User	group: demo-Grade5A	S3, S4
demo-learner2	Nextcloud	User	group: demo-Grade5A	S3 (2 avatars in collab doc)
demo-operator	Console	Console viewer (no fixture login today; **TO CONFIRM** whether Console has accounts at all)	—	C1–C3, ch. 2

Kolibri objects created during capture (also prefixed): class `demo-Grade5A`; lesson `demo-Fractions lesson` (3 videos + 3 exercises from Khan EN-US "Add and subtract fractions"); quiz `demo-Fractions quiz`. Nextcloud objects: folders `demo-Worksheets` (view-only share to demo-Grade5A), `demo-Drop Zone` (file request link), document `demo-Lesson plan` (edit share to demo-teachers + demo-Grade5A). Notes: demo-learner3 exists but is NOT enrolled before K2. Kolibri Coach vs Facility coach role: **TO CONFIRM** with Atlas (class coach is enough for K1? creating a class needs Admin → if so demo-teacher1 = Facility admin, **TO CONFIRM**). Cleanup by Atlas: delete all demo-* users, class, lesson, quiz, folders, shares and the file request link.